Regulatory Capture by Superstars: Lobbying, Innovation, and Industry Concentration*

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Abstract

A long-established concern about corporate lobbying is regulatory capture: incumbent firms shaping regulation in their favor. We study which regulations firms lobby for and how this affects concentration, innovation, and welfare. Using LLMs, we classify federal lobbying reports of public firms and trade associations, assigning each request to a policy tool from a new comprehensive taxonomy, with the direction and scope of the requested change. Lobbying for firm-specific benefits such as tax decreases and government contracts is one-sided: nearly every firm asks for more. Lobbying over regulatory costs is instead contested: the largest firms are the most likely to ask for higher costs, while trade associations mostly ask for lower ones. To quantify the consequences of lobbying, we build a dynamic general equilibrium model of oligopolistic competition and innovation in which firms lobby for firm-specific transfers and over industry-wide regulatory costs, set in a contest between strategic incumbents and an association of the competitive fringe. In the model estimated on US data, lobbying raises regulatory costs and lowers welfare through industry composition: each incumbent innovates more, but industries become more concentrated, lowering output and growth. Almost all the welfare loss comes from lobbying over regulation rather than for benefits.

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1 Introduction

Americans rank the role of money in politics among the country's most serious problems (Pew Research Center, 2026). One reason for concern is regulatory capture: firms can use money to shape regulation in their favor at the expense of other firms and the public interest. For example, large AI companies are asking the government to regulate their industry. Such regulation may make AI safer, but it may also raise the costs of smaller competitors and protect the market shares and profits of the largest firms. Concerns about regulatory capture extend well beyond AI. In 2015–2016, Altria and Reynolds lobbied on a bill that would have exempted the vaping products already on sale from premarket review by the Food and Drug Administration (FDA). In 2020–2021, both lobbied for FDA regulation of synthetic-nicotine products. H&R Block, the largest listed tax-preparation firm, lobbied for minimum competency standards for paid tax preparers. C.H. Robinson, the largest listed freight broker, lobbied to raise the bond that brokers must post to guarantee their payments to shippers and truckers. In 2012, Congress raised it from \$10,000 to \$75,000, the same amount for every broker. The cost of each of these requirements does not rise in proportion to a firm's sales, so it weighs most on smaller rivals and entrants.

Firms shape government policy mainly through lobbying and political donations. Lobbying is the larger expense. From 2010 to 2022, public firms spent 16 times as much on lobbying as their political action committees (PACs) contributed to candidates.[1] Unlike donations, lobbying is documented in mandatory reports that name the policies firms discussed with the government. These policies fall into groups with very different economic effects. One group benefits the lobbying firm without directly harming other firms, such as higher subsidies, appropriations, and contracts and lower tax rates. Another group changes the regulatory costs of operating in an industry or in the whole economy, such as compliance rules, safety and certification standards, data privacy rules, and licensing. These requirements act mainly as fixed costs, so they weigh more on smaller firms, raise barriers to entry, and increase industry concentration (Ryan, 2012; Johnson et al., 2023; Kalmenovitz et al., 2025).

This paper studies what firms lobby for and how lobbying affects industry concentration, innovation, growth, and welfare. Existing work documents how much firms spend on lobbying, how persistent lobbying is, and which broad topics it covers, but not which specific policy instruments firms lobby for, which side they take on each, or how often.

The effect of lobbying on welfare is not obvious, because it works through both static

[1] Our calculation from the lobbying reports and the Federal Election Commission filings of corporate PACs matched to Compustat firms, election cycles 2010–2022, in 2015 dollars: \$15.4 billion of lobbying against \$0.97 billion of PAC contributions to candidates.

and dynamic channels. Higher regulatory costs push marginal small firms out of the market and so raise the profits of large firms. This can reduce welfare through lower supply and misallocation. However, higher profits raise the value of innovating, so higher regulatory costs could accelerate growth and raise welfare. Regulation can also raise welfare by correcting externalities. We abstract from these benefits and study the effects of regulatory costs on consumption and labor supply.

Data and facts. We classify the federal lobbying reports filed from 2008 to 2022 with a large language model (LLM). For each client, the firm or association on whose behalf the lobbying is done, we extract every request from the free text of its reports and assign it to a specific policy instrument, such as corporate taxes, subsidies, appropriations, R&D credits, privacy regulation, or licensing, from a new taxonomy of 87 instruments. For each request, we record the direction of the requested change, whether the client supports it, and which firms and industries the policy applies to. We focus on public firms, which lobby the most and whose size, costs, and innovation we observe. We also cover trade associations, the nonprofits that represent firms, which report their revenues and dues in their tax filings. Together, US-headquartered public firms and trade associations account for 58% of federal lobbying spending.

Firms agree on direct benefits and disagree on regulatory costs. Almost all firms and associations ask for more appropriations and lower taxes. On regulatory costs they disagree. Among requests that take a side, 63% of those of the largest 5% of firms in each four-digit industry ask for higher costs, against 47% for all public firms and 36% for associations. Firms lobby most on appropriations and on regulatory costs such as privacy rules, reporting rules, and rules on how transactions are conducted. As in Huneeus and Kim (2024), the share of firms that lobby rises steeply with size, from 6% of the firm-years below the industry's median sales to 63% among the largest 5%, and nine in ten firms that lobby on an instrument in one quarter lobby on it in the next.

Motivated by these findings, we model lobbying on three policies: regulatory costs (such as privacy, licensing, and compliance requirements), the corporate tax rate, and procurement contracts. We choose these policies for three reasons. First, they let us study the industry-wide contest over regulatory costs alongside lobbying for taxes and procurement, the direct benefits that the existing literature studies. Second, they let us compare the welfare consequences of the two kinds of lobbying. Third, their outcomes are measurable. For each policy, we measure how lobbying participation varies with firm size and how persistent it is. For taxes and procurement, we also measure the return to lobbying in effective tax rates and contracts. For regulatory costs, we observe their total level and how lobbying affects the passage of bills that raise or lower them.

Quantitative model. Because large firms lobby individually while small firms rarely do, and associations side with smaller firms more often than with the largest, we build a dynamic quantitative model in which each industry consists of a few large strategic firms, which we call superstars, and a mass of atomistic fringe firms. Superstars compete à la Cournot, and fringe firms are perfectly competitive. Superstars innovate to raise their quality, and fringe firms can make a breakthrough and become superstars. Fringe firms differ in their operating costs, so higher regulatory costs make the fringe firms with the highest operating costs stop producing.

Superstars lobby for lower corporate taxes and for more procurement contracts, and superstars and the fringe lobby over regulatory costs. To lobby on an instrument, a firm pays a fixed participation cost and can make its lobbying more effective by spending more. Because fringe firms are atomistic, none of them would lobby on its own, so they lobby through a trade association that acts on behalf of its members. Regulatory costs are set in a contest that aggregates the lobbying of firms that want them higher against the lobbying of firms that want them lower. Firms take into account how regulatory costs change the number of active fringe firms and how many of them become superstars.

Results. In the estimated model, superstars lobby for higher regulatory costs, while the association lobbies against them. The largest superstars favor higher regulatory costs both because their profits rise as marginal fringe firms stop producing and because higher costs protect them from breakthroughs. The smallest superstars favor them mainly as protection against being displaced by a new superstar. On net, lobbying raises regulatory costs by 3% relative to the economy without lobbying, but it raises them in 41% of industry states and lowers them in 57%. Removing lobbying raises the number of active fringe firms by 0.7%. More of them make breakthroughs, so the share of industries with four superstars rises and industries become less concentrated.

Removing lobbying raises consumption-equivalent welfare by 0.22%. Saved lobbying spending accounts for 0.09 percentage points of this gain, lower compliance costs for 0.02, and the change in industry structure for the remaining 0.11. Without lobbying, superstars innovate less, but growth rises, because more industries have four superstars and their frontier advances faster.

Lobbying over regulatory costs accounts for 99% of the welfare gain, and tax and procurement lobbying, the focus of the literature, accounts for almost none, because it mostly shifts money from the household as a taxpayer to the household as a firm owner. Thus, the most salient outcome of lobbying, large firms paying lower taxes, is not what reduces welfare.

Related literature. The empirical literature on lobbying documents that lobbying rises with firm size and is persistent over time (Huneeus and Kim, 2024). Arayavechkit et al. (2018)

measure lobbying frequency by general issue code, find that taxes are the most frequent topic, and study tax lobbying by atomistic firms. In closely related work, Macedoni and Weinberger (2026) classify firms' positions on regulations in their lobbying reports with machine learning. They find that larger firms, especially in concentrated industries, support stricter regulations, as their model predicts when regulations mainly raise fixed costs, while capital-intensive and highly leveraged firms oppose them. We contribute a systematic classification of the specific policy instruments that firms and associations lobby for, such as corporate taxes, subsidies, appropriations, R&D tax credits, privacy regulation, and licensing, along with the requested change, the client's position, and the persistence of lobbying on each instrument. We show that firms agree on direct benefits and disagree on regulatory costs. Consistent with Macedoni and Weinberger (2026), the largest firms are the most likely to ask for higher regulatory costs, and trade associations ask for them least often.

In static models of lobbying, policy is determined by aggregating the pressures of groups lobbying for or against it (Becker, 1983; Becker and Mulligan, 2003; Grossman and Helpman, 1994; Bombardini, 2008; Kang, 2016). In models with many atomistic firms, firms lobby for lower taxes or higher sales (Arayavechkit et al., 2018; Huneeus and Kim, 2024; Hossein Dad and Nguyen, 2025). However, because these firms are atomistic, they cannot move industry-wide regulation on their own. We develop a dynamic quantitative model in which firms lobby both for direct benefits, such as taxes and procurement, and over industry-wide regulation. Regulation is set in a contest between superstars and an association of fringe firms.

The model belongs to the literature on oligopoly and innovation (Ericson and Pakes, 1995; Akcigit and Ates, 2023). We build on the model of Cavenaile et al. (2026), which introduces innovating oligopolists and a competitive fringe in general equilibrium. We add lobbying and make two changes to the model. First, fringe firms differ in their operating costs, so a higher regulatory cost makes the fringe firms with the highest operating costs stop producing. Second, a fringe firm that makes a breakthrough need not become a superstar at the lowest quality. It can displace a superstar and even leapfrog the leader. This creates an additional lobbying motive: large firms lobby not only to raise their own profits and market shares, but also to protect themselves from innovative smaller firms. We contribute to this literature by studying how lobbying over regulation affects industry structure and innovation.

2 Classifying lobbying reports into policy instruments

We measure what firms lobby for from the free text of the quarterly federal lobbying reports filed under the Lobbying Disclosure Act from 2008 to 2022 (Appendix A shows a

sample report).

The raw data are lobbying reports. Each report covers one client, one registrant (a lobbying firm or the client's own lobbyists), and one quarter. We split the free text of the reports into individual requests and assign each to one of 87 policy instruments, along with the requested change relative to current law, whom the policy covers, and the client's position. For example, the request "support a decrease in the corporate tax rate" is recorded as the corporate income tax rate, a decrease, covering all firms, and supported by the client. The final dataset records, for every client, quarter, and instrument, whether the client lobbies on the instrument and the side it takes.

2.1 Reports and clients

Every registrant files a report (form LD-2) for each client every quarter. The report names the client and the registrant, states the amount spent, and, for each general issue area, describes in free text the specific issues lobbied, often citing bills, public laws, and regulations. We use every report filed from 2008 to 2022, assembled from OpenSecrets (OpenSecrets, 2024), LobbyView (Kim, 2018), and the Senate Office of Public Records. Free issue text is largely missing before 2007. Among reports with a positive amount, 21% include it in 1999–2006, against 96% in 2007 and 98% in 2008–2022. In 2008–2022, reports with issue text account for 99% of lobbying spending. We start in 2008 because the Honest Leadership and Open Government Act of 2007 replaced semiannual with quarterly reporting, so every year in our sample has the same reporting frequency.

We match clients to public firms and trade associations, which account for 65% of lobbying dollars, 36% of reports, and 52% of the issues listed in 2008–2022 (Figure 1). We choose these clients because we observe industry, size, costs, and innovation for public firms and revenues and dues for associations. Associations also represent smaller firms. We match public firms to Compustat by name. We match their subsidiaries and merged companies through the Center for Research in Security Prices (CRSP) and Securities and Exchange Commission (SEC) filings on EDGAR, and attribute a subsidiary's lobbying to its parent. We keep firms headquartered in the United States, which leaves 75,957 firm-years for 9,690 firms in 268 four-digit North American Industry Classification System (NAICS) industries. These firms and the associations account for 58% of lobbying spending. We match trade associations to the Internal Revenue Service (IRS) register of business leagues (501(c)(6) organizations) and assign each industry association, a trade association that represents one industry, to that industry.

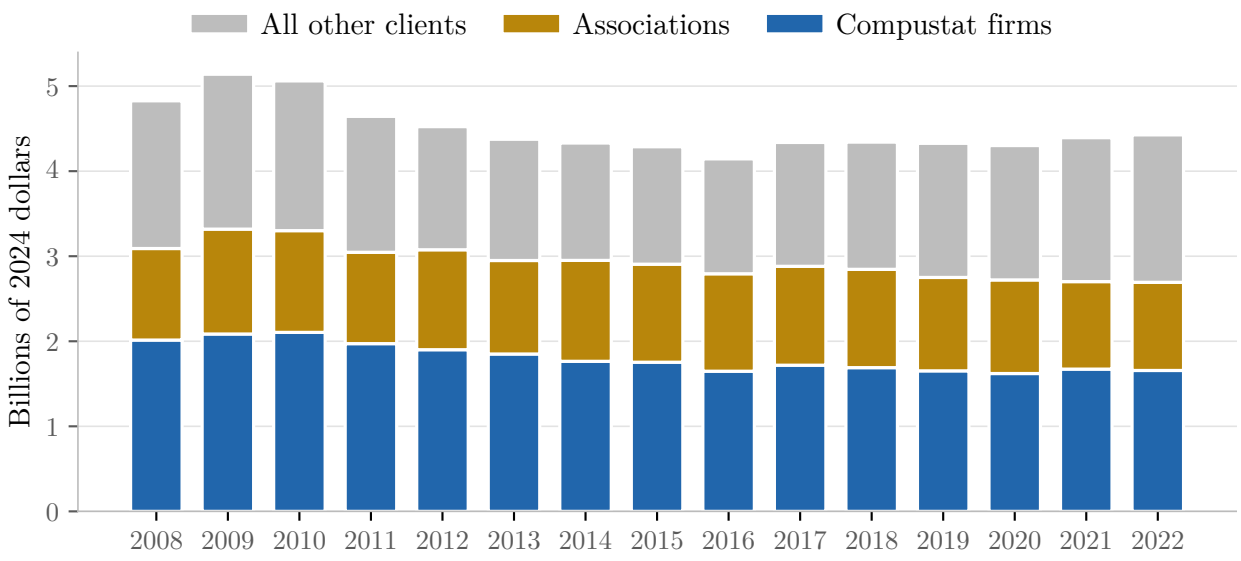


Figure 1. Who lobbies: Compustat firms and associations

Notes: Federal lobbying spending by type of client, billions of 2024 dollars, 2008–2022. Associations are clients matched to an IRS 501(c)(6) organization.

2.2 From text to policy instruments

A single issue text block often contains several requests, each on a different policy and often citing bills or regulations. We turn these blocks into instrument-level measures in five steps.

Extracting requests. We split each issue text block into requests that each address a single policy. Pattern rules split unambiguous structures, such as enumerated lists. For blocks whose boundaries require judgment, an LLM splits the text and must reproduce it verbatim (Appendix B).

Bills. We match every cited bill and public law to the congressional record, correcting mistyped numbers with the bills' titles and context. We match cited regulations to their titles and summaries on Regulations.gov. We split the summary of each matched bill into provisions and classify each provision in the same way as the issue text.

Instruments. We assign each request to an instrument in a taxonomy of 87 policy instruments. The taxonomy combines the policies that affect firms in the following inventories: the Treasury tax-expenditure list, the regulatory forms of Xie and Pérez (2018), the Harmonized Tariff Schedule and the United Nations Conference on Trade and Development (UNCTAD) classification of non-tariff measures, the federal Assistance Listings and the Federal Acquisition Regulation, and the patent, exclusivity, and immigration laws. We check its completeness against the tools-of-government catalog of Salamon (2002). We map each instrument

to the term of the firm's problem it changes, such as labor costs, output, or the cost of innovation. We call this term the instrument's wedge (Table 1). Instruments are defined by their effect on the firm rather than by whom they cover: a research credit applying to one firm and one applying to every firm are the same instrument with different eligibility (Appendix C). The instruments of the regulatory-cost wedge are requirements whose cost of compliance does not scale with the firm's output, or does so only in part.

Table 1. Wedges of the firm's problem and the instruments that affect them

Wedge	**Instruments (examples)**	**Number**
Demand, s^d	Appropriations and earmarks, procurement contracts, buyer purchase subsidies, tariffs and quotas, antidumping duties	14
Regulatory cost, s^R	Licenses and charters, product certification, pre-market approval, facility permits, recurring reporting, labeling, data privacy and cybersecurity, liability rules	18
Output, s^y	Per-unit production credits, excise taxes, emission charges, performance standards with per-unit costs, quotas	17
Profit tax, s^T	The corporate income tax, deductions, reduced rates on classes of income, foreign income, losses	7
Innovation cost, s^i	Research credits and expensing, research grants, patents, product exclusivity	5
Capital cost, s^k	Investment credits, accelerated depreciation, tax-exempt bonds, federal lending	6
Labor cost, s^w	Hiring credits, payroll taxes, wage floors, worker protections, visas	8
Set prices, s^p	Government payment rates, mandatory discounts, price ceilings and floors, rate regulation	5
Competition, s^A	Merger control, platform regulation, antitrust exemptions, mandated access	4
Lump-sum transfers, s^t	Grants and direct payments, regulatory user fees	2
Place-based policies	Enterprise zones	1

Notes: The taxonomy's 87 instruments are grouped by the wedge of the firm's problem they affect. The regulatory-cost wedge includes requirements whose cost scales with output only in part (liability and conduct rules); requirements whose cost scales fully with output belong to the output wedge. Symbols are defined in Appendix C. Appendix C lists the instruments.

Direction, coverage, and side. For each request, the classifier records the change requested relative to current law (for example introduce, raise, lower, widen, narrow, extend, fund, exempt, delay, or repeal, with the magnitude when the text states it), whom the policy covers (industry, geography, product, legal form, and thresholds on firm characteristics), and the client's position on it (support, oppose, amend, monitor, or unstated). The classifier also reports its confidence in each assignment as high, medium, or low. For each instrument, the taxonomy records whom it favors or burdens (buyers and sellers of the good, domestic and

foreign firms) and how its cost scales with the firm's output (proportionally, quasi-fixed, or mixed). A client's side on a request is its position times the direction of the change. For example, supporting a bill that raises a compliance requirement and opposing its repeal both place the client on the side of higher regulatory costs. However, reports rarely state a position. Clients support or oppose the policy in only 6% of firms' matched issues and 16% of associations' (Table 2).

From requests to client measures. We record a client as lobbying on an instrument in a quarter if one of its reports meets either of two conditions. First, it names the instrument in the client's own words, classified at high or medium confidence. Second, it cites a bill, a section or title of a bill, or a regulation all of whose cited provisions, classified at high or medium confidence, concern this instrument. The bill's name plays no role. A citation whose provisions concern several instruments counts for none of them, even when the instruments share a wedge. Otherwise, citing a defense authorization bill would count a firm as lobbying on procurement, export controls, and cybersecurity. In Appendix E, we instead count every request, giving each instrument of a citation the share of its provisions on that instrument. A client's share of requests for a higher level of an instrument in a year is computed over all reports filed for the client that year. It counts only requests stated in the client's own words at high or medium confidence, and citations that state support or opposition and whose provisions concern a single instrument.

Coverage and validation. Table 2 reports the coverage of the classification, which uses Claude Opus 5.5. Among the issues listed in the reports, 46% for firms and 47% for associations are matched to a policy instrument, either in the text itself or through a cited bill whose section or side the text names. About a third are bare citations of a bill or rule, which enter through the provisions of the legislation. The rest contain no firm policy instrument or are too vague to classify, such as "climate change" or "tax reform". We drop low-confidence issues (10% of firms' and 11% of associations' matched issues). Classifying the same text twice gives the same instruments in 94% of cases when neither classification has low confidence, and the same position in 99%. A second language model, Claude Fable 5.1, run independently on the same taxonomy, assigns the same instrument to 83% of a sample of 1,000 high- and medium-confidence issues of industry associations in 2008–2018, and to 89% of a sample of 1,000 bill provisions. The positions we record on bills agree with those recorded by MapLight, a database of organizations' positions on bills, for 95% of the 492 client–bill pairs both cover (93% of the 55 firm–bill pairs). We are hand-coding a validation sample to debias the estimates following Carlson and Dell (2025). With these instrument-level measures, we turn to what firms lobby for.

Table 2. Classification: coverage and confidence

	Compustat firms 2008–2022	Associations 2008–2022
Reports with issue text	169,612	125,556
Issues listed in the reports	1,298,944	1,557,060
matched to a policy instrument	46%	47%
bare citation of a bill or rule	31%	35%
no policy instrument	11%	8%
unclear (e.g., "climate change", "tax reform")	13%	10%
Confidence of the matched issues:		
high	51%	45%
medium	37%	40%
low (dropped)	10%	11%
cites a bill's section or states a side, no own words	2%	4%
Position stated in the matched issues:		
support or oppose	6%	16%
monitor	5%	4%
unstated	89%	80%

Notes: Compustat firms headquartered in the United States and trade associations, 2008–2022. Shares of the issues listed in the reports.

3 What firms lobby for

Firms lobby most on demand and on regulatory costs. They agree on direct benefits and disagree on regulatory costs. Lobbying rises steeply with firm size and is persistent over time.

Participation in lobbying. The share of firms and associations that lobby rises steeply with size. Figure 2 reports the share of firm-years and association-years with a lobbying report, by size: for firms, the sales percentile within the industry, and for associations, the revenue percentile among 501(c)(6) organizations. Of the firm-years below their industry's median sales, 6.1% have a report, against 19% between the median and the 80th percentile, 35% between the 80th and the 90th, 48% between the 90th and the 95th, and 63% among the largest 5%. Associations show the same pattern: among 501(c)(6) organizations, the share with a lobbying report rises from 0.3% below the median revenue to 1.9% between the median and the 80th percentile, 6.1% between the 80th and the 90th, 12% between the 90th and the 95th, and 32% among the largest 5%.

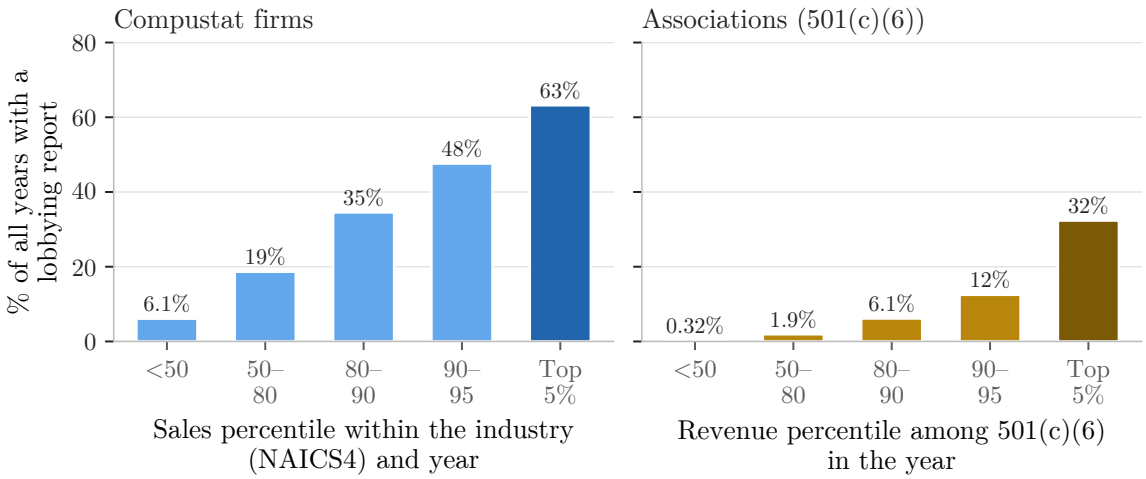


Figure 2. Lobbying by firm size

Notes: Share of all firm-years with a lobbying report (a report with a positive amount that names a classified instrument), by the firm's sales percentile within its industry (four-digit NAICS) and year; 75,957 firm-years of Compustat firms headquartered in the United States. Associations: share of the years in which a 501(c)(6) organization filing an IRS Form 990 or 990-EZ with positive revenue (30,112 a year on average) is a lobbying client with a positive amount, whether or not its reports name a classified instrument, by its revenue percentile among these organizations in the year. 2008–2022.

3.1 Fact 1: firms lobby most on demand and on regulatory costs

Demand and the regulatory-cost wedge are the most lobbied wedges. Figure 3 reports, for each wedge, the share of a client's lobbying quarters in which it lobbies on that wedge. Firms that lobby do so on demand, which covers government purchases, buyer subsidies, and trade protection, in 63% of their lobbying quarters and on the regulatory-cost wedge in 50%. For associations, the shares are 66 and 49%. The profit tax, the subject of a large literature on corporate lobbying (Richter et al., 2009; Arayavechkit et al., 2018), appears in about a fifth of the quarters (20% for firms, 15% for associations). The largest 5% of firms lobby on every wedge more often, on demand in 75% of their quarters and on the regulatory-cost wedge in 66%.

At the level of instruments, program funding and appropriations lead for every group (32% of firm-quarters, 38% of association-quarters, and 43% for the largest 5% of firms). For firms, they are followed by data privacy and cybersecurity, the most lobbied instrument of the regulatory-cost wedge (21%), technology and performance standards (20%), procurement contracts (19%), buyer subsidies and purchase mandates (19%), and public infrastructure (16%). Associations lobby more than firms on standards, reporting duties, and payment rates, and far less on procurement (8 versus 19%). Table 3 lists the three most lobbied instruments in six sectors. Firms lobby on what they sell to the government and on the rules their regulator sets. An instrument of the regulatory-cost wedge is among the three in five

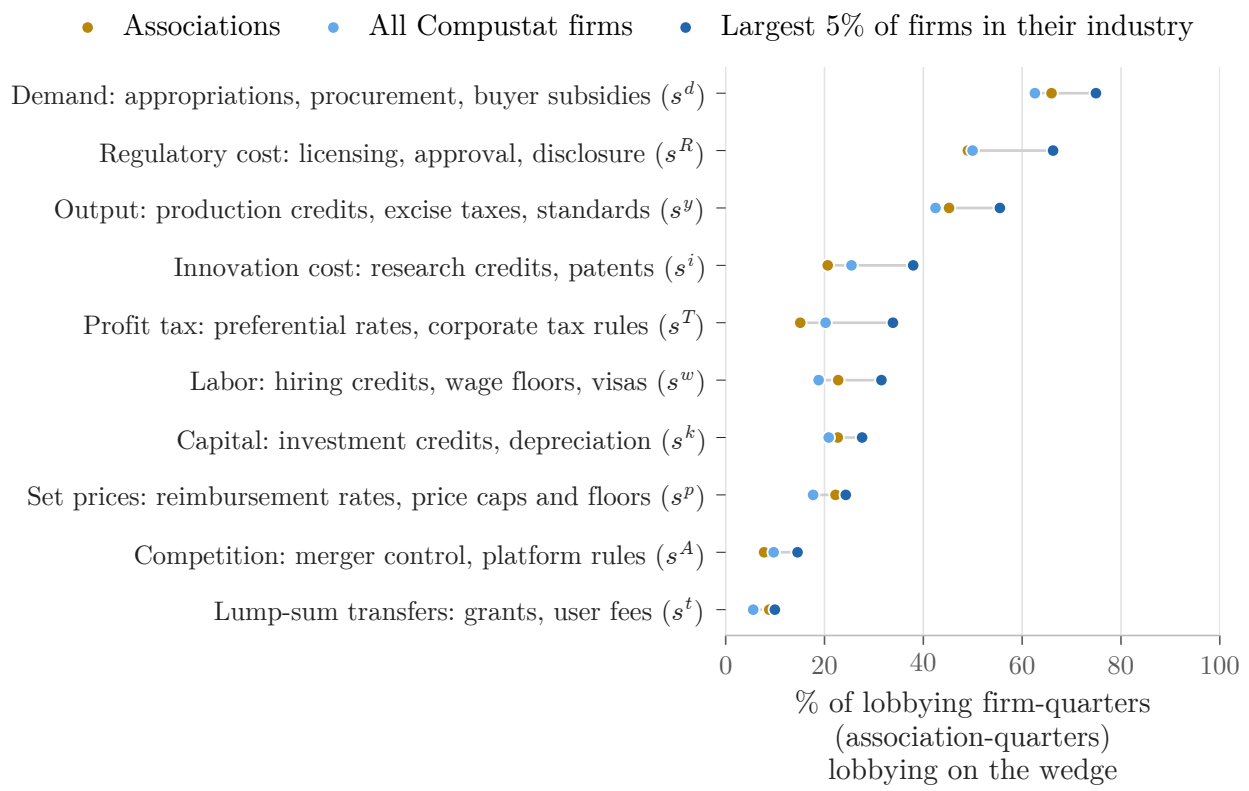


Figure 3. Lobbying by wedge of the firm's problem

Notes: Share of lobbying firm-quarters (association-quarters) in which the client lobbies on the wedge. 52,079 Compustat firm-quarters, of which 14,970 of the largest 5% of firms in their industry by sales; 60,596 association-quarters; 2008–2022.

of the six sectors: conduct rules on transactions in health care services and in finance, premarket approval in pharmaceuticals and devices, privacy in technology, and facility permits in energy and utilities.

Three instruments. The model has three instruments: the firm's own profit tax, its own federal procurement contracts, and regulatory costs. From here on, regulatory costs denote four instruments of the regulatory-cost wedge whose cost grows less than proportionally with the firm's size: licenses, recurring reporting duties, labeling, and privacy and cybersecurity rules. Reports that include the three instruments account for 61% of firms' lobbying dollars. Tax and procurement are direct benefits, and regulatory costs are contested.

3.2 Fact 2: firms agree on direct benefits and disagree on regulatory costs

Firms agree on instruments that benefit them directly. Figure 4 reports, for each instrument, the mean share of requests for more of it among the firm-years and association-years

Table 3. What each sector lobbies on most

	Most common	Second	Third
Aerospace and defense	Procurement (s^d)	Appropriations (s^d)	R&D grants (s^i)
Health care services	Payment rates (s^p)	Appropriations (s^d)	Conduct rules (s^R)
Pharma and devices	Appropriations (s^d)	Payment rates (s^p)	Pre-market approval (s^R)
Technology	Appropriations (s^d)	Procurement (s^d)	Privacy (s^R)
Finance and insurance	Conduct rules (s^R)	Buyer subsidies (s^d)	Privacy (s^R)
Energy and utilities	Technology standards (s^y)	Facility permits (s^R)	Buyer subsidies (s^d)

Notes: The three instruments lobbied in the largest share of the sector's lobbying firm-years, the profit-tax instruments counted as one; Compustat firms, 2008–2022. Conduct rules: rules on how a transaction is made.

that take a side. Between 96 and 100% of requests are for more funding, contracts, grants, and credits, and 97% of requests on the profit tax are for a lower rate. The exception is patents and copyright (53%), which favor firms that hold intellectual property and burden those that use it. On the regulatory-cost wedge, firms split. Depending on the instrument, between 6% (facility permits) and 66% (labeling) of the requests of firms that take a side are for higher costs. Across the four regulatory costs, the mean share is 47% for firms and 36% for associations.

The share of requests for higher regulatory costs rises with firm size. Counting each firm-year once, it is 27% among firms below the 80th percentile of their industry's sales, 43% between the 80th and 95th percentiles, and 63% among the largest 5%. Counting every request, which gives more weight to firms with more requests, the shares are 21, 41, and 72% (Figure 5). Associations ask for higher regulatory costs less often than the largest firms, at 36% (37% counting every request). Within the same industry and year, the associations' share is 27 percentage points below that of the largest 5% of firms (standard error 10) and close to that of firms below the 80th percentile (10 points above, standard error 10).[2] On privacy, the share is 81% among the largest firms and 50% among all firms. The size gradient rests on few observations: only 278 firm-years take a side on regulatory costs, and only 11 of 53 industries have firms stating a side both above and below the industry's 80th percentile of sales (Appendix F). We therefore target the overall shares of sides in estimation and treat the size gradient as an untargeted pattern.

A natural concern is that the sides we observe are selected, because firms state a side on only 1.9% of their requests on regulatory costs. Three checks in Appendix F show that the stated sides are representative, with one exception: the 27 firms that state a side on regulatory costs most often ask for higher costs less often than other firms.

[2]The regression compares the industry associations of the 53 industries in which firms take a side with those firms and includes industry and year fixed effects. Standard errors are clustered by client.

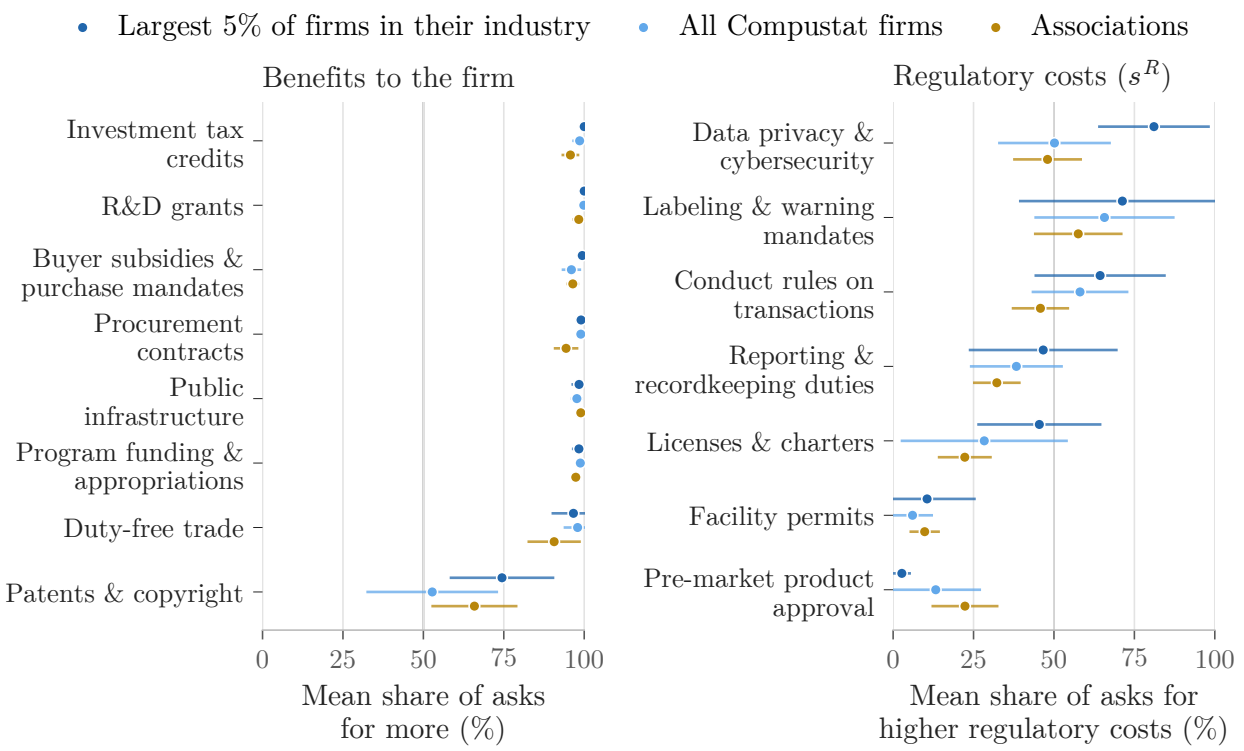


Figure 4. The side firms take: benefits and regulatory costs

Notes: Left: mean share of requests for more of the benefit among the firm-years (association-years) taking a side. Right: mean share of requests for higher regulatory costs. A client-year's share is the share of its requests on the instrument in the year that point to a higher level (Section 2.2), and each filer-year counts once. 95% confidence intervals clustered by filer. 2008–2022.

3.3 Fact 3: lobbying on taxes, procurement, and regulatory costs rises steeply with firm size

Participation on each of the three instruments rises steeply with size (Figure 6). The share of firm-years that lobby on regulatory costs rises from 0.8% below the industry's median sales to 35% among the largest 5%, on the profit tax from 0.4 to 25%, and on procurement from 1.1 to 17%. As Figure 2 shows, most of this gradient is whether a firm lobbies at all. Among the years in which a firm lobbies, the shares still rise with size, from 13 to 55% on regulatory costs, from 6 to 39% on the profit tax, and from 18 to 27% on procurement (Appendix Figure 15). Within an industry and year, a lobbying firm with sales one log unit higher is 9.1 percentage points more likely to lobby on regulatory costs, 7.8 points more likely to lobby on the profit tax, and 2.7 points more likely to lobby on procurement, against averages of 36, 23, and 22%.[3] Among the four largest firms of each industry, participation continues to rise with rank. In a given quarter, 24% of industry leaders lobby on regulatory

[3]Standard errors 0.4; 13,669 lobbying firm-years, 2008–2022.

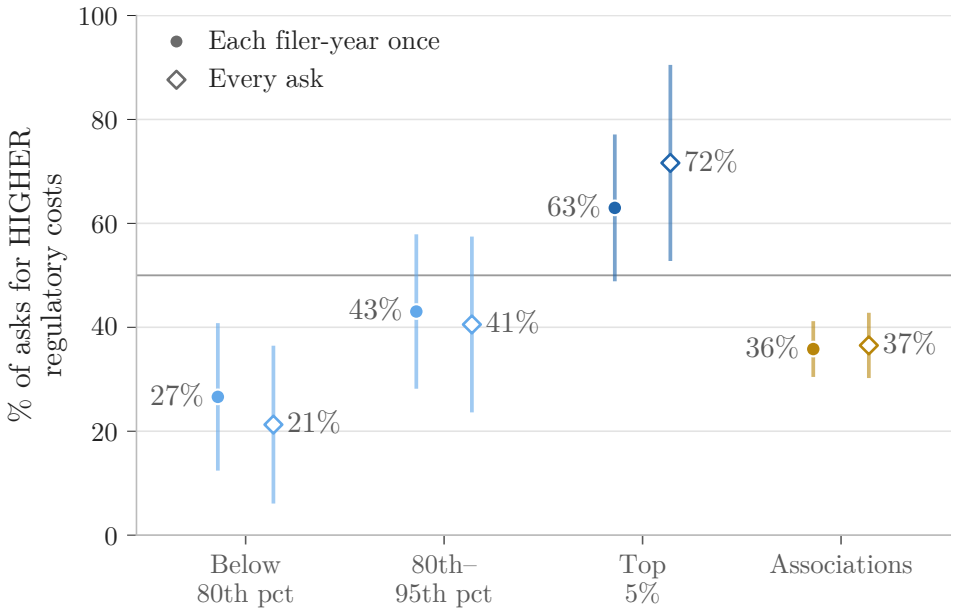


Figure 5. Share asking for higher regulatory costs, by firm size

Notes: Share of requests on regulatory costs (licenses, reporting duties, labeling, privacy) that are for higher regulatory costs, by the firm's sales percentile within its industry and year. Circles: the mean over firm-years (association-years) of each year's share of requests, each year counted once. Diamonds: requests for higher costs over all requests of the group, so that clients with more asks weigh more. 278 firm-years of 112 firms (1,284 asks) and 945 association-years of 264 associations (4,858 asks) take a side; 2008–2022. 95% confidence intervals clustered by client.

costs, against 19, 15, and 12% of the second through fourth firms (Table 4).

Table 4. Participation and persistence of the four largest firms

	Lobbying in a quarter, by rank				Next quarter	
Instrument	Leader	#2	#3	#4	Stay	Join
Regulatory costs	0.236	0.192	0.153	0.122	0.924	0.018
Profit tax	0.181	0.131	0.100	0.082	0.904	0.015
Procurement	0.133	0.090	0.059	0.058	0.897	0.010

Notes: The four largest Compustat firms headquartered in the United States in each four-digit NAICS industry and year, all 268 industries, 1,658 firms. Regulatory costs (licenses, reporting duties, labeling, privacy): a firm lobbies in a quarter if any of its reports in the quarter lobbies on them, 2008–2022, 49,832 firm-quarters. Profit tax (all corporate income tax instruments) and procurement contracts: lobbying in the last quarter of the fiscal year, fiscal years 2008–2022, 12,325 firm-years. Monitoring counts. Stay: probability of lobbying on the instrument next quarter given lobbying this quarter; join: given not lobbying.

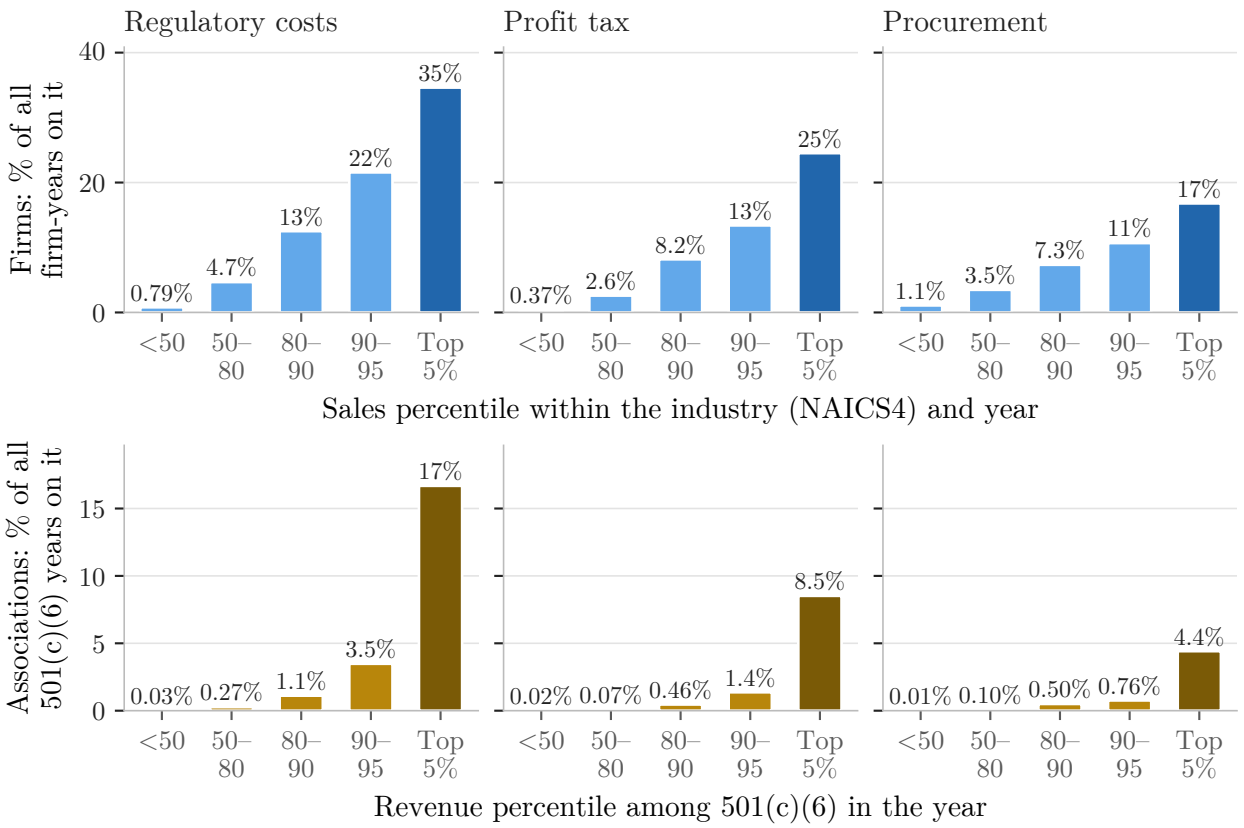


Figure 6. Participation in lobbying by firm size

Notes: Share of all firm-years in which the firm lobbies on the instrument, by the firm's sales percentile within its industry (four-digit NAICS) and year, 75,957 firm-years of Compustat firms headquartered in the United States; for associations, the share of the years of 501(c)(6) organizations filing an IRS Form 990 or 990-EZ with positive revenue (30,112 a year on average) in which the association lobbies on it, by its revenue percentile among these organizations in the year. 2008–2022. Appendix Figure 15 reports the shares among the years with a lobbying report.

3.4 Fact 4: lobbying on each instrument is persistent

Nine in ten firms that lobby on an instrument in one quarter lobby on it in the next: 91% on regulatory costs, 89% on the profit tax, and 89% on procurement among all Compustat firms, and 93, 91, and 91% among the largest 5%. Few firms start lobbying on an instrument in a given quarter: 0.4 to 0.6% of all firms and 1.6 to 3.4% of the largest 5%. Among the four largest firms of each industry, 90 to 92% stay and 1.0 to 1.8% start in the next quarter (Table 4). Associations are as persistent (92% on regulatory costs, among consecutive quarters in which the association has a report). Thus, firms appear to revise the decision to lobby on an instrument only infrequently.

3.5 Associations lobby on regulatory costs more often than the largest firms

Industry associations lobby on regulatory costs more often than the four largest firms of their industry, and mostly without them. We take all 268 four-digit NAICS industries and every quarter of 2008–2022, 16,080 industry-quarters. We say the association lobbies in a quarter when any industry association of the industry lobbies on regulatory costs, and likewise for the four largest firms. The association lobbies without these firms in 25% of industry-quarters, these firms lobby without the association in 9%, both lobby in 20%, and neither in 45%. The association lobbies in 46% of quarters, and in 83% of the quarters in which anyone in the industry lobbies on regulatory costs. Thus, in most industries the association, not the largest firms, is the main lobbyist on regulatory costs.

3.6 From facts to the model

The facts shape the model in four ways. Because tax and procurement lobbying is one-sided, the model treats it as each firm's own instrument, with an outcome observed in the data, the firm's tax rate and its contracts. Of the demand wedge, the model carries procurement contracts, whose return to lobbying we observe for each firm. Because lobbying over regulatory costs is two-sided and the rule applies to every firm in an industry, the model determines regulatory costs in a contest between firms with opposing interests. Each firm's marginal value of a higher regulatory cost, and hence its side, is an equilibrium object. Because large firms lobby individually, and associations side with smaller firms more often than with the largest, each industry has a few strategic superstars that lobby individually and a competitive fringe that lobbies only through its association. Because participation rises with size, lobbying on an instrument requires a fixed participation cost. Because participation is persistent, firms revise this decision only infrequently. Participation costs are partly shared across instruments, and the overlap of participation across instruments identifies the sharing (Appendix K.2). We compute the same moments in the model and in the data: participation by rank, persistence, side shares, association activity, and the outcomes of tax and procurement lobbying.

4 The model

We extend the economy of Cavenaile et al. (2026) with lobbying on three policy instruments and with fringe firms that differ in their operating costs.

4.1 Environment

Time is continuous, indexed by $t \in \mathbb{R}_+$. The economy is composed of a representative household, a unit mass of industries indexed by $j \in [0,1]$, a unit mass of risk-neutral entrepreneurs, and a government.

Household. The lifetime utility of the representative household depends on consumption of the final good and labor supply:

$$U = \int_0^\infty e^{-\rho t} \left[\ln C_t \; - \; \xi \, \frac{L_t^{\,1+1/\epsilon_L}}{1+1/\epsilon_L} \right] dt, \tag{1}$$

where ρ is the discount rate, ξ scales the disutility of labor supply L_t, ϵ_L is the Frisch elasticity of labor supply, and total consumption C_t is private consumption C_t^p, chosen by the household, plus government purchases of the final good G_t, which the household takes as given.

Final good. The final good Y_t is produced competitively by aggregating over the continuum of industries $j \in [0,1]$:

$$\ln Y_t \;=\; \int_0^1 \ln(y_{jt})\, dj. \tag{2}$$

We normalize the price of the final good to one. With Cobb–Douglas aggregation, expenditure on each industry equals Y_t.

Industry structure. Each industry j has $N_{jt} \le \bar{N}$ superstar firms, where $\bar{N}$ is the maximum number of superstars. Superstars produce differentiated varieties and compete in quantities. Each industry also has a continuum of competitive fringe firms with total mass m_t. The output of industry j is a CES aggregate of the superstar varieties and the fringe variety:

$$y_{jt} \;=\; \left(\sum_{i=1}^{N_{jt}} y_{ijt}^{\frac{\eta-1}{\eta}} + \tilde{y}_{fjt}^{\frac{\eta-1}{\eta}} \right)^{\frac{\eta}{\eta-1}}, \qquad \eta > 1, \tag{3}$$

where η is the elasticity of substitution between varieties and $\tilde{y}_{fjt} = \int_0^{m_t} y_{fkjt}\, dk$ is aggregate fringe output. Fringe firms produce perfect substitutes, so the fringe supplies a single variety.

Technology. Every firm, superstar or fringe, produces output with labor under decreasing returns to scale:

$$y_{ijt} \;=\; q_{ijt}\, l_{ijt}^{\alpha}, \qquad \alpha \in (0,1). \tag{4}$$

where q_{ijt} is the quality of firm i, l_{ijt} is its labor input, and α is the returns to scale. Decreasing returns keep fringe firms finite in size and give them a surplus over their operating cost.

Each superstar's quality follows a quality ladder. Denote by $n_{ijt} \in \{0, 1, \ldots, \bar{n}\}$ the quality gap of superstar i relative to the leader in industry j (the leader's gap is zero), where $\bar{n}$ is the largest gap at which a firm remains a superstar. Superstar i's quality is

$$q_{ijt} \;=\; q_{jt}^{\text{leader}} \cdot (1+\lambda)^{-n_{ijt}}, \tag{5}$$

where q_{jt}^{leader} is the quality of the leader in industry j and $\lambda > 0$ is the quality step.

All fringe firms in industry j have the same quality q_{fjt}, which diffuses from the industry's superstars:

$$q_{fjt} \;=\; \zeta\Big(\sum_{i=1}^{N_{jt}} q_{ijt}^{\gamma}\Big)^{1/\gamma}, \tag{6}$$

where $\zeta > 0$ scales fringe quality relative to the superstars and $\gamma > 0$ determines how concentrated the spillovers are. The elasticity of fringe quality with respect to the quality of superstar i is $q_{ijt}^{\gamma} / \sum_{i'=1}^{N_{jt}} q_{i'jt}^{\gamma}$, so the fringe learns the most from the most productive superstars. Each superstar internalizes that its innovation raises fringe quality and so erodes part of its advantage over the fringe. As $\gamma \to \infty$, fringe quality tracks the leader's quality, as in Cavenaile et al. (2026), and a lower γ spreads the spillovers more evenly across the superstars.

Every producing firm pays a compliance cost proportional to the industry's regulatory cost R_{jt}, which lobbying determines (Section 4.3). A fringe firm pays $R_{jt}Y_t$, and superstar i pays $c_{ijt}R_{jt}Y_t$, where $c_{ijt} = \bar{c}\,(1+\lambda)^{-\gamma_R n_{ijt}}$. The parameters $\bar{c} > 0$ and $\gamma_R \geq 0$ make a superstar's compliance cost rise with its size with elasticity $e_R < 1$, as in the definition of regulatory costs in Section 3 (Appendix K.3 gives the mapping). We interpret regulation as administrative friction: compliance uses resources and produces nothing. Each fringe firm k that produces also pays an operating cost $\varepsilon_k Y_t$, with ε_k drawn at the firm's birth from

$$\varepsilon_k \;\sim\; \text{LogN}(\mu_\varepsilon, \sigma_\varepsilon^2), \qquad \mu_\varepsilon = \ln \bar{\varepsilon} - \sigma_\varepsilon^2/2, \tag{7}$$

so that $\mathbb{E}[\varepsilon_k] = \bar{\varepsilon}$. The cumulative distribution and density of ε are $F(\cdot)$ and $f(\cdot)$. A fringe firm can stop and restart producing at no cost. At every instant, it produces if and only if $\varepsilon_k \leq \varepsilon_{jt}^*$, where the production cutoff ε_{jt}^* is derived in Section 4.5. We call the producing fringe firms active. Their mass is $m_{jt}^{\text{a}} = m_t\, F(\varepsilon_{jt}^*)$. The operating cost is the only source of heterogeneity among fringe firms. Thus, a higher R_{jt} lowers the cutoff ε_{jt}^*, and the fringe firms with the highest operating costs stop producing.

Entry and exit. A unit mass of entrepreneurs creates fringe firms. An entrepreneur who pays $\psi e_t^2 Y_t$ creates new fringe firms at Poisson rate e_t. New fringe firms are randomly

allocated across industries, so every industry has the same pool of fringe firms, active or not, of mass m_t. Entrepreneurs sell each new firm to the household at its expected value before the draw of ε_k and remain in the entrepreneur population. New firms enter only the fringe. A new superstar arises only from a breakthrough by a fringe firm. Fringe firms die at Poisson rate δ, whether or not they produce, and firms that are not producing resume production whenever the industry state makes it profitable.

As in Cavenaile et al. (2026), modeling entry into and exit from the fringe explicitly lets us map fringe entry in the model to new business creation in the data. This contrasts with models in which entrants immediately become industry leaders or are otherwise large enough to interact strategically with incumbents. In our model, each industry has a large pool of fringe firms, only some of which produce at any time. Few of them make a breakthrough and become superstars.

4.2 Innovation

Innovation occurs through two technologies: incremental innovation by superstars on the quality ladder, and breakthroughs, through which active fringe firms become superstars.

Superstar incremental innovation. Each superstar i chooses the Poisson arrival rate $z_{ijt} \geq 0$ of a one-step quality improvement, at cost

$$\text{cost}_i^{\text{R\&D}} \;=\; \chi z_{ijt}^{\phi} Y_t, \qquad \phi > 1. \tag{8}$$

where χ scales the cost and ϕ is its convexity. Upon success, the firm's quality rises by the factor $1+\lambda$. If a follower innovates, its gap falls by one. If the leader innovates, every other superstar's gap rises by one, so a superstar already $\bar{n}$ steps behind exits.

Fringe breakthroughs. Each active fringe firm k chooses the arrival rate $x_{fkjt} \geq 0$ of a breakthrough that creates a new superstar, at cost

$$\text{cost}_k^{\text{breakthrough}} \;=\; \chi_f \, x_{fkjt}^{\phi_f} Y_t, \qquad \phi_f > 1. \tag{9}$$

where χ_f scales the cost and ϕ_f is its convexity. A breakthrough creates a new superstar. The innovating fringe firm owns it and keeps producing in the fringe. The new superstar starts $h \geq 0$ steps above the entry margin with probability

$$\Pr(h) \;=\; (1-u)\, u^h, \tag{10}$$

where $u \in (0,1)$ is the probability of climbing one more step. In an industry with $\bar{N}$ superstars, the entry margin is the position of the least productive superstar, which the new superstar displaces. Otherwise it is $\bar{n}$ steps behind the leader. A new superstar that climbs past the leader becomes the new leader, at most $\bar{n}$ steps above the former leader, and superstars that end up more than $\bar{n}$ steps behind it exit. Unlike in Cavenaile et al. (2026), where new superstars join $\bar{n}$ steps behind the leader and only industries with fewer than $\bar{N}$ superstars admit them, new superstars in our model also join full industries and can land anywhere above the margin.

4.3 Lobbying

Denote the three instruments by $I \in \{T, P, R\}$: the superstar's profit tax (T), its procurement contracts (P), and the industry's regulatory cost (R). A superstar that lobbies on instrument I chooses an intensity $b_{Iijt} \geq 0$ at cost $\kappa_I \, b_{Iijt}^{\phi_I} \, Y_t$, with $\phi_I > 1$. Lobbying costs are not tax deductible.[4] The indicator $a_{Iijt} \in \{0,1\}$ equals one if superstar i lobbies on instrument I.

Profit tax (T). Lobbying on the tax instrument lowers the firm's own effective profit tax rate below the statutory rate τ_0:

$$\tau_{ijt} \;=\; \frac{\tau_0}{1 + a_{Tijt} \, b_{Tijt}}. \tag{11}$$

A firm that does not lobby pays the statutory rate τ_0, and each additional unit of intensity lowers its rate by less than the previous one.

Procurement contracts (P). Superstar i holds government contracts worth $\Gamma_{ijt} \, Y_t$, with

$$\Gamma_{ijt} \;=\; \left(\Gamma_0 + a_{Pijt} \, b_{Pijt}\right) (1+\lambda)^{-\kappa_\Gamma n_{ijt}}, \tag{12}$$

where Γ_0 is every superstar's baseline contract, b_{Pijt} is the increment from lobbying, and $\kappa_\Gamma \geq 0$ determines how fast contracts fall with the firm's gap to the leader, so larger superstars hold larger contracts. Contracts are cost-plus. The share $1 - \theta_\Gamma$ of the payment $\Gamma_{ijt} Y_t$ covers the cost of the goods the firm delivers to the government. The remaining share θ_Γ is the firm's fee and is taxed as profit.

Regulatory cost (R). The regulatory cost is set as in a legislative process, in which superstars and the association lobby for or against bills that raise or lower it. A superstar i that lobbies on R chooses an intensity b_{Rijt}, and an organized association chooses an intensity

[4] Internal Revenue Code §162(e) disallows the deduction of lobbying expenditures for federal income tax purposes.

b_{Ajt}. Each participant lobbies either for a higher regulatory cost or against it. The influence for a higher cost, B_{jt}^{pro}, is the sum of b^θ over the participants that lobby for it, and the influence against, B_{jt}^{anti}, is the same sum over those that lobby against it, with $\theta \in (0, 1)$. The regulatory cost is

$$R_{jt} \;=\; R_0 \,\frac{1 + \beta^{\text{pro}}\, B_{jt}^{\text{pro}}}{1 + \beta^{\text{anti}}\, B_{jt}^{\text{anti}}} \tag{13}$$

All participants know this influence function. Influence for a higher cost raises R_{jt} above R_0, and influence against lowers it. Because $\theta < 1$, each participant faces diminishing returns to its own intensity, and β^{pro} and β^{anti} measure how effective lobbying is for and against a higher cost. Without lobbying, $R_{jt} = R_0$. To first order, equation (13) is the steady state of a model of legislation in which bills that add or remove rules pass with a probability that rises with the influence pushing them and falls with the influence blocking them (Appendix M). Its ratio form keeps R_{jt} positive and lets either side move it.

Participation in lobbying. Lobbying on an instrument requires a commitment, such as retaining lobbyists and maintaining relationships with legislators. A superstar that lobbies on instrument I pays a flow participation cost $\Psi_{Ii}\, Y_t$, where

$$\ln \Psi_{Ii} \;\sim\; N(\mu_I, \sigma_I^2), \qquad \text{i.i.d. across firms and instruments.} \tag{14}$$

Participation is sticky. The firm draws a new cost at Poisson rate δ_I and can switch its status a_{Ii} only when a new draw arrives. In between, a participating firm keeps paying its current draw, which we call its locked cost. This timing makes lobbying status persistent, as in the data (Section 3.4). Because the arrivals are independent across firms and instruments, firms revise their participation one at a time.

Participation costs are partly shared across instruments. A firm that lobbies on instrument I pays $\Psi_{Ii} \prod_{J \neq I} (1 - \iota_{IJ})^{a_{Ji}}$ on it. Each other instrument J on which the firm also lobbies lowers the cost of I by the share $\iota_{IJ} = \iota_{JI}$. This share measures the economies of scope between the two instruments, which arise because lobbyists, relationships, and information serve both.

Trade association. Superstars and fringe firms differ in their incentive to lobby (Olson, 1965, 1982). Each superstar can move the regulatory cost by itself, so it has an incentive to lobby individually. For fringe firms, however, lobbying is a public good. Each fringe firm is atomistic, so it would rather free ride on the lobbying of other fringe firms that share its interests than lobby itself. As in Olson (1965), a trade association solves this collective action problem. Our model of the association is a reduced form of the by-product theory of Olson (1965), in which associations sell private benefits, such as trade publications and

conventions, to individual firms and produce lobbying as a complement to those sales.

The association decides whether to organize with the same timing as a superstar's participation decision. At Poisson rate δ_A, it draws a flow organizing cost $\Psi_A Y_t$, with $\ln \Psi_A \sim N(\mu_A, \sigma_A^2)$. If it organizes ($a_{Ajt} = 1$), it pays this cost until the next draw. An organized association enters the contest (13) as a single participant and chooses its side and intensity to maximize the total value of the industry's fringe firms. The association's intensity b_{Ajt} costs $\kappa_A\, b_{Ajt}^{\phi_A} Y_t$, with $\phi_A > 1$. Active fringe firms fund the association's organizing and intensity costs through dues proportional to their surplus from producing. An active fringe firm with operating cost ε pays $d_{jt}(\varepsilon)\, Y_t$, where

$$d_{jt}(\varepsilon) \;=\; \frac{\varepsilon_{jt}^* - \varepsilon}{m_t \int_0^{\varepsilon_{jt}^*} \left(\varepsilon_{jt}^* - \varepsilon'\right) dF(\varepsilon')} \left[\kappa_A\, b_{Ajt}^{\phi_A} + \Psi_A\right], \qquad \varepsilon \le \varepsilon_{jt}^*. \tag{15}$$

Fringe firms that do not produce pay nothing. A producing firm's surplus is $(1-\tau_0)(\varepsilon_{jt}^* - \varepsilon)\, Y_t$, so every member pays the same fraction of its surplus, and dues cover the association's spending exactly. Because the marginal producer pays nothing, dues do not affect which firms produce, provided the association's spending does not exceed the members' after-tax surplus, $(1-\tau_0)\, m_t \int_0^{\varepsilon_{jt}^*} (\varepsilon_{jt}^* - \varepsilon)\, dF(\varepsilon)$.

4.4 Government budget

The government buys goods through procurement contracts, rebates corporate tax revenue, pays lump-sum transfers, and finances the balance with a labor income tax. Its spending on contracts is $G_t^S = Y_t \int_0^1 \sum_i \Gamma_{ijt}\, dj$, so a contract that a superstar wins by lobbying is an additional purchase. Contracts held by other firms, including fringe firms, are outside the model.[5] Because the share $1 - \theta_\Gamma$ of each payment covers the cost of the goods delivered, the household values government purchases at $G_t = (1 - \theta_\Gamma)\, G_t^S$. The government rebates to households, lump sum, the revenue that the statutory rate τ_0 raises on every firm's tax base. It also pays households lump-sum transfers $\bar{T} Y_t$, a constant share of output. The labor income tax τ_w adjusts to finance the transfers, the contracts, and the revenue lost to tax lobbying, so that the government budget balances at each instant:

$$\tau_w\, w_t\, L_t \;=\; \bar{T}\, Y_t \;+\; G_t^S \;+\; \int_0^1 \sum_{i=1}^{N_{jt}} \left(\tau_0 - \tau_{ijt}\right) \Pi_{ijt}\, Y_t\, dj, \tag{16}$$

where w_t is the wage and Π_{ijt} is the tax base of superstar i in units of Y_t (equation (21)).

[5] Federal contracts are about one to two percent of the sales of listed fringe firms.

4.5 Decisions in the decentralized economy

Households. The representative household owns all firms. It chooses private consumption and labor supply to maximize utility (1), taking government purchases as given, subject to

$$\dot{A}_t \;=\; r_t A_t + (1-\tau_w)\, w_t L_t + T_t - C_t^p,$$

where A_t is the value of all firms, r_t the return on it, w_t the wage, and T_t the government's lump-sum payments, the corporate tax rebate plus the transfers $\bar{T} Y_t$. The household receives firms' profits as the return on A_t. Profits include the superstars' contract fees and are net of R&D, operating, compliance, and lobbying costs and of taxes. The Euler equation and the labor supply condition are

$$\frac{\dot{C}_t}{C_t} \;=\; r_t - \rho, \qquad \xi\, L_t^{1/\epsilon_L} \;=\; \frac{(1-\tau_w)\, w_t}{C_t}. \tag{17}$$

Private consumption and government purchases are both shares of output, so on the balanced growth path (BGP) both grow at the growth rate of output, g (equation (42)), and $r_t = \rho + g$. Combining the household's budget with the government's budget (16) and the firms' cash flows gives the resource constraint in Section 4.6, so the goods market clears. Because all firm payoffs are proportional to Y_t, firm decisions do not depend on the wage, labor supply, or the labor tax. The wage, labor supply, and labor tax affect only the level of output and welfare.

Final good producer. The competitive final good producer's demand for each variety gives each superstar's inverse demand and the price of fringe output. Because every industry's sales equal Y_t under Cobb–Douglas aggregation (2), these are

$$p_{ijt} = \frac{y_{ijt}^{-\frac{1}{\eta}} Y_t}{\sum_{i'=1}^{N_{jt}} y_{i'jt}^{\frac{\eta-1}{\eta}} + \tilde{y}_{fjt}^{\frac{\eta-1}{\eta}}}, \qquad p_{fjt} = \frac{\tilde{y}_{fjt}^{-\frac{1}{\eta}} Y_t}{\sum_{i'=1}^{N_{jt}} y_{i'jt}^{\frac{\eta-1}{\eta}} + \tilde{y}_{fjt}^{\frac{\eta-1}{\eta}}}. \tag{18}$$

Product market. Superstars compete à la Cournot. Each superstar chooses y_{ijt} to maximize operating profit $\pi_{ijt} = p_{ijt}\, y_{ijt} - w_t\, l_{ijt}$, taking rivals' quantities and the fringe aggregate $\tilde{y}_{fjt}$ as given, subject to (4), (5) and (18). Fringe firms take p_{fjt} as given. Appendix G derives the best responses and profits. A superstar's markup and operating profit are functions of its share of industry sales $\omega_{ijt} = p_{ijt} y_{ijt} / Y_t$:

$$M_{ijt} \;=\; \frac{\eta}{(\eta-1)(1-\omega_{ijt})}, \qquad \pi_{ijt} \;=\; \omega_{ijt}\Big(1 - \frac{\alpha}{M_{ijt}}\Big)\, Y_t, \tag{19}$$

and each active fringe firm earns

$$\pi_{fjt} \;=\; \frac{(1-\alpha)\,\omega_{fjt}\,Y_t}{m^{\mathrm{a}}_{jt}}, \tag{20}$$

where ω_{fjt} is the fringe's share of industry sales. The markup rises with the sales share because a larger superstar faces a less elastic residual demand, and operating profit is sales net of the labor bill, which is the fraction α/M_{ijt} of sales. Fringe firms price at marginal cost, but because of decreasing returns they earn positive operating profits. All sales shares depend only on the industry's quality gaps and the active fringe mass. Both superstar and fringe profits fall with the active fringe mass, $\partial\pi_{ijt}/\partial m^{\mathrm{a}}_{jt} < 0$ and $\partial\pi_{fjt}/\partial m^{\mathrm{a}}_{jt} < 0$.

Tax base. A superstar's tax base, in units of Y_t, is its operating profit plus its contract fee, minus compliance and R&D costs:

$$\Pi_{ijt} \;=\; \frac{\pi_{ijt}}{Y_t} \;+\; \theta_\Gamma\,\Gamma_{ijt} \;-\; c_{ijt}R_{jt} \;-\; \chi z^{\phi}_{ijt}. \tag{21}$$

Losses are fully offset, so a superstar whose base is negative receives $\tau_{ijt}|\Pi_{ijt}|Y_t$. Fringe firms pay the statutory rate τ_0, with the same loss offset, on operating profit minus compliance costs, their operating cost ε_k, and their breakthrough R&D costs. Participation costs and association dues, like lobbying costs, are not deductible.

States. The technology state of an industry is $\Theta = (N, \mathbf{n})$, where N is the number of superstars and $\mathbf{n} = (n_1, \ldots, n_{N-1})$, with $0 \le n_1 \le \cdots \le n_{N-1} \le \bar{n}$, is the vector of follower gaps behind the leader. The industry state adds every participation status: $S = (\Theta, \mathbf{a}_T, \mathbf{a}_P, \mathbf{a}_R, a_A)$, where $\mathbf{a}_I = (a_{I1}, \ldots, a_{IN})$ collects the superstars' statuses on instrument I and a_A is the association's status. A superstar's own state $s_i = (i, \mathbf{\Psi}_i)$ consists of its rank i in S, which gives its gap and its statuses, and the participation costs $\mathbf{\Psi}_i = (\Psi_{Ti}, \Psi_{Pi}, \Psi_{Ri})$ it has locked on the instruments it lobbies on. Locked costs are private, so a rival's policies enter a firm's problem through their expectation given S under the stationary distribution μ of industry states and locked costs. We write firm values in units of output, $v := V/Y_t$, where V is a value in units of the final good. On the balanced growth path, normalized values are discounted at $r - g = \rho$. We drop time and industry subscripts where this creates no ambiguity.

Transitions. The industry state changes with the following events. Innovation by superstar i moves the industry to state S^{+i}, by the ladder rules of Section 4.2. A breakthrough whose new superstar lands h steps above the margin moves it to $S^{X,h}$. After a redraw of superstar i's cost on instrument I, the state is $S^{I,i,a}$, with its status on I set to $a \in \{0, 1\}$. After a redraw of the association's organizing cost, the state is $S^{A,a}$. A new superstar starts

with every lobbying status off, and a superstar's own state after an event, such as s_i^{+i}, is its new rank with the same locked costs. Appendix J gives the rules for ties.

Superstar problem. A superstar chooses its R&D intensity and its lobbying intensities on the instruments it participates in at every instant, and its participation on instrument I whenever its cost on that instrument is redrawn. Its Hamilton–Jacobi–Bellman (HJB) equation is

$$\begin{aligned}
\rho\, v(S,s_i) \;=\; & \max_{z_i,\, b_{Ti},\, b_{Pi},\, b_{Ri}} \Big\{ (1-\tau_i)\,\Pi_i \;-\; \sum_I a_{Ii}\Big[\kappa_I\, b_{Ii}^{\phi_I} + \Psi_{Ii} \prod_{J\neq I} (1-\iota_{IJ})^{a_{Ji}}\Big] \\
& + z_i \left[v(S^{+i}, s_i^{+i}) - v(S,s_i) \right] \;+\; \sum_{i'\neq i} \bar{z}_{i'}(S) \left[v(S^{+i'}, s_i^{+i'}) - v(S,s_i) \right] \\
& + X(S) \sum_h P_h(S) \left[v(S^{X,h}, s_i^{X,h}) - v(S,s_i) \right] \\
& + \sum_{i'\neq i} \sum_I \delta_I \Big[\bar{\mathcal{P}}_{Ii'}(S)\, v(S^{I,i',1}, s_i) + \big(1 - \bar{\mathcal{P}}_{Ii'}(S)\big)\, v(S^{I,i',0}, s_i) - v(S,s_i) \Big] \\
& + \delta_A \Big[\mathcal{P}_A(S)\, v(S^{A,1}, s_i) + \big(1 - \mathcal{P}_A(S)\big)\, v(S^{A,0}, s_i) - v(S,s_i) \Big] \\
& + \sum_I \delta_I \Big(\mathbb{E}_{\Psi'} \max \big\{ v(S^{I,i,1}, s_i^{I,\Psi'}),\; v(S^{I,i,0}, s_i^{I,0}) \big\} - v(S,s_i) \Big) \Big\},
\end{aligned} \tag{22}$$

subject to (13). The first line is the after-tax flow net of lobbying and locked participation costs. The second line contains innovation by the firm and by its rivals, whose expected rates given the industry state are $\bar{z}_{i'}(S) := \mathbb{E}[z_{i'} \mid S]$. The third line contains breakthroughs, which arrive at rate $X(S)$. A new superstar lands h steps above the margin with probability $P_h(S)$, the probability (10) truncated at $\bar{n}$ steps above the leader (Appendix J gives the value when several superstars are tied at the margin). The fourth line contains rivals' redraws, after which rival i' is on instrument I with probability $\bar{\mathcal{P}}_{Ii'}(S)$, given after equation (26). The fifth line contains the association's redraws, after which it is organized with probability $\mathcal{P}_A(S)$ (equation (34)). The last line contains the firm's own redraws. The firm either joins instrument I and locks the new draw Ψ', moving to own state $s_i^{I,\Psi'}$, or stays out, moving to $s_i^{I,0}$. A firm that exits has value zero. The first-order conditions for innovation, procurement, and tax lobbying are

$$(1-\tau_i)\,\chi\phi\, z_i^{\phi-1} \;=\; v(S^{+i}, s_i^{+i}) - v(S,s_i), \tag{23}$$

$$(1-\tau_i)\,\theta_\Gamma\,(1+\lambda)^{-\kappa_\Gamma n_i} \;=\; \kappa_P\,\phi_P\, b_{Pi}^{\phi_P - 1}, \tag{24}$$

$$\frac{\tau_0\,\Pi_i}{(1+b_{Ti})^2} \;=\; \kappa_T\,\phi_T\, b_{Ti}^{\phi_T - 1}, \tag{25}$$

where the tax intensity is zero when the base Π_i is not positive. Each condition equates the marginal cost of an intensity to its marginal benefit. For R&D, the benefit is the value of one step up the ladder. For procurement lobbying, it is the after-tax fee on an additional contract, which depends on the state only through the firm's gap. For tax lobbying, it is the tax saved on the base. A higher locked cost lowers the firm's value at every date until its next redraw on the instrument, so the value after joining falls with the new draw. At a redraw, the firm therefore joins instrument I if and only if its draw is below a cutoff:

$$a'_{Ii} = 1 \iff \Psi' \le \Psi^*_I(S, s_i), \qquad v\big(S^{I,i,1}, s_i^{I,\Psi^*_I}\big) = v\big(S^{I,i,0}, s_i^{I,0}\big). \tag{26}$$

A rival's probability of being on the instrument after a redraw is therefore $\bar{\mathcal{P}}_{Ii'}(S) = \mathbb{E}\big[\Phi\big((\ln \Psi^*_I(S, s_{i'}) - \mu_I)/\sigma_I\big) \mid S\big]$, where Φ is the standard normal distribution function.

Fringe problem. A fringe firm's value depends on its operating cost ε and, through its dues (15), on the organizing cost Ψ_A that an organized association has locked. The HJB equation of a fringe firm with operating cost ε is

$$\begin{aligned}(\rho+\delta)\, v^f(S, \Psi_A, \varepsilon) = \max\Big\{0,\ (1-\tau_0)\Big[\frac{\pi_f(S)}{Y} - R(S) - \varepsilon\Big] - a_A\, d(S, \Psi_A, \varepsilon) \\ + \max_{x \ge 0}\big[x\, v^{\text{new}}(S) - (1-\tau_0)\,\chi_f x^{\phi_f}\big]\Big\} \\ + \sum_{S' \ne S} Q(S, S')\,\big[\mathbb{E}\, v^f(S', \Psi'_A, \varepsilon) - v^f(S, \Psi_A, \varepsilon)\big],\end{aligned} \tag{27}$$

where $d(S, \Psi_A, \varepsilon)$ is the dues (15) at state S. The firm produces when the second term in the braces is positive, and only producing firms can attempt breakthroughs, at rate x. The industry's transition rates $Q(S, S')$ are induced by superstars' innovation, breakthroughs, and the redraws of the superstars and the association. The expectation is over the new organizing cost Ψ'_A, which changes only when the association reorganizes at a redraw. The expected value of the new superstar to the innovating firm is

$$v^{\text{new}}(S) = \sum_h P_h(S)\, v\big(S^{X,h}, \text{rank of the new superstar}\big). \tag{28}$$

The first-order condition gives the same breakthrough intensity for every active firm in the industry,

$$(1-\tau_0)\,\chi_f\,\phi_f\, x_f^{\phi_f - 1} = v^{\text{new}}(S), \tag{29}$$

so breakthroughs arrive in the industry at rate $X(S) = m^{\text{a}}(S)\, x_f(S)$. Because switching is costless and ε is permanent, the decision to produce affects only the current flow. Because

the marginal producer pays no dues, a fringe firm produces if and only if $\varepsilon \le \varepsilon^*(S)$, where

$$\varepsilon^*(S) \;=\; \frac{\pi_f(S)}{Y} - R(S) + \frac{\Omega(S)}{1-\tau_0}, \qquad \Omega(S) \;=\; \max_{x \ge 0} \left\{ x\, v^{\text{new}}(S) - (1-\tau_0)\, \chi_f x^{\phi_f} \right\}. \tag{30}$$

The marginal producer's operating cost equals its operating profit net of compliance, plus the value $\Omega(S)$ of the breakthrough option that only producing firms have, expressed before tax. Because π_f depends on $m^{\text{a}} = mF(\varepsilon^*)$, equation (30) defines $\varepsilon^*(S)$ as a fixed point at each state. The fixed point is unique, since π_f falls with m^{a}.

Let $\bar{v}^f(S, \Psi_A) := \mathbb{E}_\varepsilon[v^f(S, \Psi_A, \varepsilon)]$ be the value of a firm in the industry's fringe pool. No decision depends on the organizing cost the association has locked, and each pool firm pays on average a share $1/m$ of it until the association's next redraw or the firm's death, so

$$\bar{v}^f(S, \Psi_A) \;=\; \bar{v}_0^f(S) \;-\; \frac{a_A\, \Psi_A}{m\,(\rho + \delta + \delta_A)}, \tag{31}$$

where the pool value gross of the current organizing cost solves

$$\begin{aligned} (\rho+\delta)\, \bar{v}_0^f(S) \;=\; & (1-\tau_0)\, \mathbb{E}\big[(\varepsilon^*(S) - \varepsilon)^+\big] \;-\; \frac{a_A\, \kappa_A\, b_A(S)^{\phi_A}}{m} \;-\; \frac{\delta_A\, \mathbb{E}\big[\Psi_A\, \mathbf{1}\{\Psi_A \le \Psi_A^*(S)\}\big]}{m\,(\rho + \delta + \delta_A)} \\ & + \sum_{S' \neq S} Q(S, S') \left[\bar{v}_0^f(S') - \bar{v}_0^f(S)\right]. \end{aligned} \tag{32}$$

Its flow terms are the producers' after-tax surplus, the dues that pay for lobbying, and the members' share of any organizing cost committed at the next redraw. Entrepreneurs sell new firms at the pool value averaged over the stationary distribution μ. Because fringe firms die at rate δ, a constant pool requires entry $e = \delta m$. Free entry sets the marginal cost of creating a firm, $2\psi e = 2\psi\delta m$, equal to this average value:

$$2\psi\delta\, m \;=\; \sum_S \mu(S) \left[\bar{v}_0^f(S) - \frac{a_A\, \bar{\Psi}_A(S)}{m\,(\rho + \delta + \delta_A)}\right], \tag{33}$$

where $\bar{\Psi}_A(S)$ is the expected cost, under μ, that an association organized at state S has locked at its last redraw.

Association problem. At each redraw, the industry's current fringe firms decide whether the association organizes. They organize if their gain covers their share of the new cost:

$$\Psi_A \;\le\; \Psi_A^*(S) \;=\; (\rho + \delta + \delta_A)\, m \left[\bar{v}_0^f(S^{A,1}) - \bar{v}_0^f(S^{A,0})\right], \qquad \mathcal{P}_A(S) \;=\; \Phi\Big(\frac{\ln \Psi_A^*(S) - \mu_A}{\sigma_A}\Big), \tag{34}$$

Firms that join the pool later pay the rest of the cost, and the value of a new firm in the free entry condition (33) is net of this payment.

Lobbying on the regulatory cost. In a Markov-perfect equilibrium, each agent takes rivals' policies, the fringe's breakthrough intensity, and continuation values as given, but internalizes the effect of its lobbying on R and, through R, on which fringe firms produce. A superstar that lobbies on R chooses b_{Ri} to satisfy

$$\Delta_i(S, s_i)\,\frac{\partial R(S)}{\partial b_{Ri}} \;=\; \kappa_R\,\phi_R\,b_{Ri}^{\phi_R-1}, \tag{35}$$

where Δ_i is the marginal value of a higher regulatory cost to the firm:

$$\Delta_i \;=\; \underbrace{-(1-\tau_i)\,c_i}_{\text{(i) own compliance}} \;+\; \underbrace{(1-\tau_i)\,\frac{\partial(\pi_i/Y)}{\partial m^{\mathrm{a}}}\,\frac{dm^{\mathrm{a}}}{dR}}_{\text{(ii) fewer active fringe firms}} \;+\; \underbrace{\frac{dX}{dR}\sum_h P_h\left[v(S^{X,h}, s_i^{X,h}) - v(S, s_i)\right]}_{\text{(iii) fewer breakthroughs}}. \tag{36}$$

The marginal value Δ_i is the key object of the paper. Channel (i), the firm's own compliance cost of c_i per unit of R, is negative and largest in magnitude for the leader. Channel (ii), the Cournot gain when marginal fringe firms stop producing, is positive, since $dm^{\mathrm{a}}/dR < 0$ and $\partial\pi_i/\partial m^{\mathrm{a}} < 0$. Channel (iii), the gain from fewer breakthroughs, which would lower the firm's rank or displace it, is positive, since $dX/dR < 0$ and, at the estimate, a breakthrough lowers the value of every superstar in every state.

The only fringe firm whose status responds to R is the marginal one at ε^*, so

$$\frac{dm^{\mathrm{a}}}{dR} \;=\; m\,f(\varepsilon^*)\,\frac{d\varepsilon^*}{dR}, \qquad \frac{d\varepsilon^*}{dR} \;=\; \frac{-1}{1 - m\,f(\varepsilon^*)\,\dfrac{\partial(\pi_f/Y)}{\partial m^{\mathrm{a}}}}, \qquad \frac{dX}{dR} \;=\; x_f\,\frac{dm^{\mathrm{a}}}{dR}. \tag{37}$$

The denominator of $d\varepsilon^*/dR$ exceeds one, since $\partial\pi_f/\partial m^{\mathrm{a}} < 0$. When a higher R makes marginal fringe firms stop producing, the remaining producers earn higher profits, which offsets part of their higher compliance cost. While organized, the association chooses b_A at every instant to maximize the total value of the industry's fringe firms net of its lobbying cost, taking rivals' intensities as given:

$$m^{\mathrm{a}}\,\Delta_A\,\frac{\partial R}{\partial b_A} \;=\; \kappa_A\,\phi_A\,b_A^{\phi_A-1}, \tag{38}$$

where Δ_A is the marginal value of a higher R per producing member:

$$\Delta_A \;=\; \underbrace{-(1-\tau_0)}_{\text{own compliance}} \;+\; \underbrace{(1-\tau_0)\,\frac{\partial(\pi_f/Y)}{\partial m^{\mathrm{a}}}\,\frac{dm^{\mathrm{a}}}{dR}}_{\text{fewer active fringe firms}} \;+\; \underbrace{\frac{m}{m^{\mathrm{a}}}\,\frac{dX}{dR}\sum_h P_h\left[\bar{v}_0^f(S^{X,h})-\bar{v}_0^f(S)\right]}_{\text{fewer breakthroughs}}. \quad (39)$$

The first two terms sum to $(1-\tau_0)\,d\varepsilon^*/dR<0$ (equation (30)). When marginal rivals stop producing, a member's operating profit rises by less than its compliance cost. Only the last term can make Δ_A positive. It is the change in the value of every firm in the pool when breakthroughs become rarer, and it is positive when a breakthrough by another firm lowers the value of a firm in the pool.

A participant lobbies for a higher regulatory cost if its marginal value is positive and against it if its marginal value is negative. The superstars on each side are

$$\mathcal{N}^{\mathrm{pro}}(S)=\{i: a_{Ri}=1,\ \Delta_i>0\},\qquad \mathcal{N}^{\mathrm{anti}}(S)=\{i: a_{Ri}=1,\ \Delta_i<0\}, \quad (40)$$

and the association lobbies on the side given by the sign of Δ_A. Given the influence function (13), the marginal influence of b_{Ri} is

$$\frac{\partial R(S)}{\partial b_{Ri}} \;=\; \begin{cases} \dfrac{R_0\,\beta^{\mathrm{pro}}\,\theta\,b_{Ri}^{\theta-1}}{1+\beta^{\mathrm{anti}}B^{\mathrm{anti}}} & \text{if } i\in\mathcal{N}^{\mathrm{pro}}(S),\\[2ex] -\dfrac{R\,\beta^{\mathrm{anti}}\,\theta\,b_{Ri}^{\theta-1}}{1+\beta^{\mathrm{anti}}B^{\mathrm{anti}}} & \text{if } i\in\mathcal{N}^{\mathrm{anti}}(S), \end{cases} \quad (41)$$

and $\partial R/\partial b_A$ takes the same form. Against a higher cost, the firm's own intensity enters both R and B^{anti}, so the intensities at each state solve a fixed point (Appendix J gives the closed forms). Interior intensities require $\phi_R>\theta$. When the marginal values of a higher R differ in sign, for example when some superstars gain from a higher cost while the association loses, the two sides push R in opposite directions. $R(S)$ and all intensities jointly solve equations (35) and (38).

4.6 Equilibrium

Growth. The frontier of an industry advances when its leader, or a firm tied with it, innovates, and when a new superstar lands above the leader. The aggregate growth rate is

$$g \;=\; \ln(1+\lambda)\,\mathbb{E}_\mu\Big[\sum_{i:\,n_i=0} z_i \;+\; X(S)\sum_h P_h(S)\,n_h^+(S)\Big], \quad (42)$$

where the expectation $\mathbb{E}_\mu$ is over industry states and locked costs, and $n_h^+(S) \geq 0$ is the number of steps by which a new superstar that lands h steps above the margin passes the former leader.

Labor market and output. Each producer's labor demand follows from the product market equilibrium and is proportional to Y/w. Labor market clearing requires

$$\int_0^1 \left(\sum_{i=1}^{N_{jt}} l_{ijt} + \int_0^{m_t} l_{fkjt}\, dk \right) dj \;=\; L_t, \tag{43}$$

with L_t given by the household's labor supply (17). Output then follows from (2).

Resource constraint. Final-good output is allocated among private consumption, government purchases, R&D, entry, fringe operating costs, compliance costs, and lobbying costs. Dividing by Y_t and aggregating across industries with μ gives the resource constraint on the balanced growth path,

$$1 \;=\; \frac{C_t^p}{Y_t} \;+\; (1-\theta_\Gamma)\,\mathbb{E}_\mu\Big[\sum_i \Gamma_i\Big] \;+\; \mathcal{R}_{\text{R\&D}} \;+\; \psi\,(\delta m)^2 \;+\; \mathcal{R}_{\text{fixed}} \;+\; \mathcal{R}_{\text{comp}} \;+\; \mathcal{L}, \tag{44}$$

where

$$\begin{aligned}
\mathcal{R}_{\text{R\&D}} \;&=\; \mathbb{E}_\mu\Big[\sum_i \chi\, z_i^{\phi}\Big] \;+\; \sum_S \mu(S)\, m^{\text{a}}(S)\, \chi_f\, x_f(S)^{\phi_f}, \\
\mathcal{R}_{\text{fixed}} \;&=\; m \sum_S \mu(S)\, \mathbb{E}\big[\varepsilon\, \mathbf{1}\{\varepsilon \leq \varepsilon^*(S)\}\big], \\
\mathcal{R}_{\text{comp}} \;&=\; \sum_S \mu(S) \Big[\sum_{i=1}^{N(S)} c_i(S) + m^{\text{a}}(S)\Big]\, R(S), \\
\mathcal{L} \;&=\; \mathbb{E}_\mu\Big[\sum_i \sum_I a_{Ii}\big(\kappa_I\, b_{Ii}^{\phi_I} + \Psi_{Ii} \prod_{J\neq I} (1-\iota_{IJ})^{a_{Ji}}\big) \;+\; a_A\big(\kappa_A\, b_A^{\phi_A} + \Psi_A\big)\Big].
\end{aligned}$$

Compliance costs use output and produce nothing. Taxes, contract fees, lump-sum payments, and association dues move resources between agents, so they do not appear in the resource constraint (44).

Definition 1 (*Balanced-Growth-Path Equilibrium*)**.** *A stationary Markov-perfect balanced-growth-path (BGP) equilibrium, in which superstars' strategies depend on (S, s_i), locked costs are private, and beliefs about them are formed under μ, consists of*

- superstar values $v(S, s_i)$ and policies $\{z_i, b_{Ri}, b_{Ti}, b_{Pi}\}$ with participation cutoffs $\{\Psi_I^*\}$; fringe values $v^f(S, \Psi_A, \varepsilon)$, pool values $\bar{v}_0^f(S)$, values of new superstars $v^{\text{new}}(S)$, cutoffs

$\varepsilon^*(S)$ and breakthrough intensities $x_f(S)$; the association's cutoff $\Psi_A^*(S)$ and intensity $b_A(S)$;

- the regulatory cost $R(S)$ at every industry state;
- quantities, prices and sales shares at every industry state;
- a fringe pool m, an entry rate e, and active masses $m^{\mathrm{a}}(S) = mF(\varepsilon^*(S))$;
- a stationary distribution μ over industry states, superstars' locked costs, and the association's locked cost;
- a wage ratio w/Y, labor supply L, a labor tax τ_w, private consumption C^p/Y, lump-sum payments T/Y, an interest rate r, and a growth rate g;

such that:

1. **Product market.** Given $m^{\mathrm{a}}(S)$, quantities and prices satisfy the superstars' Cournot best responses and the fringe's price-taking conditions.
2. **Superstars.** Given the intensities of rivals and the association, the fringe's breakthrough intensity x_f and the organizing probability $\mathcal{P}_A$, values and policies solve (22), where each firm internalizes through (13) the effect of its own intensity on $R(S)$ and, through $R(S)$, on $m^{\mathrm{a}}(S)$; participation follows (26).
3. **Fringe.** v^f solves (27), pool values satisfy (31) and (32), and v^{new} is given by (28); $\varepsilon^*(S)$ satisfies (30), $x_f(S)$ satisfies (29), the association's decisions satisfy (34) and (38), and dues (15) are feasible.
4. **Contest.** At every state, intensities satisfy (35) for every superstar on the R instrument, with sides (40), and (38) for an organized association, and $R(S)$ is given by (13).
5. **Free entry.** m satisfies (33), with $e = \delta m$.
6. **Consistency.** μ is the invariant distribution of the process induced by the policies, and the rivals' expected policies $\bar{z}_{i'}$ and $\bar{\mathcal{P}}_{Ii'}$ are computed under μ.
7. **Labor market and government.** The labor market clears (43) at the labor supply (17), the government budget (16) balances, and T is the rebate of corporate tax revenue plus the transfers.
8. **Growth and goods market.** g satisfies (42), $r = \rho + g$, and C^p/Y satisfies (44).

Computing the equilibrium. With $\bar{N} = 4$ and $\bar{n} = 5$, the full state has 481,040 industry states and 1.9 million superstar states. We compute the equilibrium with two simplifications. First, a firm's beliefs about its rivals' tax and procurement statuses equal their stationary frequencies among superstars of the same rank in the same technology state. The state keeps the regulation statuses, the association's status, and each firm's own statuses, which reduces it to 2,180 industry states and 33,104 superstar states. Decisions use the beliefs, while the stationary distribution, growth, and every moment use the actual statuses. The computed equilibrium thus replaces rivals' tax and procurement statuses in the state with these beliefs. Second, a superstar's policies are evaluated at the expected cost of its current period of participation, $\hat{\Psi}_I = \mathbb{E}[\Psi \mid \Psi \leq \Psi_I^*]$, so that values are affine in the locked costs and the cutoff (26) has a closed form. Spending uses the exact distribution of locked costs. Appendices H–J give the details.

4.7 Welfare and counterfactual analysis

We compare each equilibrium with the economy without lobbying, which is identical except that every participation cost and the association's organizing cost are infinite. Thus $R = R_0$, every superstar pays τ_0, and contracts equal their baseline $\Gamma_0(1+\lambda)^{-\kappa_\Gamma n_i}$. Its fringe pool m adjusts through free entry, and lump-sum transfers stay at $\bar{T}$. A counterfactual ban on an instrument sets that instrument's participation costs to infinity. We report two kinds of results for balanced growth paths. The first is the changes in output, in the share of output used by each item of the resource constraint, in growth, and in hours, none of which depends on preferences. The second is a consumption-equivalent welfare measure that aggregates them.

Output, resource use, and growth. Consumption at date zero, C_0, equals output times the share of output left after compliance costs, fringe operating costs, entry, R&D, and lobbying. Every producer's technology is homogeneous of degree α in labor, so log output is the sum of three terms:

$$\begin{aligned} \ln Y_0 &= \alpha \ln L + \ln \mathcal{A}^* + \ln E, \\ \ln \mathcal{A}^* &= \sum_S \mu(S) \frac{1}{\upsilon} \ln \Big[\sum_{i=1}^{N(S)} \mathcal{A}_i(S)^\upsilon + \mathcal{A}_f(S)^\upsilon \Big], \qquad \upsilon = \frac{\eta - 1}{\eta - \alpha(\eta - 1)}. \end{aligned} \tag{45}$$

Here $\mathcal{A}^*$ is the productivity that an efficient allocation of the same labor would achieve, given each industry's state and its active fringe firms. A superstar has $\mathcal{A}_i = q_i$, and the active fringe, taken as one producer, has $\mathcal{A}_f = (m^{\mathrm{a}})^{1-\alpha} q_f$. The term $\ln E \leq 0$ is the output lost because markups distort the allocation of labor across producers. The share of output

consumed splits by item of the resource constraint (44), and growth splits by source (42).

Consumption-equivalent welfare. On the balanced growth path, welfare is

$$W \;=\; \frac{\ln C_0}{\rho} \;+\; \frac{g}{\rho^2} \;-\; \frac{D}{\rho}, \qquad D \;=\; \xi\,\frac{L^{1+1/\epsilon_L}}{1+1/\epsilon_L}. \tag{46}$$

The three terms are the value of initial consumption, of its growth, and of the disutility of hours D. We summarize welfare by the consumption-equivalent variation (CEV) of Lucas (1987), as in Jones and Klenow (2016): the permanent percentage increase in consumption, at every date of the equilibrium with lobbying, that would make the household as well off as in the economy without lobbying. It solves

$$\ln(1+\mathrm{CEV}) \;=\; \rho\left(W^{\mathrm{ref}} - W\right) \;=\; \underbrace{\ln C_0^{\mathrm{ref}} - \ln C_0}_{\text{level}} \;+\; \underbrace{\frac{g^{\mathrm{ref}} - g}{\rho}}_{\text{growth}} \;+\; \underbrace{D - D^{\mathrm{ref}}}_{\text{hours}}, \tag{47}$$

so that CEV > 0 means that lobbying lowers welfare, or equivalently that removing lobbying raises it. Following the decomposition of Jones and Tonetti (2020) into level, privacy, and growth terms, the measure splits additively into the level of consumption, its growth rate, and hours. Preferences enter only through the weights on growth and on hours. The weight on growth is the inverse of the gap between the interest rate and the growth rate, and the weight on hours is the after-tax wage relative to consumption, so both are observable prices. The decomposition therefore depends on preferences only through $r - g$, which equals ρ with log utility (Appendix N).

5 Estimation

We estimate the model by the simulated method of moments, in two steps. The first step estimates twelve technology parameters on moments of industry structure, firm dynamics, and aggregates, in the economy with compliance costs and without lobbying. The second step holds the technology fixed and estimates thirty lobbying parameters in the equilibrium with lobbying. On taxes and procurement, we observe the return to lobbying. On regulation we do not, so we rely on who lobbies, what they ask for, and which bills pass. As in Cavenaile et al. (2026), the regressions in Sections 5.4 and 5.5 are moments the model must reproduce, computed the same way in the data and in the model, rather than causal estimates.

5.1 Strategy

Let $\mathcal{M}(\vartheta)$ be the vector of model moments at the parameter vector ϑ and $\hat{\mathcal{M}}$ the vector of their data values. The estimator minimizes the sum of squared deviations with equal weights. Element by element, the deviation is $\mathcal{M}/\hat{\mathcal{M}}-1$ for technology moments, $\ln(\mathcal{M}/\hat{\mathcal{M}})$ for lobbying moments that are positive by construction, and $(\mathcal{M}-\hat{\mathcal{M}})/\max(|\hat{\mathcal{M}}|, \text{se})$ for those that can take either sign, where se is the standard error of the data moment. There are 30 technology moments and 38 lobbying moments. Every model moment is defined as its data counterpart is. Moments of industry structure come from the stationary distribution over industry states, and moments measured on a panel of firms come from a simulated panel of superstars of the same length, with the same regressions and fixed effects. Growth must lie within 2.5 standard errors of the data, compliance costs between 0.33 and 0.96% of output, and their size elasticity between 0.40 and 0.98 (Section 5.4). The objective penalizes spending per request on taxes and procurement only outside a band around its data value, and it penalizes superstars' and associations' total lobbying spending only above a ceiling, because the data measure these objects imprecisely.

An equilibrium takes about two hours to compute and the objective is not smooth in all parameters, so both steps start from a global search over a scrambled Sobol sequence. The estimate is the point with the lowest objective among those that reproduce all eight statistics of the bill regression of Section 5.5 (Table 8) within 2.5 standard errors. Appendix K gives the identification argument for each block and every moment.

5.2 Parameters set outside the model

The model describes the private business sector, so output maps to business value added, 75.6% of GDP (Table 5). A superstar is one of the four largest firms of its industry with sales of at least 10% of the leader's, so $\bar{N}=4$. The ladder depth $\bar{n}$ is the largest gap at which the smallest superstar, in every configuration of its rivals' gaps, sells at least 10% of the leader's sales in the product market equilibrium with $\hat{m}^{\text{a}}$ active fringe firms, which is 5 at the estimate. The number of active fringe firms without lobbying or contracts, $\hat{m}^{\text{a}}$, is the Business Dynamics Statistics count in the median industry-year across 288 four-digit industries, 5,245 firms. The entry cost ψ takes the value at which free entry delivers this number. We set the lump-sum transfers $\bar{T}$ so that the labor income tax in the economy without lobbying equals 24.4%, the average US labor income tax (OECD Taxing Wages), and hold them at that share of output in every economy we compare. The returns to scale follow from the cost data alone. The Economic Census payroll ratio, 0.764, is the inverse of the superstar markup, so the markup is $1/0.764=1.309$. The superstars' ratio of sales to

cost of goods sold, 1.497, equals the markup divided by α, so $\alpha = 1.309/1.497 = 0.875$.

Table 5. Parameters set outside the model

Parameter		Value	Source or target
Discount rate	ρ	0.04	Cavenaile et al. (2026)
Frisch elasticity of labor supply	ϵ_L	0.5	standard
Weight on the disutility of labor	ξ	0.519	hours equal one without lobbying, taxes, and contracts
Maximum number of superstars	$\bar{N}$	4	superstar definition
Smallest superstar, share of the leader's sales		0.10	superstar definition
Active fringe firms per industry, no lobbying or contracts	$\hat{m}^{\mathrm{a}}$	5,245	median industry-year (BDS)
Returns to scale in labor	α	0.875	payroll ratio and sales-to-cost ratio
Capital share (labor-share mapping)		0.195	2008–2022, as in Barkai (2020)
Contractor's margin	θ_Γ	0.101	Compustat, federal contractors
Effectiveness of lobbying for a higher cost	β^{pro}	1	normalization

Notes: The entry cost ψ is set by free entry. BDS: Business Dynamics Statistics.

5.3 Technology

Table 6 reports the twelve technology estimates and the moments most informative about each. A larger quality step λ and a higher elasticity of substitution η both widen the sales gaps between superstars. Growth and the payroll ratio separate the two, because growth rises with λ and the payroll ratio, the inverse of the markup, rises with η. Growth, R&D intensity, and the within-firm elasticity of patents to R&D identify the cost of superstars' innovation. The rate at which new superstars appear, and how it varies with the sales gap to the leader, identifies the breakthrough cost. The sales share of the four largest firms (CR4) identifies fringe quality. The dispersion of operating costs, the firm exit rate, and business applications identify the fringe's operating costs and turnover. Table 14 in the appendix reports all 30 moments.

5.4 Lobbying

Table 7 reports the thirty lobbying estimates. The effective tax rate identifies τ_0. Contracts over sales and their elasticity to sales identify the baseline contract Γ_0 and its decline with the gap, κ_Γ. The return to lobbying identifies the lobbying costs. Within the firm, a year of procurement lobbying is followed by log contracts 0.22 higher. Among firms that lobby, one log unit more spending per request is followed by a tax rate on taxable income 0.42 percentage points lower over the next three years. On every instrument, participation by rank identifies the mean of the log participation cost, how steeply participation rises with

Table 6. Technology parameters

Parameter		Estimate	Most informative moments
Quality ladder and demand			
Quality step	λ	0.217	sales gaps among superstars; growth
Elasticity of substitution	η	3.97	sales gaps among superstars; payroll ratio
Superstars' innovation			
R&D cost: level	χ	655	growth; R&D intensity
R&D cost: convexity	ϕ	4.42	within-firm elasticity of patents to R&D
Breakthroughs			
Breakthrough cost: level, aggregate units	$\tilde{\chi}_f$	25.8	new superstars per industry-year
Breakthrough cost: convexity	ϕ_f	4.65	new superstars by #4/#1
Climb of new superstars	u	0.359	landing steps; new superstars' sales relative to the leader
Fringe quality			
Fringe quality, aggregate units	$\tilde{\zeta}$	0.765	CR4
Curvature of spillovers to the fringe	γ	1.23	CR4; leader's share of superstars' R&D
Fringe operating costs and turnover			
Operating cost: mean, per firm-year	$\bar{\varepsilon}$	2.04×10^{-4}	applications that become employers
Operating cost: dispersion of logs	σ_ε	1.67	dispersion of fixed costs
Death rate	δ	0.074	firm exit rate; applications per employer firm

Notes: Costs are in units of aggregate output Y_t. Parameters with a tilde are in aggregate units, which do not depend on the unit in which fringe firms are counted: the per-firm breakthrough cost level is $\chi_f = \tilde{\chi}_f \, (\hat{m}^{\mathrm{a}})^{\phi_f - 1}$ and the fringe quality scale is $\zeta = \tilde{\zeta} \, (\hat{m}^{\mathrm{a}})^{\alpha - 1}$, where $\hat{m}^{\mathrm{a}} = 5{,}245$ is the number of active fringe firms per industry in the data (Table 5).

the firm's gain from lobbying identifies its dispersion, and the persistence of participation identifies the redraw rate.

We do not observe regulatory costs by industry and year. We bound compliance costs with the estimates of Trebbi et al. (2026). Their total, 0.96% of output, is the upper bound. The part due to the four instruments we classify as regulatory costs (licenses, reporting duties, labeling, and privacy), 0.33%, is the lower bound. The baseline regulatory cost R_0 satisfies $R_0 \, (K + \hat{m}^{\mathrm{a}}) = \mathcal{C}$, where $\mathcal{C}$ is compliance costs without lobbying as a share of output and K is the average of $\sum_i c_i$ over technology states with $\bar{N}$ superstars, so that compliance costs are a share $\mathcal{C}$ of output in a full industry with the observed number of active fringe firms. We constrain the size elasticity of superstars' compliance costs, e_R, to lie between 0.40 and 0.98. Measured costs of specific rules rise less than proportionally with firm size. In our calculations from the size-class costs of California's privacy regulation and of federal tax compliance, the elasticity is 0.33 to 0.47 (Berkeley Economic Advising and Research, LLC, 2019; Contos et al., 2012). It is 0.41 to 0.46 for securities disclosure and internal-control rules (Ewens et al., 2024) and 0.69 to 0.84 for bank compliance (Dahl et al., 2018) and Canadian regulatory filings.[6] Occupation-based measures of all compliance work, from Trebbi et al.

[6]Our calculations: 0.69 to 0.70 from the Conference of State Bank Supervisors' 2014 survey of community

(2026) and from the Current Population Survey within industries, give 0.95 to 0.98, which likely overstate it. Within this band, the share of superstars' requests for higher regulatory costs identifies e_R, because the faster compliance costs rise with size, the more superstars ask for lower costs. How often the association lobbies, how long it keeps lobbying, and how often each side lobbies without the other identify its organizing cost. Upper bounds on superstars' and associations' lobbying spending restrict the costs of lobbying intensity. We identify the effect of lobbying on regulatory costs from bill passage. At the estimate, the association's spending stays below a third of its members' after-tax surplus in every organized state, so dues are feasible.

Table 7. Lobbying parameters

Parameter		Estimate
Regulation		
Compliance without lobbying, share of output	$\mathcal{C}$	0.0081
Size elasticity of superstars' compliance costs	e_R	0.82
Contest curvature	θ	0.15
Effectiveness of lobbying against a higher cost	β^{anti}	7.63
Cost of influence	κ_R	6.13×10^{-4}
Convexity of lobbying cost: superstars, association	ϕ_R, ϕ_A	2.11, 3.08
Participation cost of superstars: log mean, s.d.	μ_R, σ_R	−9.48, 2.81
Participation cost redraw rate (per year)	δ_R	0.10
Association organizing cost: log mean relative to superstars', s.d.	$\mu_A - \mu_R$, σ_A	1.96, 1.44
Association cost redraw rate (per year)	δ_A	0.10
Association lobbying cost: spending / industry sales	κ_A	2.22×10^{-4}
Profit tax		
Statutory tax rate	τ_0	0.244
Lobbying cost: leader's tax cut, share of τ_0	κ_T	4.2%
Lobbying cost convexity	ϕ_T	987
Participation cost: log mean, s.d.; redraw rate (per year)	μ_T, σ_T; δ_T	−6.56, 2.45; 1.45
Procurement		
Baseline contracts: contracts / sales	Γ_0	0.017
Decline of contracts with the gap	κ_Γ	2.34
Lobbying cost: leader's contracts due to lobbying	κ_P	1.5%
Lobbying cost convexity	ϕ_P	12.6
Participation cost: log mean, s.d.; redraw rate (per year)	μ_P, σ_P; δ_P	−6.62, 5.66; 0.63
Economies of scope in participation		
Regulation and tax; regulation and procurement; tax and procurement	ι_{RT}; ι_{RP}; ι_{TP}	0.03; 0.10; 0.09

Notes: The lobbying costs of the association and of the tax and procurement instruments are searched over, and reported, in the units of the moments they target.

banks, 0.81 to 0.83 from the 2015–2017 surveys in Dahl et al. (2018), and 0.79 to 0.84 from Statistics Canada's 2011 Survey of Regulatory Compliance Costs.

5.5 Bills

Bill passage identifies how lobbying influence translates into regulatory costs. We call a set of bills of one Congress that are the same legislation a bill. A regulation bill is a bill with a core regulatory direction and at least half of its provisions on regulation, excluding must-pass and appropriations bills. A regulation bill raises regulatory costs if most of its core provisions raise them, and lowers them otherwise. Following Kang (2016), a bill ℓ that raises regulatory costs passes with probability

$$P_\ell \;=\; \frac{P_\ell^0 + B_\ell^{\text{pro}}}{1 + B_\ell^{\text{pro}} + B_\ell^{\text{anti}}}, \qquad B_\ell^{\text{pro}} = \beta^{\text{pro}} \sum_{i \in \text{pro}} b_{Ri}^\theta \, \varsigma_{i\ell}, \qquad B_\ell^{\text{anti}} = \beta^{\text{anti}} \sum_{i \in \text{anti}} b_{Ri}^\theta \, \varsigma_{i\ell}, \tag{48}$$

and a bill that lowers regulatory costs passes with probability $(P_\ell^0 + B_\ell^{\text{anti}})/(1 + B_\ell^{\text{pro}} + B_\ell^{\text{anti}})$. Here P_ℓ^0 is the probability that the bill passes when no superstar or association lobbies on it, the sums run over the superstars and the association that lobby for and against a higher regulatory cost, $\beta^{\text{pro}} b_{Ri}^\theta$ or $\beta^{\text{anti}} b_{Ri}^\theta$ is participant i's influence, and $\varsigma_{i\ell}$ is the share of that influence it puts on bill ℓ (Appendix M). Thus participants that lobby for a higher regulatory cost push bills that raise it and block bills that lower it, and participants that lobby against do the reverse. The association lobbies against higher regulatory costs, so it blocks bills that raise them and pushes bills that lower them.

We measure the baseline passage probability P_ℓ^0 directly from bills on which no superstar and no industry association lobbied, classified with the procedure of Section 2.2. A bill counts as passed if it became law or if its core regulatory provision is in a law of the same Congress. Without lobbying by a superstar or an association, 3.42% of bills that raise regulatory costs pass, against 7.67% of bills that lower them. We then simulate every real bill. For each industry that lobbied on the bill in the data, we draw an industry state from the stationary distribution and simulate which of its firms lobby on the bill. We sum influence across industries and draw passage from equation (48). Appendix K.4 gives the details.

We run one regression on real and simulated bills (Table 8). The coefficient on the number of superstars on a bill identifies the cost of influence κ_R, because cheaper influence means more influence per firm, so passage rises faster with the number of superstars. The coefficient on association opposition identifies β^{anti}, because it measures how much blocking lowers passage. The share of the 2,190 lobbied regulation bills that pass identifies the total influence of both sides, given the baseline passage rates. By construction of the estimate, the model reproduces all eight statistics within 2.5 standard errors (Section 5.1). The repeal intensity is set to match the ratio of compliance costs removed to costs added by the laws of the period: 0.050 in the data and 0.056 in the model.

Table 8. The bill regression: data and model

	Raising bills			Lowering bills		
	Data	s.e.	Model	Data	s.e.	Model
Number of superstars on the bill	0.404	0.111	0.275	0.368	0.181	0.535
Any association on the bill	0.160	0.247	0.190	0.220	0.355	0.426
Association supports the bill	0.659	0.232		−0.109	0.324	0.145
Association opposes the bill	−0.522	0.331	−0.292	−0.140	0.766	
Any other client on the bill	0.398	0.281	0.200	−0.064	0.369	0.171
Share of bills passed	0.0801	0.0067	0.0662	0.1458	0.0152	0.1450

Notes: Fractional logit of passing, one regression with coefficients by bill type. Outcome: law, or the core provision in a law of the same Congress. Superstars: log(1 + superstars listing the bill, weighted by the share of their lobbying on regulation; each counts one in the model). Any association, supports, opposes, and any other client are indicators. Controls: tax or contract content, sponsor in the majority, sponsor chairs a referral committee, bipartisan sponsorship, log cosponsors. Fixed effects: number of levers, Congress, lowering bill. Sample: all regulation bills, 2008–2022, 2,360 rows, 3,248 bills after sample weights, 255 passed rows. Standard errors clustered by bill family (2,015). Share passed: the 2,190 bills in lobbying reports. In the model the association never supports a raising bill or opposes a lowering bill, so these two cells are empty. Any other client is not targeted.

5.6 Fit

Table 9 reports the main moments in the data and the model, and Table 14 in Appendix K reports all of them. Figure 7 shows superstars' participation by rank in the model and the data. The model fits growth, the structure of sales among superstars, the arrival of new superstars, the tax rate, contracts over sales, and the return to tax lobbying. It reproduces the gap between superstars and the association in the share of requests for higher regulatory costs (67 against 27% in the model, 53 against 34% in the data; Section 6.1 explains how the model association's share is counted), and both compliance moments lie inside their bands. The model overstates how often firms and associations lobby on regulation. The leader lobbies in 48% of quarters, against 33% in the data, and the association lobbies in 90% of industry-quarters, against 56%. Persistence is 0.91 against 0.60. The model's return to procurement lobbying is 0.006, against 0.217 in the data, and the model understates the overlap of participation across instruments (Appendix K.2). Having shown that the estimated model reproduces who lobbies, on which side, and which bills pass, we turn to the consequences of lobbying.

Table 9. Main moments: data and model

Moment	Data	Model
Technology		
Growth of business output per hour	0.0159	0.0172
CR4	0.213	0.255
Superstars per industry	3.64	3.71
New superstars per industry-year	0.132	0.144
Sales / cost of goods sold, superstars	1.497	1.686
Labor share	0.596	0.644
Regulation		
Compliance costs / output	0.003 – 0.010	0.0082
Size elasticity of compliance costs	0.40 – 0.98	0.81
Leader lobbying in a quarter	0.328	0.484
Still lobbying a year later, given the state	0.600	0.905
Superstars' share of requests for higher costs	0.533	0.671
Association lobbies in an industry-quarter	0.558	0.897
Association's share of requests for higher costs	0.343	0.273
Tax and procurement		
Effective tax rate, all firms	0.222	0.242
Return to tax lobbying	−0.0042	−0.0040
Contracts / sales	0.019	0.017
Return to procurement lobbying	0.217	0.006

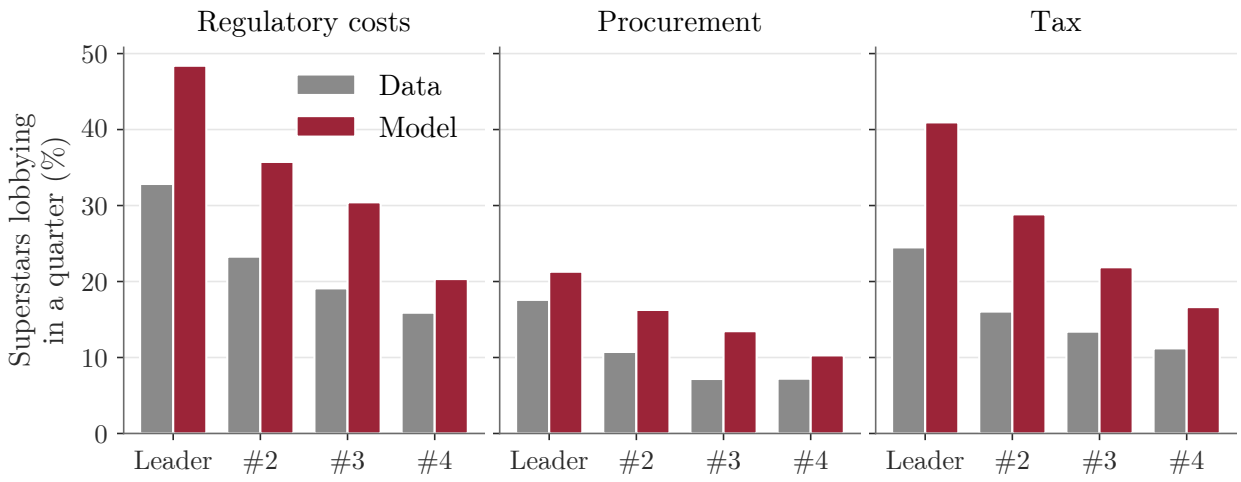


Figure 7. Who lobbies: superstars' participation by rank, model and data

Notes: Share of superstars lobbying in a quarter, by rank in the industry. Data: US-headquartered superstars, 2008–2022, weights equal across industries.

6 The consequences of lobbying

Lobbying raises regulatory costs by 3% on average, and removing it raises consumption-equivalent welfare by 0.225%. Lobbying over regulation accounts for almost all of the gain. We compare the estimated economy with the economy without lobbying of Section 4.7.

6.1 Who lobbies, on which side, and why

Superstars lobby for higher regulatory costs. In the estimated model, a superstar's marginal value of a higher regulatory cost, Δ_i in equation (36), is positive for 99.9% of superstars across industry states, so superstars lobby for a higher regulatory cost. Figure 8 shows the three terms of this marginal value by rank. Each term is measured per unit of R, the yearly compliance cost of an active fringe firm in units of output. One more unit of R costs the leader 173 after tax and the fourth superstar 70, because a superstar pays $c_i R$ and c_i rises with size. The gains exceed the compliance cost at every rank. A higher R makes marginal fringe firms stop producing, which raises superstars' profits and lowers the rate of fringe breakthroughs, while compliance costs rise less than proportionately with size ($e_R = 0.82$). The source of the gain varies with rank. The leader's gain from higher profits as marginal fringe firms stop producing (188) is about as large as its gain from fewer fringe breakthroughs (215), each of which could add a competitor or leapfrog it. The main gain of the fourth superstar is fewer breakthroughs (138 against 38 from profits), since a new superstar could push it out of the top four.

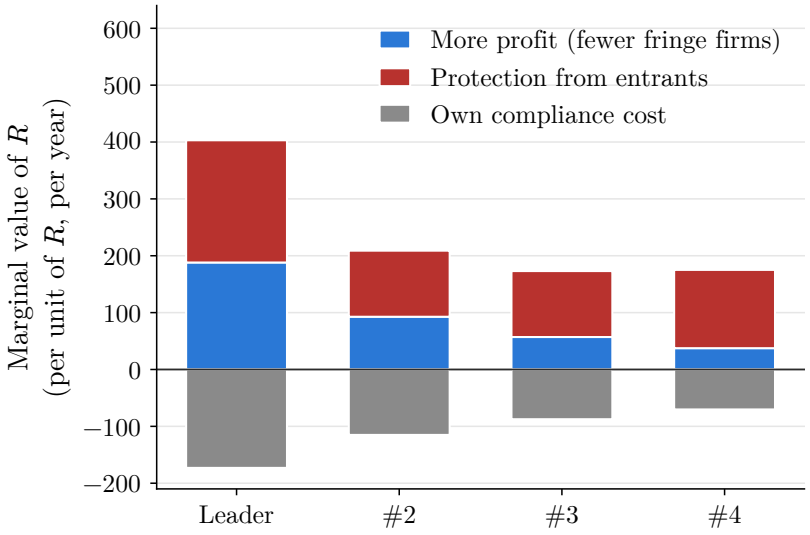


Figure 8. Why superstars lobby for higher regulatory costs

Notes: The three components of a superstar's marginal value of a higher regulatory cost (36), by rank, per unit of R (the yearly compliance cost of an active fringe firm, in units of output), averaged over the stationary distribution of industry states at the estimate. Blue: Cournot gain as marginal fringe firms stop producing. Red: value of fewer fringe breakthroughs. Grey: own compliance cost after tax.

The association lobbies against higher regulatory costs. The association lobbies against higher regulatory costs in every industry state, because its marginal value of a higher regulatory cost, Δ_A in equation (39), is negative everywhere. A fringe firm bears the compliance cost with little gain. In the units of Figure 8, per producing member, the

own compliance cost is -0.756 after tax, the gain as marginal rivals stop producing is $+0.285$, and the change in the pool's value from fewer breakthroughs is -0.004 on average. Thus $\Delta_A = -0.476$ on average, and it is negative in every organized state. The first two terms sum to the after-tax change in the production cutoff, which is negative because the survivors' higher profits only partly offset the compliance cost. Because the association always lobbies against, its share of requests for higher regulatory costs in Table 9, 0.27, counts the channel of Δ_A that favors a higher cost, on average $0.285/(0.756 + 0.285 + 0.004)$ (Appendix K.3).

Lobbying raises regulatory costs in some industries and lowers them in others. Superstars and the association lobby together in 72% of industry-quarters, and the balance of the two sides differs across industries. On average, regulatory costs are 1.03 times R_0. They are above R_0 in 41% of industry states and below it in 57%, and no one lobbies on regulation in the remaining 2%. Across states, the cost is 0.50 times R_0 at the 10th percentile, 0.86 at the median, and 1.53 at the 90th percentile. Regulatory costs are lowest where few superstars compete: on average 0.88 times R_0 in industries with one superstar, 0.99 with two, 1.06 with three, and 1.03 with four.

6.2 The welfare gain from removing lobbying

Removing lobbying raises consumption-equivalent welfare by 0.225%. Saved lobbying spending accounts for 0.094 percentage points of this gain, lower compliance costs for 0.016, and the change in industry structure, net of the operating and entry costs of the additional fringe firms, for the remaining 0.115. By equation (47), the gain is the sum of four terms,

$$\ln(1 + \mathrm{CEV}) \;=\; \Delta \ln Y_0 \;+\; \Delta \ln(C_0/Y_0) \;+\; \Delta g/\rho \;-\; \Delta D,$$

where Δ is the economy without lobbying minus the economy with lobbying. The four terms are output today, the share of output consumed, growth, and the disutility of hours. Table 10 reports each term and its parts.

Removing lobbying raises the expected value of a firm in the fringe pool, so the pool grows by 0.9% under free entry and the mass of active fringe firms by 0.7%. More active fringe firms mean more breakthroughs, so the stationary distribution of industries shifts toward states with more superstars. Figure 9 shows the distribution of industries with lobbying, and Figure 10 shows how it changes without lobbying, together with the change in active fringe firms in each type of industry.

Output rises by 0.125%. Output combines what the superstars produce, what the fringe produces, and the gain from buying from several producers, the love of variety. Figure 11 reports each part, first holding every industry's state fixed and then letting the distribution

Table 10. The welfare gain from removing lobbying, by term

	With → without lobbying	Welfare gain (%)
Output today, $\Delta \ln Y_0$		**0.125**
Same industry states		0.048
Superstars		−0.015
Fringe		0.086
Love of variety		−0.024
Industries adjust		0.077
Superstars		0.000
Fringe		0.077
Love of variety		−0.001
Share of output consumed, $\Delta \ln(C_0/Y_0)$	91.46 → 91.52	**0.069**
Lobbying spending	0.09 → 0.00	0.094
Compliance costs	0.82 → 0.80	0.016
Fringe operating costs	4.74 → 4.75	−0.015
Entry costs	1.34 → 1.37	−0.028
R&D	1.56 → 1.56	0.001
Growth, $\Delta g/\rho$	1.72465 → 1.72595	**0.032**
Same industry states	change −0.00025	−0.006
Industries adjust	change +0.00155	0.039
Hours, $-\Delta D$	change +0.001	**−0.001**
Total, $\ln(1 + \text{CEV})$		**0.225**

Notes: From the economy with lobbying to the economy without lobbying, at the estimate. Welfare gain: the term's contribution to $100 \ln(1 + \text{CEV})$ in equation (47), percent of consumption at every date. Output today: change in log output. Same industry states keeps every industry's state, weighted by the distribution of states with lobbying; industries adjust adds the shift in that distribution. Superstars and fringe: quantities added up, with log-mean quantity weights; love of variety: the difference between the industry aggregate and that sum. Share of output consumed and its items: percent of output (resource constraint (44)); an item's welfare gain is minus its change divided by the share of output consumed. Growth: percent a year, changes in percentage points, capitalized at $r - g = \rho = 0.04$. Hours: percent change. Parts add up to the totals before rounding.

of states adjust. In the same industry states, more active fringe firms raise fringe output by 0.10%, which adds 0.086% to total output. The fringe draws sales and labor from the superstars. Their output falls by 0.10%, which lowers total output by 0.015%. The love-of-variety term lowers output by 0.024%, because production shifts toward the fringe, already the largest producer (Appendix Figure 18). Together, output rises by 0.048% in the same industry states. When the distribution of industry states adjusts, fringe output rises by a further 0.09%, which adds 0.077% to total output. Almost all of this rise is higher fringe quality, because fringe quality (6) rises with the number and quality of superstars. Superstars' output and the love-of-variety term barely change. Combining the two steps, output rises by $0.048 + 0.077 = 0.125\%$, the first term in Table 10.

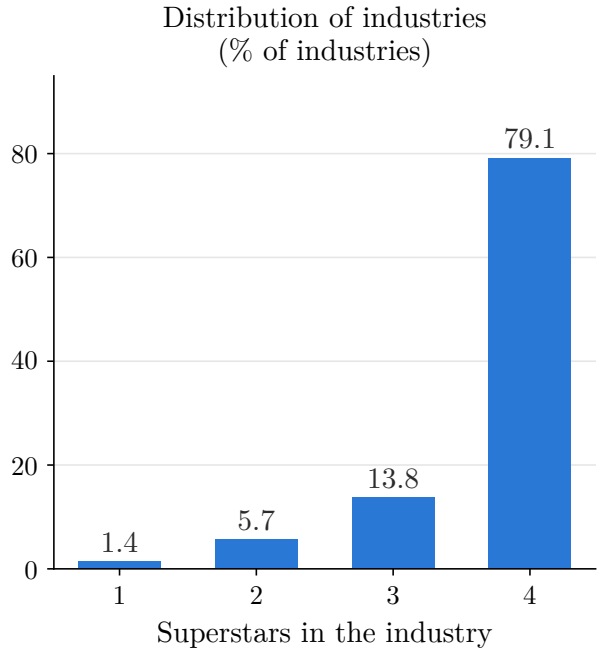


Figure 9. Industries by number of superstars

Notes: Share of industries by number of superstars in the economy with lobbying, percent. At the estimate.

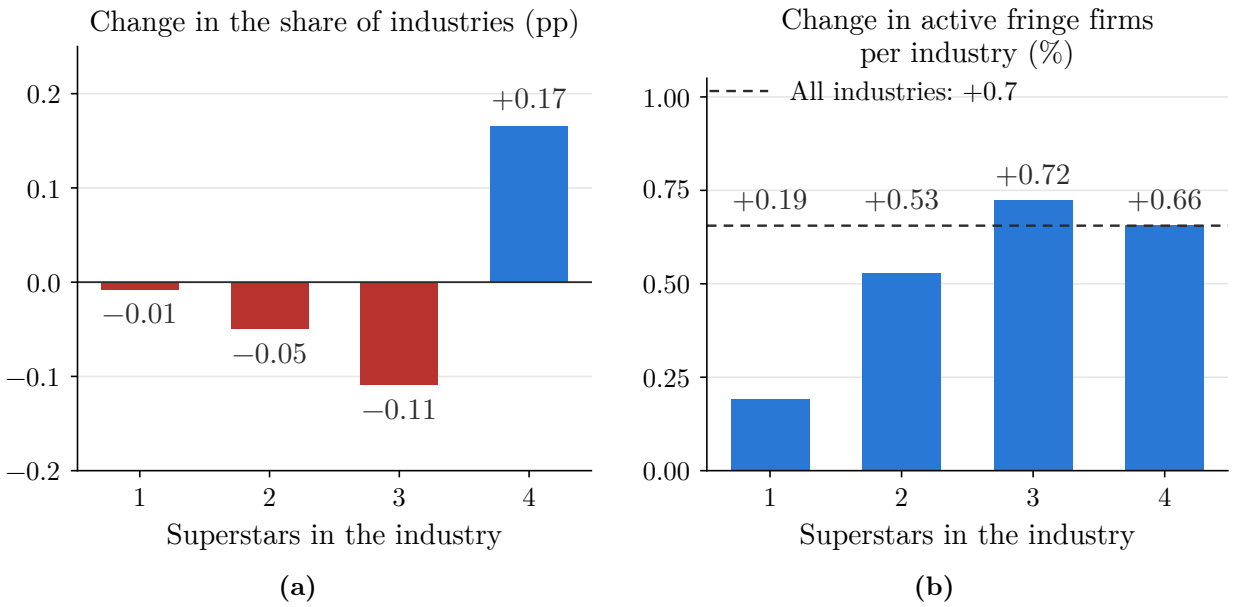


Figure 10. Industry composition without lobbying

Notes: Change from the economy with lobbying to the economy without lobbying, by number of superstars in the industry. Panel (a): change in the share of industries, percentage points. Panel (b): change in the average number of active fringe firms per industry among industries with that number of superstars, percent; the dashed line is the change across all industries. At the estimate.

The share of output consumed rises by 0.069%. All costs in the model are in units of output, so the share of output consumed, C_0/Y_0, is one minus the shares of output used

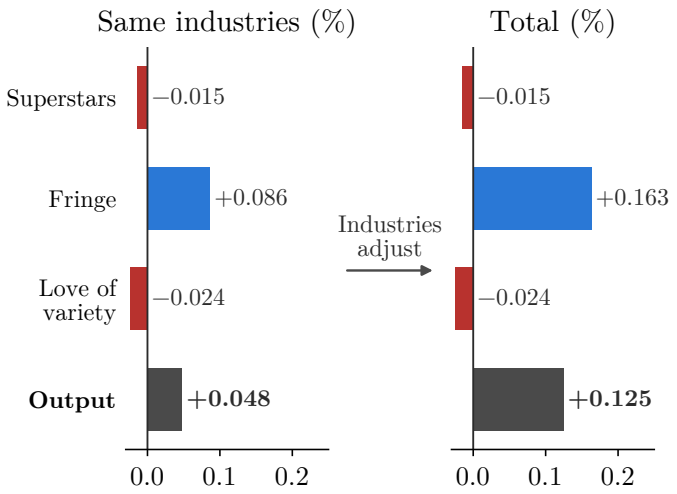


Figure 11. Output without lobbying, by part

Notes: Effect on the log of output of removing lobbying, percent, at the estimate. Same industries: every industry keeps its state, weighted by the distribution of states with lobbying. Total: including the shift in the distribution of states. Superstars and fringe: quantities added up, with log-mean quantity weights. Love of variety: the difference between the industry aggregate and that sum. The parts add up before rounding.

for R&D, entry, fringe operating costs, compliance, and lobbying (equation (44)). Without lobbying, lobbying spending of 0.086% of output disappears, which is worth 0.094% of consumption. The association's organizing cost is 77% of this spending. Compliance costs fall from 0.82 to 0.80% of output, which adds 0.016%. The additional fringe firms raise fringe operating costs from 4.74 to 4.75% of output and entry costs from 1.34 to 1.37%, which subtract 0.015 and 0.028%. R&D spending barely changes. Together, the share of output consumed rises from 91.46 to 91.52%, which adds 0.069% to welfare, the second term in Table 10 (Figure 12).

Growth rises through industry composition. Removing lobbying raises growth by 0.0013 percentage points, from 1.72465 to 1.72595% a year, even though superstars innovate less, because industries shift toward states with four superstars. In a given industry state, the total innovation rate of an industry's superstars is lower without lobbying in industries with two, three, and four superstars, by 0.06% in industries with four (Figure 13), because the value of a step up the quality ladder is lower without lobbying. Each fringe firm also attempts fewer breakthroughs, because becoming a superstar is worth 0.5% less when more fringe firms compete for it. The larger number of fringe firms nevertheless raises the total breakthrough rate. If the distribution of industry states stayed the same, growth would fall by 0.0003 percentage points. Superstars would lower growth by 0.0007 percentage points, and fringe breakthroughs would raise it by 0.0004. The change in industry composition raises growth by 0.0016 percentage points, so growth rises overall. Industries with four superstars

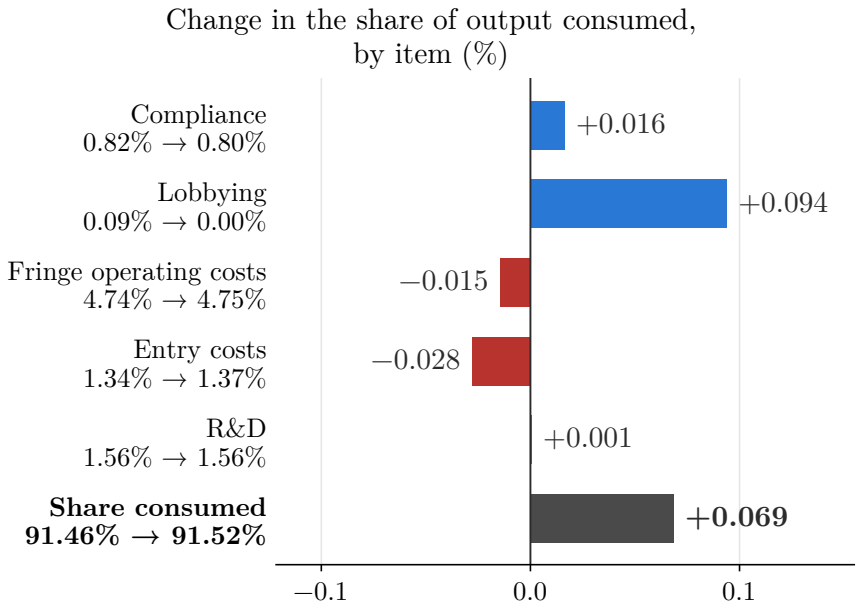


Figure 12. The share of output consumed without lobbying

Notes: Each bar is the item's contribution to the change in the log of the share of output consumed, percent, which is the second term of the welfare decomposition (Table 10). Labels on the left: spending with and without lobbying, percent of output. At the estimate.

contribute the most growth (Figure 13), because their top superstars are more likely to be tied, so more of them advance the frontier when they innovate. Without lobbying, the share of industries with four superstars rises from 79.1 to 79.3% (Figure 10). Small as it is, the rise in growth is worth 0.032% of consumption, because it is capitalized at $r - g = \rho = 4\%$. This is the third term in Table 10.

Hours barely change. Without lobbying, hours rise by 0.001%, at a cost in disutility of labor of 0.001% of consumption. Hours depend only on the after-tax wage relative to consumption (equation (17)). A higher wage makes each hour pay more, while higher consumption makes each extra unit of income worth less. A 1% rise in this ratio raises hours by 0.5%, the Frisch elasticity ϵ_L. Removing lobbying raises output by 0.125%, which raises both the after-tax wage and consumption. Consumption rises by a further 0.07%, because a larger share of output is consumed, and the after-tax wage by a further 0.07%, because the labor share rises and the labor tax falls. Their ratio therefore rises by only 0.003%, so hours barely move. This is the fourth term in Table 10.

Combining the four terms. Adding the four terms, $0.125 + 0.069 + 0.032 - 0.001$, gives the welfare gain of 0.225%. The direct savings appear in the share of output consumed, through lobbying spending (0.094) and compliance costs (0.016). Everything else, 0.115, is the change in industry structure. It consists of higher output and growth, because industries

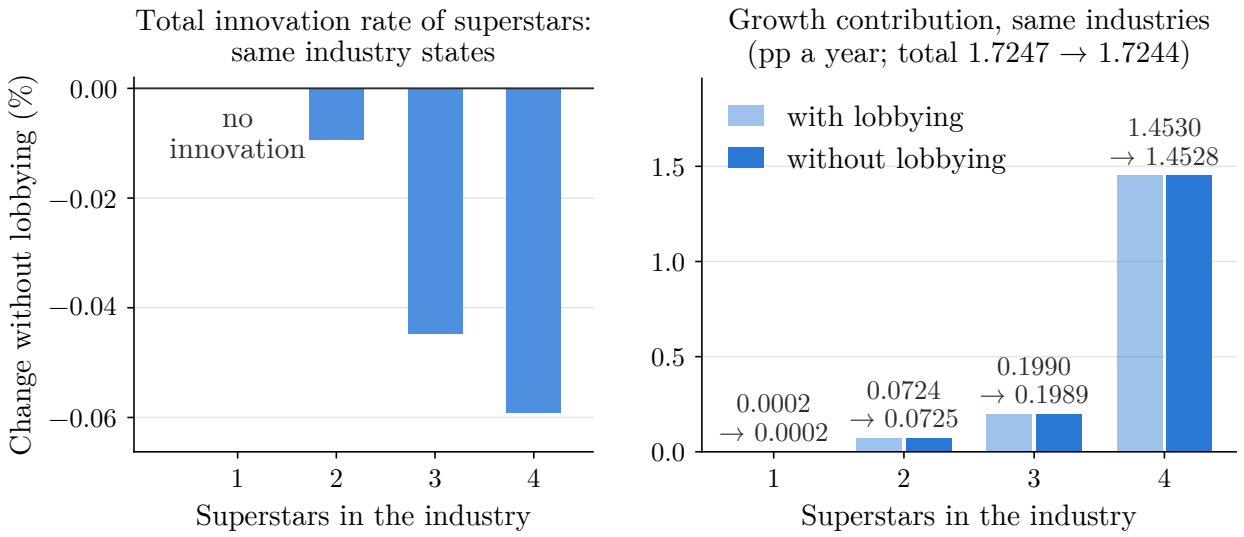


Figure 13. Innovation and growth within industry states

Notes: Left: change in the total innovation rate of an industry's superstars, the sum of their innovation rates, without lobbying, in the same industry state, by number of superstars, percent; industries weighted by their share with lobbying. Right: contribution of the industries with each number of superstars to growth, percentage points per year: their share of industries with lobbying times their growth rate, with lobbying and without lobbying in the same industry states. At the estimate.

have more superstars and the fringe learns from a stronger set of superstars, net of the operating and entry costs of the additional fringe firms and of the extra hours.

Benefits of regulation. Some regulations affected by lobbying may protect consumers. We ask how valuable these benefits would have to be for removing lobbying to leave welfare unchanged. Suppose each unit of output spent on compliance yields benefits to households worth a fixed number of units of consumption, leaving firms' decisions unchanged. Removing lobbying raises welfare unless each unit of compliance spending yields benefits worth more than 15.1 units of consumption. If each unit of compliance spending returned exactly one unit of benefits, compliance would no longer be a cost, yet removing lobbying would still raise welfare by 0.21% of lifetime consumption, about the 0.225% gain less the 0.016 points from compliance costs, because lobbying spending absorbs resources and higher regulatory costs reduce the number of active fringe firms. Direct welfare evidence on the rules in R is scarce. Occupational licensing, the clearest case, raises prices and wages, and its welfare costs exceed its benefits (Kleiner and Soltas, 2023). The large benefit-cost ratios reported for federal regulation come mainly from air-quality rules (Office of Management and Budget, 2013), which are not among the regulatory costs we model.

6.3 Lobbying over regulation accounts for almost all of the welfare gain

Removing only the lobbying over regulation, by setting its participation cost to infinity, delivers almost the entire gain: 0.223% of lifetime consumption, against 0.225% when all lobbying is removed (Table 11). Regulatory costs return to R_0. The remaining tax and procurement lobbying costs only 0.0022% of consumption, almost all of it through tax lobbying. Removing tax and procurement lobbying while lobbying over regulation continues delivers 0.012%, more than the 0.0022% that tax and procurement lobbying costs on its own, because regulatory costs then also fall, from 1.032 to 1.026 times R_0.

Table 11. The welfare cost of each kind of lobbying

Lobbying allowed	$\mathbb{E}[R]/R_0$	Growth	Welfare loss (%)
All three (estimated economy)	1.032	1.7247%	0.2248
Regulation only	1.026	1.7245%	0.2124
Tax and procurement	1.000	1.7262%	0.0022
Tax only	1.000	1.7262%	0.0022
Procurement only	1.000	1.7260%	0.0000
None	**1.000**	**1.7259%**	**0**

Notes: Each row solves the equilibrium with the participation cost of the instruments not allowed set to infinity, at the estimated parameters. $\mathbb{E}[R]/R_0$: regulatory costs relative to their level without lobbying. Welfare loss: consumption-equivalent variation relative to the economy without lobbying, which equals the welfare gain from removing the lobbying allowed in the row.

Tax and procurement lobbying on its own costs little. Tax and procurement lobbying on its own costs only 0.0022% of consumption, because both instruments mainly transfer resources within the household. In the model, a tax cut due to lobbying is a transfer from the household as taxpayer to the household as owner of the firm, financed by a slightly higher labor tax. A contract obtained by lobbying is an additional government purchase. The household consumes the goods the contractor delivers, and the government pays their cost plus the contractor's fee. The fee is a transfer financed by the same labor tax. Besides the effect on innovation, the efficiency costs are lobbying resources (0.010% of output) and the distortion of labor supply from a labor tax that is 0.036 percentage points higher. Removing this lobbying raises welfare by 0.0006% through output and by 0.0146% through the share of output consumed, mostly by saving lobbying spending. However, growth falls, at a cost of 0.0059%, because tax cuts and contracts won by lobbying make innovating slightly more valuable. Hours rise as the labor tax falls, at a cost of 0.0071% in disutility of labor. The total gain is 0.0022%.

6.4 Policy implications

The counterfactuals have two policy implications. First, lobbying reforms should target lobbying over regulation rather than lobbying in general, because banning it delivers 99% of the gain from removing all lobbying. Tax and procurement lobbying on its own costs almost nothing. Second, the welfare effect depends on whether lobbying raises or lowers regulatory costs. The direction is determined by the balance between the superstars and the association. Measures that lower small firms' cost of organizing an association might therefore lower regulatory costs without restricting political activity.

7 Conclusion

This paper studies what firms lobby for and how lobbying affects industry concentration, innovation, growth, and welfare. Using an LLM classification of federal lobbying reports, we construct a new dataset of the specific policy instruments firms lobby for, from a taxonomy of 87 instruments. Almost all firms and associations ask for more appropriations and lower taxes, but they disagree on regulatory costs. Among requests that take a side, 63% of those of the largest 5% of firms in each four-digit industry ask for higher costs, against 36% for associations.

To quantify the consequences of lobbying, we build a dynamic general equilibrium model of oligopolistic competition and innovation in which firms lobby for lower taxes and procurement contracts and over industry-wide regulatory costs. In the estimated model, superstars lobby for higher regulatory costs, while the trade association lobbies against them. The leader favors higher regulatory costs both for higher profits as marginal fringe firms stop producing and for protection against breakthroughs. The smallest superstars favor them mainly for that protection. On net, lobbying raises regulatory costs by 3%, but it raises them in 41% of industry states and lowers them in 57%. Removing lobbying raises the number of active fringe firms by 0.7% and the share of industries with four superstars. Superstars innovate less, but growth rises through the change in industry composition. Overall, removing lobbying raises consumption-equivalent welfare by 0.22%. Saved lobbying spending accounts for 0.09 percentage points of this gain, lower compliance costs for 0.02, and the change in industry structure for the remaining 0.11.

Lobbying over regulatory costs accounts for 99% of the welfare gain from removing lobbying, while tax and procurement lobbying, the focus of the literature, mostly shifts money from the household as a taxpayer to the household as a firm owner. Thus, not all lobbying is the same: a ban on lobbying over regulation alone delivers almost all of the gain from ban-

ning all lobbying. Removing lobbying raises welfare unless each unit of compliance spending yields benefits worth more than 15.1 units of consumption.

Because our classification records how the compliance cost of each instrument scales with firm size, it can be used to study whether the largest firms take different sides on rules that cost all firms about the same and on rules whose costs grow with firm size.

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Appendix

A Example of the lobbying report

Clerk of the House of Representatives	Secretary of the Senate
Legislative Resource Center	Office of Public Records
135 Cannon Building	232 Hart Building
Washington, DC 20515	Washington, DC 20510
http://lobbyingdisclosure.house.gov	http://www.senate.gov/lobby

LOBBYING REPORT

Lobbying Disclosure Act of 1995 (Section 5) - **All Filers Are Required to Complete This Page**

1. Registrant Name ☑ Organization/Lobbying Firm ☐ Self Employed Individual
APPLE INC.

2. Address
Address1 700 K Street NW Address2
City WASHINGTON State DC Zip Code 20001 Country USA

3. Principal place of business (if different than line 2)
City Cupertino State CA Zip Code 95014 Country USA

4a. Contact Name: Mr. Timothy Powderly | b. Telephone Number: 2027729500 | c. E-mail: gholger@apple.com | 5. Senate ID#: 4152-12

7. Client Name ☑ *Self* ☐ *Check if client is a state or local government or instrumentality*
APPLE INC. | 6. House ID#: 314500000

TYPE OF REPORT 8. Year 2026 Q1 (1/1 - 3/31) ☑ Q2 (4/1 - 6/30) ☐ Q3 (7/1 - 9/30) ☐ Q4 (10/1 - 12/31) ☐
9. Check if this filing amends a previously filed version of this report ☐
10. Check if this is a Termination Report ☐ Termination Date _____ 11. No Lobbying Issue Activity ☐

INCOME OR EXPENSES - YOU MUST complete either Line 12 or Line 13

12. Lobbying

INCOME relating to lobbying activities for this reporting period was:
Less than $5,000 ☐
$5,000 or more ☐ $ _____

Provide a good faith estimate, rounded to the nearest $10,000, of all lobbying related income for the client (including all payments to the registrant by any other entity for lobbying activities on behalf of the client).

13. Organizations

EXPENSE relating to lobbying activities for this reporting period were:
Less than $5,000 ☐
$5,000 or more ☑ $ 2,480,000.00

14. REPORTING Check box to indicate expense accounting method. See instructions for description of options.

☑ **Method A.** Reporting amounts using LDA definitions only

☐ **Method B.** Reporting amounts under section 6033(b)(8) of the Internal Revenue Code

☐ **Method C.** Reporting amounts under section 162(e) of the Internal Revenue Code

Signature Digitally Signed By: Timothy Powderly, Senior Director, Government Affairs **Date** 4/20/2026 10:26:49 PM

LOBBYING ACTIVITY. Select as many codes as necessary to reflect the general issue areas in which the registrant engaged in lobbying on behalf of the client during the reporting period. Using a separate page for each code, provide information as requested. Add additional page(s) as needed.

15. General issue area code CPT

16. Specific lobbying issues

General patent policy, including issues related to the Patent Trial and Appeal Board, S. 1553/H.R. 3160 the PREVAIL Act, and S. 708/H.R. 1574 the RESTORE Patent Rights Act
Issues related to standard essential patents
Issues related to anti-counterfeit policy
General trademark and copyright policy
Issues related to patent proceedings and the International Trade Commission, including the Advancing Americas Interests Act
General overview of music licensing legal framework

Figure 14. Sample lobbying report

B Issue-text splitting: examples

Each example shows a raw LD-2 issue-text block and the subissues produced by the splitting stage. Deterministic pattern rules handle unambiguous structure (bill-led segments, enumerated lists, blank-line groups); blocks whose boundaries require semantic judgment are routed to the language-model stage. Original filers' typos are preserved.

Example 1: bill-led block, one subissue per bill.

> "S. 1816 and H.R. 3852: Chesapeake Reauthorization, H.R. 5088: America's Commitment to Clean Water Act, H.R. 5509: Chesapeake Bay Program Reauthorization and Improvement Act"

Split into three subissues: (1) S. 1816 / H.R. 3852, Chesapeake Reauthorization (a cross-chamber companion pair kept as one subissue); (2) H.R. 5088, America's Commitment to Clean Water Act; (3) H.R. 5509, Chesapeake Bay Program Reauthorization and Improvement Act.

Example 2: semicolon list mixing instruments.

> "Issues and incentives related to biodiesel, and the Renewable Fuel Standard; S. 1277/H.R. 2238, Biodiesel Tax Incentive Reform and Extension Act; S. 1294, Oil Independence for a Stronger America Act, Title V – Alternative Transportation Fuels; S. 1220, Fulfilling U.S. Energy Leadership Act, Title IIII – Alternative Fuel Deployment. Implementation of Budget Control Act. Extension of biodiesel tax credit. S. 3240, Agriculture Reform, Food and Jobs Act."

Split into seven subissues. The block mixes a regulatory quantity mandate (the Renewable Fuel Standard), a tax-credit companion pair (S. 1277 / H.R. 2238), title-restricted positions on two energy bills, statute implementation, a free-standing tax-extender request, and a farm bill. Keeping it whole would force a single label onto instruments from different wedges.

Example 3: language-model split required (no reliable delimiters).

> "Issues pertaining to intellectual property rights including, but not limited to, patent protection and enforcement, copyright protection and enforcement in international markets, and antitrust, S. 3197 – Platform Competition and Opportunity Act of 2021, S. 2992 – American Innovation and Choice Online Act"

Commas here separate both topics and bills, so pattern rules cannot place boundaries. The language model produces four subissues: (1) patent protection and enforcement; (2) copyright protection and enforcement in international markets; (3) antitrust, S. 3197; (4) antitrust, S. 2992. The intellectual-property topics and the antitrust bills belong to different wedges.

C Policy-instrument taxonomy and model crosswalk

The policy objects of the model, the industry-wide regulatory fixed cost $R(S)$, the firm's own profit tax τ_i, and its own procurement contracts Γ_i, must be measured from text that refers to none of them. A lobbying report names a statute, a rule, a bill, or a program; it does not say whether the policy at issue raises a fixed cost of operating or a cost that scales with production. Asking a language model to supply that judgment directly would place the central economic distinction of the paper inside a single step that can be neither inspected nor revised. We therefore separate the reading of the text from its economic interpretation. A closed codebook of policy instruments, each defined by the statutory or regulatory language that identifies it rather than by what it does to costs, is written in advance, and a crosswalk from instruments to the terms of the firm's problem is fixed before any text is classified. The language model assigns text to instruments; the assignment of instruments to economic channels is ours, and takes the form of a fixed table.

C.1 Design

Classification proceeds in three stages. In the first, a language model reads a single text segment and returns, for each distinct request the segment contains, the identifier of a codebook instrument, the conditions defining whom the policy covers, the direction and, where the text states one, the magnitude of the change requested relative to current law, and the position the filer takes. Each of these is a statement about what the text says. In the second stage the identifier is matched to the codebook, which supplies the economics: the wedge the instrument moves, the government account on which it falls, the payoff it typically delivers to a firm that qualifies for it, the base with which that payoff scales, and, for instruments delivered through the tax code, the Treasury provision numbers. The match is mechanical. The third stage derives the quantities the analysis requires: the breadth of the request, whether it favors the firms it covers, the side of the contest on which the filer stands, and the firm's portfolio across wedges.

Two consequences of this division matter. Because the economic content of the classification is held in a table written in advance, it can be examined, disputed, and replaced without reclassifying a single filing, and every result can be recomputed under an alternative crosswalk. Because the classifier's errors and the crosswalk's errors are of distinct kinds, they can also be measured separately. The unit of classification is the provision rather than the bill or the filing: an omnibus statute moves many wedges at once, so a bill enters the panel as a vector of classified provisions, and a subissue containing several distinct requests yields one observation for each. A segment consisting only of a bill reference is recorded as

such together with the resolved bill number, and its instruments arrive through the bill's provisions; a segment that names no policy object, or names a policy area too vaguely to place, is recorded as uninformative and excluded. Neither is forced onto an instrument.

C.2 The codebook and its sources

The codebook is intended to be exhaustive over the policy space on which firms lobby, and its completeness is borrowed rather than asserted: every instrument traces to an official or established inventory of its domain. We draw on the Treasury tax-expenditure inventory, whose provision numbers the codebook retains; the regulatory-forms taxonomy of Xie and Pérez (2018); the UNCTAD International Classification of Non-Tariff Measures; the Harmonized Tariff Schedule of the United States and Title 19 of the U.S. Code; the federal Assistance Listings; the Federal Acquisition Regulation; the patent statute and the exclusivity provisions administered by the Food and Drug Administration; the Immigration and Nationality Act; and, as a check on the inventory as a whole, the tools-of-government catalog of Salamon (2002). Coverage failures are not random: they concentrate in the domains a researcher knows least well and bias the measured portfolio toward whatever the codebook covers. Deriving the list from external inventories makes omissions visible to any reader who consults the same sources.

Instruments are organized by what they do to the firm's problem, not by whom they reach. A research credit written for a single firm and one available to the whole economy are the same instrument, distinguished only by the eligibility conditions the classifier reads from the text. Breadth is thus a property of the observation rather than of the codebook, which keeps the observed breadth of lobbying falsifiable.

C.3 Wedges and the crosswalk to the model

A wedge names the term of the firm's problem an instrument moves. Table 12 lists the ten wedges, the margin each moves, and the object of the model to which each corresponds. Three objects of the model are measured from the classification: regulatory fixed costs are the industry-wide burden $R(S)$ of the contest (13); the profit-tax wedge is the object of tax lobbying (11); and procurement contracts, an instrument of the demand wedge, are the object of procurement lobbying (12). The other wedges are recorded rather than discarded, which allows the classification to be reused for extensions and for counterfactuals defined on individual instruments.

Instruments whose incidence is conditional on concentration, among them merger control, conduct rules for dominant platforms, and regulation triggered by size thresholds, are kept

Table 12. Wedges and the crosswalk to the model

Wedge	Margin moved	Model object
R_j	Industry-wide quasi-fixed cost of operating, entering, or introducing a product	$R(S)$: **regulatory costs lobbying**
b_T	Profit-tax level; scales after-tax profit with no first-order effect on marginal prices	τ_i: **tax lobbying**
s^d	Demand facing the firm: government purchases, buyer subsidies, trade protection	Γ_i (procurement contracts): **procurement lobbying**; other instruments recorded
s^y	Per-unit wedge on marginal cost or marginal revenue	recorded; extension
s^r	Price of R&D and the private return to innovation	R&D cost χz^{ϕ}; extension
s^k	User cost of capital, including financing	recorded; no capital in the model
s^w	Effective wage bill	labor cost; extension
s^p	Price the firm may charge	markup; extension
$A(\Theta)$	Concentration-conditional rules: act on the industry state or switch on above a size threshold	transitions of Θ; extension
s^i	Lump-sum transfer to, or fee from, the firm	profit level; extension

separate from R_j, since their burden depends on a firm's position in the industry rather than falling on every active firm alike; they are the natural object of an extension in which lobbying acts on the transitions of the industry state directly. A requirement enters R_j only if its compliance cost is quasi-fixed; any component of the same statute whose cost scales with output is assigned to the output wedge.

C.4 What the classifier records

Table 13 lists what the classifier records, one observation for each request it identifies.

Eligibility records only the conditions the text adds beyond the instrument's own definition, and because it is recorded in standard classification systems it also serves as the key that joins an observation to the firm panel. Conditions that bind on quantities the firm chooses are recorded against a closed list of variables that map into the model: output, revenue, price, employment, wages, capital, investment, research spending, profit, market share and concentration, merger events, and firm age. The direction of the requested change is one of a small set of operations, distinguishing the creation of an instrument from its expansion, reduction, extension, or repeal; magnitudes are recorded when the text states them and never imputed. Positions are recorded as the filer states them, and a filing that reports monitoring an issue is kept distinct from one that states no position.

Table 13. What the classifier records for each request

Field	Content
Instrument	The codebook instrument the text concerns, or a label marking the segment as a bare bill reference or as uninformative
Legislation	Bill and public-law numbers named in the text, resolved to the Congress in which they were introduced
Granularity	Whether the text is finer than the nearest instrument and, if so, a verbatim statement of the finer target
Eligibility	Whom the policy covers, over and above what the instrument itself implies: industry, geography, product, occupation, legal form, and certified status, recorded in standard classification systems
Conditions	Thresholds the policy places on quantities the firm chooses, expressed as comparisons against a closed list of model variables
Effect	The direction of the change requested relative to current law and, when the text states them, its magnitude and starting level
Position	The position the filer takes: support, opposition, either with an amendment, the pursuit of a specific change, monitoring, or none stated
Confidence	The classifier's confidence in the instrument assignment; low-confidence observations are dropped

C.5 From classified text to model objects

Besides the wedge, each instrument carries attributes that attach to every observation assigned to it: the government account (expenditure, receipt, or a mandate whose cost is borne privately), the sign of the payoff to a qualifying firm, and its shape, which records whether the payoff scales with a flow or is a lump independent of scale. The shape enforces the rule that only quasi-fixed burdens are regulatory costs.

The scope of a request is determined by the width of its eligibility conditions: a named entity yields a firm-specific request, an industry code with qualifiers a narrow one, an industry branch an industry-wide one, and unrestricted eligibility an economy-wide one. Whether a request favors the firms it covers follows from its direction and the instrument's sign. Let $\text{dir} \in \{-1, 0, +1\}$ denote the requested change in the level of instrument I, with creation, increase, extension, reauthorization, and funding taking the value $+1$, reduction and repeal -1, and restatements and studies 0, and let $\text{sgn}_I \in \{-1, +1\}$ be the instrument's sign. Favorability is

$$\text{fav} \;=\; \text{dir} \cdot \text{sgn}_I. \tag{49}$$

The same quantities place an observation on one side of the contest. For an observation on a regulatory costs instrument, let $\text{st} = +1$ when the filer supports the request and $\text{st} = -1$ when it opposes it. The filer would move the regulatory fixed cost in the direction $\text{st} \cdot \text{dir}$:

supporting a bill that raises a compliance requirement and opposing the repeal of one are the same political act, and the classification records them on the same side without a further judgment. The share of a firm-year's regulatory costs requests with $\text{st} \cdot \text{dir} = +1$ determines its side, and the share of positions that ask for higher regulatory costs is the moment that disciplines the elasticity of superstars' compliance costs to their size in Appendix K.3.

For a few instruments the sign is not a property of the instrument: a tariff protects the firms that produce the good and raises the costs of the firms that use it. Observations on such instruments are resolved filer by filer, first from the stated position, because supporting a duty reveals the protected side, and then from a match between the affected good and the firm's own products or inputs. Classified observations collapse to a panel of firm, quarter, instrument, wedge, and side, from which every moment of Sections 3 and 5 is computed.

C.6 The instruments

The list below groups the 87 instruments of the codebook by wedge.

Regulatory fixed costs *(wedge R_j, 18 instruments)*

- Licenses and charters required to operate; product certification and inspection; pre-market product approval; facility permitting and certificates of need.
- Recurring reporting, recordkeeping, and registration duties; labeling and warning mandates; securities disclosure and shareholder-rights rules; data privacy and cybersecurity rules; one-time outlays required by a standard; financial assurance (surety) bonds.
- Health, safety, and technical requirements on imports; services trade and data-flow restrictions; foreign investment screening; issuance of limited rights to public resources.
- Liability immunities and shields, limits on damages, expansion of liability exposure; conduct rules on how a transaction is made.

Profit tax *(wedge b_T, 7 instruments)*

- The general corporate income tax; deductions from taxable profit; reduced rates on classes of income; reduced tax on foreign and export income; exemption by legal form; deferral and basis rules; deductibility of losses and forgiven debt.

Demand *(wedge s^d, 14 instruments)*

- Government purchases: program funding, appropriations, and authorizations; procurement contracts; public infrastructure and services used by firms; tax exemption for state and local government bonds.
- Buyer purchase subsidies and purchase mandates; domestic-content requirements for public purchases; markets reserved by law to designated sellers; advertising and marketing restrictions; government provision of competing services.

– Trade: import tariffs and tariff-rate quotas; limits on import quantities; antidumping and countervailing duties; duty-free trade under preference programs; export controls and sanctions.

Output *(wedge s^y, 17 instruments)*

– Per-unit production credits and support payments; percentage depletion; excise and sales taxes; taxes on emissions and discharges; regulatory levies on revenue; public-resource royalties.
– Technology and performance standards with per-unit costs; bank capital and liquidity requirements; limits on domestic production; tradable permits; portfolio and blending standards; prohibitions of products; obligations to serve; insurance coverage mandates.
– Duty suspensions and drawback; customs procedures and rules of origin; federal insurance backstops.

Innovation cost *(wedge s^r, 5 instruments)*

– R&D tax credits and expensing; R&D grants and technology transfer; patent, copyright, trademark, and trade-secret law; market exclusivity for approved products; enforcement of intellectual property against foreign infringement.

Capital cost *(wedge s^k, 6 instruments)*

– Investment tax credits and expensing; accelerated and bonus depreciation; tax-exempt private-activity bonds; federal lending and loan guarantees; creditor rights in bankruptcy; deductibility of business interest.

Labor cost *(wedge s^w, 8 instruments)*

– Hiring and wage credits; untaxed employee benefits; tax deferral on retirement contributions; payroll taxes; wage floors, hour rules, and benefit mandates; workplace safety and worker protections; employee status and collective bargaining; restrictions on hiring foreign workers.

Set prices *(wedge s^p, 5 instruments)*

– Government-set payment rates; mandatory discounts, rebates, and negotiated prices; price ceilings; price floors; allowed rates of return and revenue caps.

Competition *(wedge $A(\Theta)$, 4 instruments)*

– Merger control and conduct rules; platform gatekeeper regulation; antitrust exemptions; mandated access and interoperability.

Lump-sum transfers and fees *(wedge s^i, 2 instruments)*

– Grants and direct payments to firms; regulatory user fees.

Place-based bundles *(wedge 1 instrument)*

– Zone regimes that combine an employment credit, a financing subsidy, and a capital-gains preference; decomposed into their components, with the geographic restriction recorded as a condition of eligibility.

C.7 Validation

Since the crosswalk is deterministic, validation concerns the classifier's output alone. Section 2.2 reports the checks: agreement with an independent classification by a second language model, agreement with the positions recorded by MapLight, and the stability of repeated classification.

D Participation: all observations and lobbying periods

The participation figures of Section 3 use two denominators. Figures 2 and 6 divide by all observations: all firm-years and, for associations, all years of 501(c)(6) organizations filing an IRS Form 990 or 990-EZ with positive revenue; they measure how often a client lobbies, the object of the model's participation targets. Figure 3 divides by the quarters with a lobbying report; it measures what lobbying clients lobby on. Figure 15 reports the shares from Figure 6 for the firm-years with a lobbying report, and for associations one bar over the 16,397 association-years with a report.

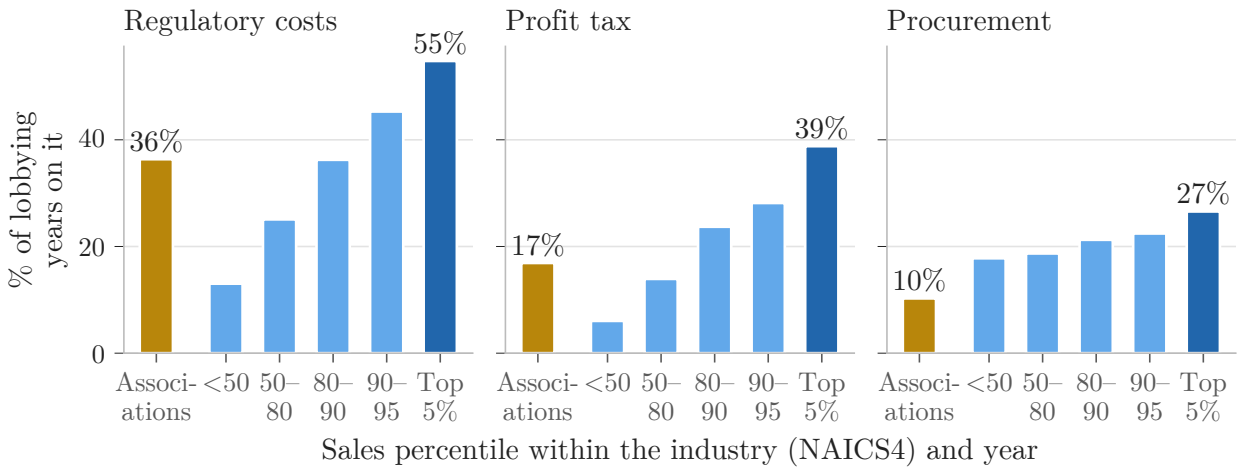


Figure 15. Participation in lobbying by firm size, among lobbying years

Notes: Share of the years with a lobbying report in which the client lobbies on the instrument: the 14,439 firm-years in which the firm lobbies and the 16,397 association-years with a report, by the firm's sales percentile within its industry (four-digit NAICS) and year. A year with a lobbying report has a report with a positive amount that names a classified instrument. 2008–2022.

E Robustness: asks that touch several instruments

The figures of Section 3 count an ask on an instrument only when the ask is about that instrument alone. An ask is one request in a report: a request in the filer's own words, a named section or title of a bill, or a bill or regulation that the filer cites. Each ask has total weight one. A request in the filer's own words counts for the instrument it names when the classification has high or medium confidence. A citation counts only when the provisions it refers to all concern a single instrument. The provisions a citation refers to are those its words point to (a named section or title, or the provisions that the readings of its words pick among the bill's provisions) and all the provisions of the bill otherwise. The bill's name plays no role. For example, a 2020 report that lists "H.R. 2, the Moving Forward Act, and the need for corrosion planning in federal bridge projects" refers to two of the bill's 41 provisions, both on public infrastructure, and counts on that instrument; a report of the same quarter that cites the bill by its number and title alone refers to all 41 provisions, on 18 instruments, and counts on none of them.

This appendix counts every ask instead. A citation divides its weight among the instruments of the provisions it refers to, in proportion to the number of provisions on each, as if each provision were a separate request. A 2019 citation of the America's Water Infrastructure Act (S. 3021, 115th Congress) refers to two provisions on procurement and thirteen on public infrastructure and gives 2/15 to procurement and 13/15 to public infrastructure. A 2020 report that asks to "Support H.R.748/S.3548 the CARES Act" refers to all 180 provisions of the Coronavirus Aid, Relief, and Economic Security Act; program funding and appropriations, the instrument of 44 of them, receives 0.24. A provision that the bill's summary states several times counts as many times, a request that names several sections of a bill divides its weight equally among them, and provisions classified at low confidence keep their share but do not count. A client lobbies on an instrument in a quarter with probability $1 - \prod(1 - w)$, where the product runs over its requests in the quarter and w is a request's weight on the instrument (an ask counted once within a report, and the weights of one request on a wedge or front summed up to one). The sides count every citation that states support or opposition and every request in the filer's own words; each provision's share goes to the side implied by the ask's stance and the provision's direction. Requests in the filer's own words keep weight one. The denominators do not change. The weights depend on how finely a bill's summary divides it into provisions. Giving each instrument of a citation the same weight, one over the number of its instruments, instead changes the participation shares of this appendix by at most 3.4 percentage points and the shares asking for higher regulatory costs by firm size by at most 1.3 points.

Counting every ask raises the shares and leaves the ranking of the fronts and of the leading wedges in place. Lobbying firms lobby on regulatory costs in 40% of their quarters with a report instead of 33%, on the profit tax in 28% instead of 20%, and on procurement in 24% instead of 19%; associations in 39, 19 and 14% instead of 33, 15 and 8%. Demand and regulatory costs remain the two most lobbied wedges for every group, and program funding and appropriations the most lobbied instrument (Figure 16). The mean share of asks for higher regulatory costs among filers taking a side is 34% below the 80th percentile of their industry's sales, 43% between the 80th and the 95th, and 62% among the largest 5%, against 27, 43 and 63% in the baseline; for associations it is 40% against 36%. Sides on single instruments rest on fewer filer-years and move more (Figure 17): the largest change is on licenses and charters among the largest 5% of firms, where the share is 63% against 45% in the baseline (23 expected filer-years with a side against 11).

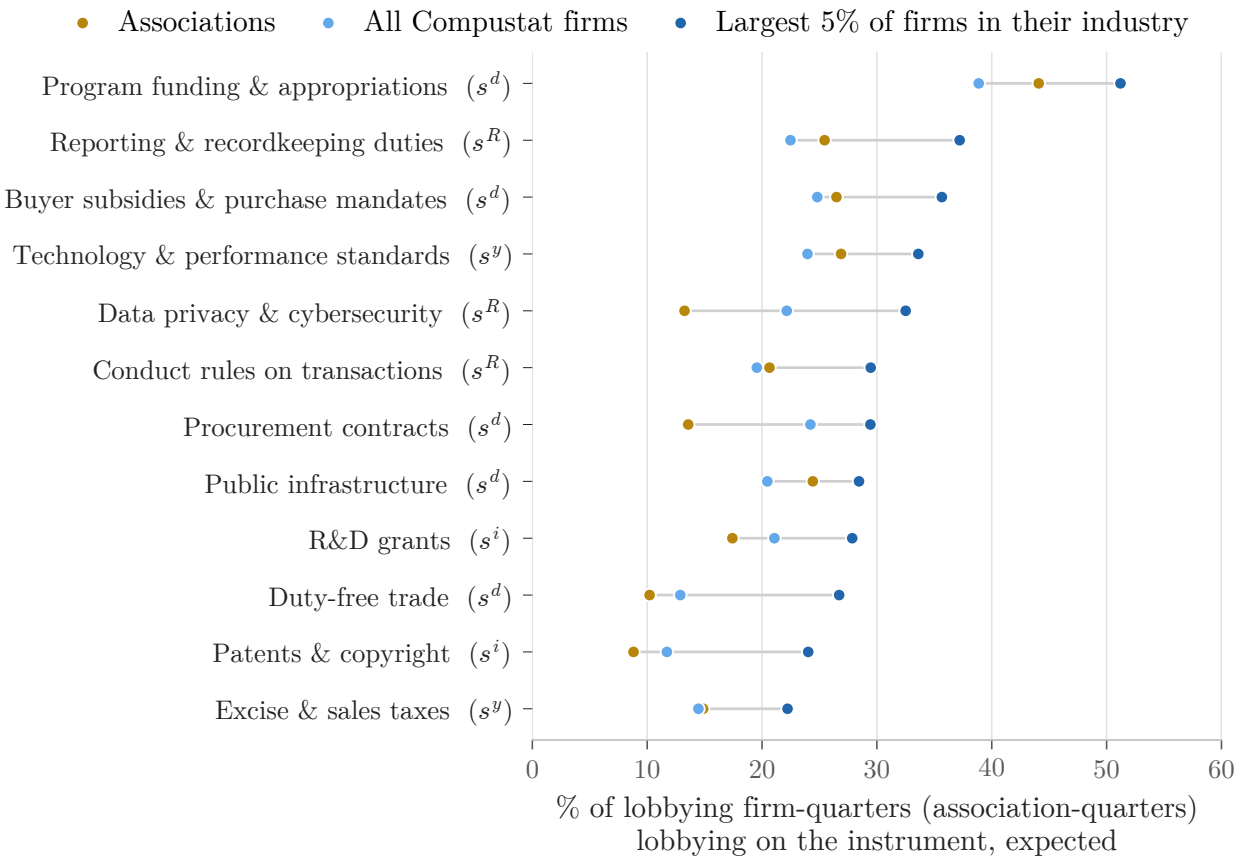


Figure 16. Lobbying by instrument, every ask weighted by provision share

Notes: Share of lobbying firm-quarters (association-quarters) in which the client lobbies on the instrument, counting every ask: a citation gives each instrument the share of its provisions on the instrument, and the share is the expected share of quarters with lobbying on the instrument. 52,079 Compustat firm-quarters, of which 14,970 of the largest 5% of firms in their industry by sales; 60,596 association-quarters; 2008–2022. The instruments are the 14 most lobbied by firms and the 10 most lobbied by associations, without government-set payment rates and investment tax credits.

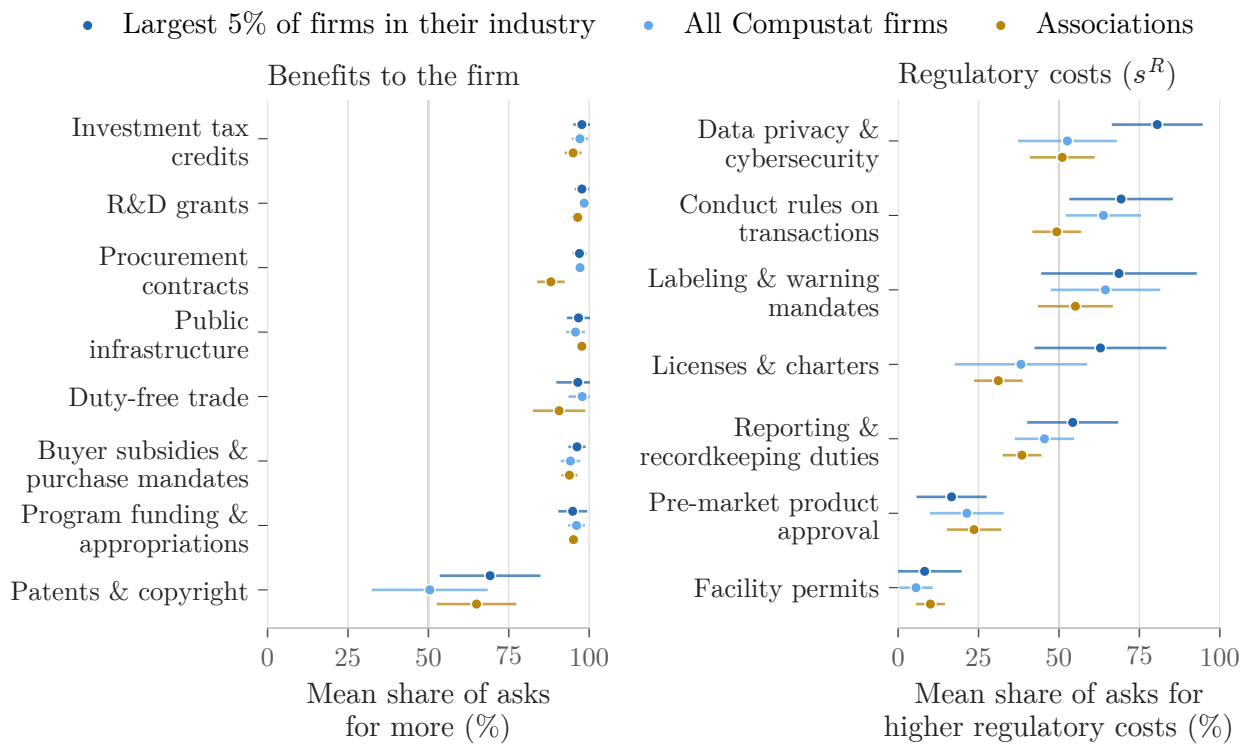


Figure 17. The side firms take, every signed ask weighted by provision share

Notes: As Figure 4, counting every ask that states support or opposition, each with its provision shares.

F Stated sides: precision and selection

Precision of the size gradient. The size gradient in the share of requests for higher regulatory costs rests on few observations: only 278 firm-years take a side on regulatory costs. Counting each firm-year once, the largest 5% of firms exceed firms below the 80th percentile of their industry's sales by 36 percentage points across all firms (standard error 10) and by 24 points within an industry and year (standard error 13). Counting every request, the gaps are 50 and 26 points (standard errors 12 and 12). Only 11 of 53 industries have firms stating a side both above and below the industry's 80th percentile of sales. Counting each firm once rather than each firm-year, the gap across all firms falls to 18 points (standard error 11) and the gap within industry-years to 10 points (standard error 16).

Selection of stated sides. Firms state a side on 1.9% of their requests on regulatory costs and 4.2% of their requests on other instruments. Only 5% of the firm-years that lobby on regulatory costs state a side. Three checks show that the stated sides are representative, with one exception. First, large firms state a side in more years only because they file more requests. The largest fifth of firms within their industry and year file a median of 101 requests a year, against 8 for the smallest fifth. Across all instruments, they state a side 17.8 percentage points more often (standard error 2.2), but no more often at the same number of requests (-1.2, standard error 2.3). On regulatory costs, they are no more likely to state a side, either unconditionally (0.0, standard error 1.5) or at the same number of requests (-4, standard error 2.0). Second, on most instruments, whether a side is written down depends mostly on the lobbying firm that writes the report rather than on the client. A client's reports state a side in 11.0% of cases when its lobbying firm states sides for its other clients and in 2.6% otherwise, and clients that state sides often do not choose lobbying firms that do (correlation 0.02). On regulatory costs, the client decides instead. For two clients of the same lobbying firm in the same year, a side is stated in 5.2% of the regulatory-cost reports of the client that states sides more often with its other lobbying firms, against 0.2% for the other client (882 lobbying-firm-years). For two lobbying firms of the same client in the same year, the shares are 1.8 and 1.3%. Third, the 27 firms that state a side on regulatory costs in more than a third of their other years ask for higher regulatory costs less often. Within an industry and year, their share of requests for higher regulatory costs is 37 percentage points below that of firms that state a side only once (standard error 14). Because the firm-year share gives these frequent statements more weight, it slightly understates support for higher regulatory costs: counting each firm once raises the mean share from 47 to 50%. We keep the firm-year share because the estimation targets and the model count the same unit: in the model, each superstar is weighted by the time it spends in each state, which corresponds

to counting each firm-year once.

G Static Cournot equilibrium under decreasing returns to scale

This appendix derives the static product-market objects used in Section 4: the superstar best responses, markups, and profits; the fringe output and profits; and the relative-output system solved at each industry state. The derivation follows Cavenaile et al. (2026), Appendix C.1, adapted to our notation. We fix an industry and a point in time and drop the j and t subscripts throughout. The industry contains N superstars with productivities $\{q_i\}_{i=1}^N$, an active fringe of mass m^{a} with common productivity q_f, the wage is w, and aggregate output is Y. Government contracts do not enter: government purchases are purchases of the final good, and a firm's contracts depend on its gap, not on its output choice.

Demand and notation. The final-good producer's problem yields the inverse demands

$$p_i \;=\; \frac{y_i^{-1/\eta}\,Y}{D}, \qquad p_f \;=\; \frac{\tilde{y}_f^{-1/\eta}\,Y}{D}, \qquad D \;\equiv\; \sum_{i'=1}^{N} y_{i'}^{\frac{\eta-1}{\eta}} + \tilde{y}_f^{\frac{\eta-1}{\eta}}. \tag{50}$$

It is convenient to define, for each superstar i and for the fringe aggregate,

$$\Sigma_i \;\equiv\; \sum_{i'=1}^{N}\left(\frac{y_{i'}}{y_i}\right)^{\frac{\eta-1}{\eta}} + \left(\frac{\tilde{y}_f}{y_i}\right)^{\frac{\eta-1}{\eta}} \;=\; \frac{D}{y_i^{\frac{\eta-1}{\eta}}}, \qquad \Sigma_{-i} \;\equiv\; \Sigma_i - 1, \qquad \Sigma_f \;\equiv\; \frac{D}{\tilde{y}_f^{\frac{\eta-1}{\eta}}}, \tag{51}$$

where Σ_i includes firm i's own term (equal to one). Revenues then satisfy $p_i y_i = Y/\Sigma_i$ and $p_f \tilde{y}_f = Y/\Sigma_f$: each producer's share of industry expenditure is the reciprocal of its own-normalized CES aggregator, $\omega_i = 1/\Sigma_i$ and $\omega_f = 1/\Sigma_f$, with $\sum_i \omega_i + \omega_f = 1$.

Superstar best response. Superstar i chooses y_i taking rivals' quantities $\{y_{i'}\}_{i'\neq i}$ and the fringe aggregate $\tilde{y}_f$ as given, internalizing the effect of y_i on D. Inverting the production function (4), $l_i = (y_i/q_i)^{1/\alpha}$, so operating profit is

$$\pi_i \;=\; \frac{y_i^{\frac{\eta-1}{\eta}}\,Y}{D} \;-\; w\left(\frac{y_i}{q_i}\right)^{1/\alpha}. \tag{52}$$

The first-order condition equates marginal revenue with marginal cost:

$$\frac{\eta-1}{\eta}\; y_i^{-1/\eta}\, Y\; \frac{D - y_i^{\frac{\eta-1}{\eta}}}{D^2} \;=\; \frac{w}{\alpha}\; y_i^{\frac{1-\alpha}{\alpha}}\; q_i^{-1/\alpha}. \tag{53}$$

The factor $(D - y_i^{(\eta-1)/\eta})/D^2$ on the left is the Cournot internalization: expanding output depresses the industry price index in proportion to the firm's own weight in D. Multiplying both sides of (53) by $(\alpha/w)\, q_i^{1/\alpha}\, y_i^{(\eta-1)/\eta}$ collects all powers of y_i into $y_i^{1/\alpha}$; raising to the power α and substituting (51) gives the best response

$$y_i \;=\; \alpha^{\alpha} \left(\frac{Y}{w}\right)^{\alpha} \left(\frac{\eta-1}{\eta}\right)^{\alpha} q_i \left[\frac{\Sigma_{-i}}{\Sigma_i^2}\right]^{\alpha}. \tag{54}$$

Markups. From (50) and (51), $p_i = Y/(y_i\Sigma_i)$. Substituting this into the right side of (53) shows that marginal cost equals $\frac{\eta-1}{\eta}\, p_i\, \Sigma_{-i}/\Sigma_i$, so the markup is

$$M_i \;=\; \frac{p_i}{\mathrm{MC}_i} \;=\; \frac{\eta}{\eta-1} \cdot \frac{\Sigma_i}{\Sigma_{-i}} \;=\; \frac{\eta}{(\eta-1)(1-\omega_i)}. \tag{55}$$

The markup formula as a function of relative outputs is identical to the constant-returns baseline of Cavenaile et al. (2026); decreasing returns alter the level of quantities but not the mapping from market shares to markups.

Superstar profits. The wage bill satisfies $wl_i = w\, y_i^{1/\alpha} q_i^{-1/\alpha} = \alpha\, \mathrm{MC}_i\, y_i$, so using (55),

$$\pi_i \;=\; p_i y_i - w l_i \;=\; \frac{Y}{\Sigma_i}\left[1 - \alpha\,\frac{\eta-1}{\eta}\,\frac{\Sigma_{-i}}{\Sigma_i}\right] \;=\; \frac{\big(\eta + [(1-\alpha)\eta + \alpha]\,\Sigma_{-i}\big)\,Y}{\eta\,\Sigma_i^2} \;=\; \omega_i\Big(1 - \frac{\alpha}{M_i}\Big)Y,$$

and the wage bill is $wl_i = \alpha\,\omega_i Y/M_i$. The term $(1-\alpha)\eta + \alpha$ multiplying Σ_{-i} captures the additional rent that decreasing returns generate on inframarginal units.

Relative outputs. Taking the ratio of the pre-exponentiation form of (53) for two superstars i and i' and raising to the power α yields

$$\left(\frac{y_i}{y_{i'}}\right)^{\frac{\eta(1-\alpha)+\alpha}{\eta}} \;=\; \frac{q_i}{q_{i'}}\left[\frac{\Sigma_i - 1}{\Sigma_i - \left(\frac{y_{i'}}{y_i}\right)^{\frac{\eta-1}{\eta}}}\right]^{\alpha}, \tag{56}$$

where both sums are normalized by y_i. Combining the superstar best response (54) with the fringe aggregate output (58) below gives the superstar-to-fringe condition

$$\left(\frac{y_i}{\tilde{y}_f}\right)^{\frac{\eta(1-\alpha)+\alpha}{\eta}} \;=\; \left(\frac{\eta-1}{\eta}\right)^{\alpha} \frac{q_i}{q_f}\,(m^{\mathrm{a}})^{\alpha-1}\left[\frac{\Sigma_{-i}}{\Sigma_i}\right]^{\alpha}. \tag{57}$$

Fringe output and profits. Each fringe firm is a price taker and equates price to marginal cost, $p_f = \frac{w}{\alpha}\, y_{fk}^{\frac{1-\alpha}{\alpha}}\, q_f^{-1/\alpha}$. Using $p_f = Y/(\tilde{y}_f\Sigma_f)$ and $\tilde{y}_f = m^{\mathrm{a}}\, y_{fk}$, all powers of y_{fk} again collect

into $y_{fk}^{1/\alpha}$, giving per-firm and aggregate fringe output

$$y_{fk} \;=\; \frac{\alpha^{\alpha}\, Y^{\alpha}\, q_f}{w^{\alpha}\,(m^{\mathrm{a}})^{\alpha}\,\Sigma_f^{\alpha}}, \qquad \tilde{y}_f \;=\; m^{\mathrm{a}}\, y_{fk} \;=\; \frac{(m^{\mathrm{a}})^{1-\alpha}\,\alpha^{\alpha}\, Y^{\alpha}\, q_f}{w^{\alpha}\,\Sigma_f^{\alpha}}. \tag{58}$$

Since $wl_{fk} = \alpha\, p_f\, y_{fk}$, per-firm fringe profit is

$$\pi_{fk} \;=\; (1-\alpha)\, p_f\, y_{fk} \;=\; \frac{(1-\alpha)\, Y}{m^{\mathrm{a}}\,\Sigma_f}. \tag{59}$$

Fringe firms charge no markup, yet decreasing returns ($\alpha < 1$) leave them positive operating profits. This is essential for the lobbying extension: it is these profits, net of the regulatory cost $R(S)$ and the operating cost ε, that give the cutoff $\varepsilon^*(S)$ its content.

Solving the static block. Productivities enter only through ratios. By (5) and (6),

$$\frac{q_i}{q_f} \;=\; \frac{(1+\lambda)^{-n_i}}{\zeta\left(\sum_{i'=1}^{N}(1+\lambda)^{-\gamma n_{i'}}\right)^{1/\gamma}},$$

where $n_{i'}$ is the gap of superstar i' behind the leader. Given the technology state Θ and the active mass m^{a}, equations (56)–(57) form a system in the N relative outputs $\{y_i/\tilde{y}_f\}_{i=1}^{N}$, which we solve by Newton's method with the analytic Jacobian. Its solution determines the aggregators $\{\Sigma_i, \Sigma_f\}$, hence sales shares, normalized profits π_i/Y and π_f/Y, and markups (55); these depend only on (Θ, m^{a}), not on the wage, which the dynamic solver exploits. The derivatives $\partial(\pi_i/Y)/\partial m^{\mathrm{a}}$ and $\partial(\pi_f/Y)/\partial m^{\mathrm{a}}$ in the stakes (36) and (39) follow from the implicit function theorem on the same system. Labor demands $l_i = (y_i/q_i)^{1/\alpha}$ and $l_{fk} = (y_{fk}/q_f)^{1/\alpha}$ are each proportional to Y/w, and enter only labor-market clearing (43).

H Participation in lobbying

This appendix gives the closed form of the participation cutoff (26) in the computed model of Section 4.6, in which a superstar's policies are evaluated at the expected costs of its current spells. Fix a superstar with own state $s_i = (i, \mathbf{\Psi}_i)$ in industry state S. The durations D_I, the values v^0, and the cutoffs Ψ_I^* below depend on s_i only through the rank i, not through the locked costs. A redraw does not change the rank, so after joining or staying out of instrument I at a redraw the firm is at $(S^{I,i,1}, s_i)$ or $(S^{I,i,0}, s_i)$.

Durations. For each instrument I on which the firm is active, $D_I(S, s_i)$ is the expected discounted time over which its current draw is paid, weighted by the scope discount. It

solves, on states where $a_{Ii} = 1$,

$$\begin{aligned}\Big(\rho + \sum_{I'} \delta_{I'} + Q(S, s_i)\Big)\, D_I(S, s_i) \;&=\; \prod_{J \neq I} \big(1 - \iota_{IJ}\big)^{a_{Ji}} \;+\; \sum_{(S', s_i')} Q(S, s_i; S', s_i')\, D_I(S', s_i') \\ &+ \sum_{I' \neq I} \delta_{I'} \Big[\big(1 - \mathcal{P}_{I'}(S, s_i)\big)\, D_I\big(S^{I', i, 0}, s_i\big) + \mathcal{P}_{I'}(S, s_i)\, D_I\big(S^{I', i, 1}, s_i\big)\Big],\end{aligned} \tag{60}$$

where $Q(S, s_i; S', s_i')$ are the rates of the events in (22) other than the firm's own redraws (innovation, breakthroughs, and the redraws of rivals and of the association), $Q(S, s_i) = \sum_{(S', s_i')} Q(S, s_i; S', s_i')$, $\mathcal{P}_{I'}(S, s_i) = \Phi\big((\ln \Psi^*_{I'}(S, s_i) - \mu_{I'})/\sigma_{I'}\big)$ is the probability of being on instrument I' after its next redraw, and $D_I = 0$ after the firm exits. The firm's own redraw on I ends the spell, so it appears in the discount rate with no continuation term.

Values. When the policies do not depend on the locked costs, the value is affine in them, $v(S, s_i) = v^0(S, s_i) - \sum_I a_{Ii}\, \Psi_{Ii}\, D_I(S, s_i)$, where v^0 solves (22) without the locked costs in the flow and with the own-redraw terms

$$\begin{aligned}\delta_I \Big[&\big(1 - \mathcal{P}_I(S, s_i)\big)\, v^0\big(S^{I, i, 0}, s_i\big) + \mathcal{P}_I(S, s_i)\, v^0\big(S^{I, i, 1}, s_i\big) \\ &- D_I\big(S^{I, i, 1}, s_i\big)\, \mathbb{E}\big[\Psi\, \mathbf{1}\{\Psi \leq \Psi^*_I(S, s_i)\}\big] - v^0(S, s_i)\Big].\end{aligned} \tag{61}$$

The expectation is the partial mean of the lognormal distribution (14),

$$\mathbb{E}\big[\Psi\, \mathbf{1}\{\Psi \leq \Psi^*\}\big] \;=\; e^{\mu_I + \sigma_I^2/2}\, \Phi\Big(\frac{\ln \Psi^* - \mu_I - \sigma_I^2}{\sigma_I}\Big).$$

Cutoffs and policies. Comparing the values after joining and after staying out, with the new draw subtracted, gives the cutoff

$$\Psi^*_I(S, s_i) \;=\; \frac{v^0(S^{I, i, 1}, s_i) - v^0(S^{I, i, 0}, s_i) - \sum_{I' \neq I} a_{I'i}\, \Psi_{I'i}\, \big[D_{I'}(S^{I, i, 1}, s_i) - D_{I'}(S^{I, i, 0}, s_i)\big]}{D_I(S^{I, i, 1}, s_i)}. \tag{62}$$

The policies (z_i, b_{Ii}), the cutoffs, and the stake (36) are evaluated at the firm's expected locked costs, $\hat{\Psi}_I(S, s_i) = \mathbb{E}[\Psi \mid \Psi \leq \Psi^*_I(S, s_i)]$, so that the locked costs need not be tracked as a continuous state. Spending and resource costs use the exact distribution of locked costs: on each instrument, the stationary mass of locked costs at every state solves a linear balance in which a spell starts with a draw accepted at the cutoff of the state where it starts and moves with the firm until its next redraw or its exit.

I Beliefs about rivals' tax and procurement statuses

This appendix describes the beliefs that replace rivals' tax and procurement statuses in the computed model of Section 4.6. The reduced industry state $S^{\text{red}} = (\Theta, \mathbf{a}_R, a_A)$ keeps the technology state, the superstars' statuses on regulation, and the association's status, and each superstar's own state keeps its own tax and procurement statuses $\mathbf{a}^{TP} = (a_T, a_P)$. Let $\mu(S^{\text{red}}, i, \mathbf{a}^{TP})$ be the stationary measure of superstars at rank i of reduced state S^{red} with own statuses $\mathbf{a}^{TP}$. The belief about the statuses of the superstar at rank i in technology state Θ is

$$\mathcal{B}(\mathbf{a}^{TP} \mid \Theta, i) \;=\; \frac{\sum_{S^{\text{red}} \in \Theta} \mu(S^{\text{red}}, i, \mathbf{a}^{TP})}{\sum_{S^{\text{red}} \in \Theta} \sum_{\mathbf{a}'} \mu(S^{\text{red}}, i, \mathbf{a}')}, \tag{63}$$

where $S^{\text{red}} \in \Theta$ runs over every configuration of the regulation statuses and the association's status in technology state Θ. Decision objects use the beliefs: rivals' innovation rates and redraw probabilities in (22) are averages of rivals' policies at the actual reduced state over (63), and so are the fringe's pool value, the association's cutoff, and the contest stakes. A superstar's own stake is averaged over its own tax and procurement statuses with the same weights, so that its intensity on regulation, and the regulatory cost, depend on S^{red} alone. The process uses the actual conditional mix, $\mu(\mathbf{a}^{TP} \mid S^{\text{red}}, i) = \mu(S^{\text{red}}, i, \mathbf{a}^{TP}) / \sum_{\mathbf{a}'} \mu(S^{\text{red}}, i, \mathbf{a}')$: the stationary distribution, growth, the locked-cost balances, and every moment. Because no participation status enters the conditioning, a firm's participation choice cannot change what it predicts about its rivals.

J Computing the equilibrium

At the estimated ladder ($\bar{n} = 5$), the computed model has 84 technology states, 2,180 reduced industry states once the regulation statuses and the association's status are included, and 33,104 superstar states once each firm's rank and own tax and procurement statuses are included. The equilibrium is computed by three nested fixed points.

1. **Inner**, given the pool m and the contest outcome at every state: superstars' values, durations, policies and participation cutoffs; the fringe cutoffs, active masses and breakthrough intensities; the association's cutoffs; the beliefs; and the stationary distribution.

2. **Middle**, given m: the stakes from the inner solution, the contest at every state (the regulatory cost, dues and lobbying costs), iterated to a fixed point with Anderson acceleration (Walker and Ni, 2011).

3. **Outer**: the pool m that satisfies free entry (33), by a secant method.

An equilibrium is accepted only if one further evaluation at full precision certifies it: the contest residual is below 10^{-6}, the free-entry residual below 10^{-7} (relative), and the inner fixed point is solved to 10^{-9}. At the estimate, the lobbying equilibrium takes about two hours on one core, and the certificate gives a contest residual of 8.4×10^{-7} and a free-entry residual of 7.7×10^{-10}. The economy without lobbying is computed with the same code, with every participation cost set to infinity. The global search of Section 5.1 solves each point in a fresh process with the same certificate; points whose fixed points do not certify within the time limit are discarded.

Ranks and ties. Statuses and locked costs move with their firms when ranks change. Tied superstars keep their order, and a new superstar ranks below the superstars it ties. A new superstar that would land more than $\bar{n}$ steps above the former leader is placed at $\bar{n}$ steps, and when several superstars are tied at the entry margin, the value after a breakthrough is averaged over which of them is displaced.

Intensities on the regulatory cost. By equations (35) and (41), a superstar that lobbies for a higher cost chooses $b_{Ri} = \left[\Delta_i R_0 \beta^{\text{pro}} \theta / (\kappa_R \phi_R (1 + \beta^{\text{anti}} B^{\text{anti}}))\right]^{1/(\phi_R - \theta)}$, and one that lobbies against chooses $b_{Ri} = \left[|\Delta_i| R \beta^{\text{anti}} \theta / (\kappa_R \phi_R (1 + \beta^{\text{anti}} B^{\text{anti}}))\right]^{1/(\phi_R - \theta)}$. Against a higher cost, R and B^{anti} depend on the firm's own intensity, so the intensities at each state are solved as a fixed point.

K Estimation: identification and moments

This appendix gives the identification argument of each block and reports every moment.

K.1 Technology

We estimate twelve parameters on the 30 moments of Table 14. Every parameter affects every moment, but some moments are especially informative about each parameter.

1. *Quality step and demand (λ, η).* A larger step and a higher elasticity both widen the sales gaps between superstars, so the distribution of sales gaps pins down a combination of the two. Growth equals $\ln(1+\lambda)$ times the rate at which the frontier advances (42), and the payroll ratio is the inverse of the superstar markup, which falls with η. Together they separate the two parameters.

2. *Superstars' innovation (χ, ϕ).* Growth and R&D intensity measure the level of innovation and its cost. By (23), a superstar's innovation rises with the value of an innovation

with elasticity $1/(\phi - 1)$. Patents are proportional to innovation, so the within-firm elasticity of patents to R&D equals $1/\phi$ and measures the convexity directly. The leader's share of superstars' R&D shows how the incentive to innovate differs between leaders and followers, which separates the cost level from the fringe's parameters.

3. *Breakthroughs ($\tilde{\chi}_f$, ϕ_f, u).* The rate at which new superstars appear pins down a combination of the level and convexity of the breakthrough cost. The convexity also sets how breakthroughs respond to the value of entry (29), and this value differs across industries; entry by the previous year's #4/#1 ratio carries that information. A share $(1-u)u^h$ of new superstars lands h steps above the entry margin (10), so the landing shares and the entrants' sales relative to the leader identify u.

4. *Fringe quality ($\tilde{\zeta}$, γ).* A more productive fringe takes a larger share of industry sales, so CR4 pins down $\tilde{\zeta}$. The curvature γ determines whether the fringe learns mostly from the leader or from all superstars (6). It changes how much of an innovation's value superstars keep, which the leader's R&D share reflects.

5. *Fringe costs and turnover ($\bar{\varepsilon}$, σ_ε, δ).* The dispersion of fixed costs measured from selling, general, and administrative expenses pins down σ_ε. Given σ_ε, the mean $\bar{\varepsilon}$ sets the share of new firms that produce, measured by the share of business applications that become employers. Applications per employer firm equal the death rate divided by the producing share of the pool, and the exit rate adds the firms that stop producing; together they pin down δ. The exit premium of loss-making firms measures how close producers are to the cutoff in (30).

We hold out moments the model is not built to match. The gap dynamics of pairs of superstars are checks, because small changes in sales ratios in the data are permanent drift that a ladder of discrete steps cannot produce. The share of firms with losses is a check, because the fringe's gross margin is $1 - \alpha$ by construction.

K.2 Tax and procurement

The block has sixteen parameters: the statutory rate τ_0; the baseline contract Γ_0 and its decline with the firm's gap κ_Γ; for each instrument, the lobbying cost κ_I, its convexity ϕ_I, and the mean, dispersion, and redraw rate of the participation cost $(\mu_I, \sigma_I, \delta_I)$; and the three economies of scope ι_{IJ}. Table 15 lists the moments, which answer six questions about each instrument.

Table 14. Technology moments: data and model

Moment	Data	s.e.	Model	Source
Sales among superstars				
Industries with fewer than four superstars	0.229	0.028	0.209	Compustat
Superstars per industry	3.64	0.05	3.71	Compustat
Leader's share of superstars' sales, median	0.458	0.010	0.469	Compustat
Leader's share of superstars' sales, 90th percentile	0.735	0.035	0.670	Compustat
#2 / #1 sales, 10th percentile	0.235	0.017	0.250	Compustat
#2 / #1 sales, median	0.613	0.019	0.527	Compustat
Industries with #2 / #1 below 0.3	0.162	0.021	0.127	Compustat
Industries with #2 / #1 above 0.8	0.252	0.021	0.188	Compustat
#4 / #1 sales, median	0.288	0.014	0.223	Compustat
Costs and R&D of the largest firms				
Sales / cost of goods sold, superstars	1.497	0.029	1.686	Compustat
Payroll per dollar of sales, four largest firms relative to the rest	0.764	0.029	0.662	Economic Census
Leader's share of superstars' R&D	0.484	0.034	0.455	Compustat
Within-firm elasticity of patents to R&D	0.262	0.104	0.226	Compustat, patents
New superstars				
New superstars per industry-year	0.132	0.013	0.144	Compustat
lowest tercile of the previous year's #4 / #1	0.105	0.013	0.136	Compustat
highest tercile of the previous year's #4 / #1	0.171	0.025	0.181	Compustat
Landing at the entry bar	0.636	0.032	0.480	Compustat
Landing one step above the bar	0.247	0.024	0.273	Compustat
Landing two steps above the bar	0.069	0.016	0.075	Compustat
Log sales relative to the leader, mean	−1.255	0.069	−1.223	Compustat
Log sales relative to the leader, standard deviation	0.639	0.029	0.545	Compustat
Fringe				
CR4	0.213	0.012	0.255	Economic Census
Exit premium of loss-making fringe firms	0.035	0.003	0.029	Compustat
Dispersion of log fixed costs, active fringe firms	0.648	0.030	0.608	Compustat
Firm exit rate	0.079	0.003	0.075	BDS
Business applications per employer firm	0.242	0.008	0.258	BFS, BDS
Applications that become employers within two years	0.273	0.006	0.288	BFS
Aggregates				
Growth of business output per hour	0.0159	0.0040	0.0172	Fernald (2014), 2008–2022
Labor share	0.596	0.004	0.644	Karabarbounis and Neiman (2014)
R&D funded by business / business value added	0.0261	0.0015	0.0125	NSF, BEA

Notes: Superstars are the four largest Compustat firms of a four-digit NAICS industry-year with at least 10 percent of the leader's sales, in industry-years with at least six Compustat firms, 2008–2022. CR4 is the median industry-year in the Economic Census. In the data, a step of the ladder is a sales ratio of 1.27. Standard errors from a bootstrap; growth: Newey-West. BDS: Business Dynamics Statistics; BFS: Business Formation Statistics. Appendix L defines each moment. The model's labor share and R&D are in the units of the data, with a capital share of 0.195.

1. *What happens without lobbying?* The effective tax rate of all US-headquartered firms identifies τ_0. Contracts over sales identify Γ_0, and the elasticity of contracts to sales across firms identifies κ_Γ.

2. *What does lobbying buy?* This identifies the lobbying cost κ_I. A year of procurement lobbying is followed by contracts 0.22 log points higher within the firm. On taxes, among firms that lobby, a log point more spending per ask is followed by a tax rate on taxable income 0.42 percentage points lower over the next three years, before the 2017 tax reform.

3. *Who lobbies?* Participation by rank sets the mean of the participation cost. Its slope in the firm's stake within an industry-year sets the dispersion: when the cost varies little across firms, the stake decides who lobbies and the slope is steep.

4. *How long do they stay?* Persistence, net of the industry state, identifies the redraw rate: given the state, a firm stays in or out because its participation cost persists.

5. *How much do they spend?* At an interior optimum, spending on an instrument equals the gain from lobbying divided by ϕ_I, so spending relative to the stake identifies the convexity. The data bound this ratio rather than measure it, because a report's amount covers all of its asks; the bounds come from two ways of splitting the amount.

6. *Which instruments go together?* A discount on the participation cost of a second instrument makes a firm that lobbies on one more likely to lobby on another, net of rank, industry, and year. These links identify the economies of scope.

K.3 Regulation

The block has fourteen parameters: the compliance level $\mathcal{C}$, which sets R_0; the elasticity e_R of superstars' compliance costs to their size; the contest curvature θ and the effectiveness of the anti side β^{anti}; the cost of influence κ_R and the convexities ϕ_R and ϕ_A; the participation cost of superstars $(\mu_R, \sigma_R, \delta_R)$; and the association's organizing cost $(\mu_A, \sigma_A, \delta_A)$ and lobbying cost κ_A. Table 16 lists the moments.

Compliance costs and size. Both $\bar{c}$ and γ_R follow from one parameter, the elasticity $e_R \in [0, 1)$ of compliance costs to size. Let $a + b\,n$ be the least-squares line of the log of a superstar's sales relative to one fringe firm on its gap n, over every superstar seat of every technology state, in the product market equilibrium with $\hat{m}^{\text{a}}$ active fringe firms. Then $\bar{c} = e^{e_R a}$ and $\gamma_R = -e_R b / \ln(1+\lambda)$, so that c_{ijt} is approximately the superstar's sales relative to one fringe firm raised to the power e_R.

Table 15. Tax and procurement moments: data and model

	Tax			Procurement			
Moment	Data	s.e.	Model	Data	s.e.	Model	Identifies
Effective tax rate, all firms	0.222	0.006	0.242				τ_0
Contracts / sales				0.019	0.006	0.017	Γ_0
Elasticity of contracts to sales				0.705	0.234	1.242	κ_Γ
Payoff: outcome after lobbying	−0.0042	0.0028	−0.0040	0.217	0.098	0.006	κ_I
Lobbying in a quarter, leader	0.245	0.025	0.409	0.176	0.021	0.213	μ_I
Lobbying in a quarter, ranks 2–4	0.135	0.016	0.225	0.084	0.012	0.133	μ_I
Lobbying on log stake, within industry-year	0.062	0.009	0.145	0.036	0.005	0.054	σ_I
Still lobbying a year later, given the state	0.591	0.024	0.283	0.626	0.029	0.547	δ_I
Spending per ask / stake, lower bound ($\times 10^{-5}$)	0.88		0.99	3.9		8.8	ϕ_I
Spending per ask / stake, upper bound ($\times 10^{-5}$)	7.4		0.99	38		8.8	ϕ_I
Across instruments, in a quarter							
Tax on regulation	0.252	0.026	0.019				ι_{RT}
Procurement on regulation				0.119	0.020	0.029	ι_{RP}
Tax on procurement	0.119	0.028	0.028				ι_{TP}

Notes: Superstars headquartered in the United States, fiscal years 2008–2022, weights equal across industries, unless stated. Effective rate: total tax over pretax income, all US-headquartered firms with positive income (46,235 firm-years). Tax payoff: three-year cash taxes over sales on taxable income over sales, interacted with lobbying last year and log dollars per ask; windows ending by December 2017. Procurement payoff: log prime contracts on last year's share of quarters lobbied, with firm and year fixed effects. Stake: pretax income (tax), last year's contracts (procurement). Persistence: lobbying in the fiscal fourth quarter of next year on this year's, with industry, rank, and year fixed effects and controls for the industry state. Across instruments: linear probability model of one instrument on the other within a quarter, with rank, industry, and year fixed effects. Appendix L gives the definitions.

1. *Compliance costs ($\mathcal{C}$, e_R).* We bound the level from Trebbi et al. (2026). Their measure of compliance labor in private establishments, as a share of output, is 0.96 percent, and we use it as an upper bound; the share of that cost serving our four regulatory instruments, 0.33 percent, is the lower bound. No study measures how the cost of these instruments rises with firm size, so e_R lies in the band from 0.40 to 0.98. The lower end reflects measured costs of specific rules, 0.33 to 0.84; the upper end reflects occupation-based measures across all compliance work, 0.95 to 0.98, which likely overstate it (Section 5.4 gives the sources).

2. *Sides (e_R).* Within this band, the share of positions asking for higher costs helps locate e_R. A superstar gains from higher costs through rivals' costs and deterred entry, and pays its own compliance bill. The faster that bill rises with size, the more its own cost weighs against those gains, and the more superstars ask for lower costs. In the data 53 percent of superstars' positions and 34 percent of associations' positions ask for higher costs. In the model, a position's share is the part of the stake coming from channels that favor higher costs.

3. *Who lobbies (μ_R, σ_R, δ_R).* As on the other instruments, participation by rank and its slope in the firm's share of industry sales identify the mean and dispersion of the participation cost, and persistence identifies the redraw rate.

4. *Force and spending (θ, ϕ_R, κ_R).* How fast spending rises with size within an industry disciplines the curvature θ and the convexity ϕ_R. Superstars' spending per lobbying quarter is held below a ceiling, and what spending buys is identified on bills (Section 5.5).

5. *The association (μ_A, σ_A, δ_A, κ_A).* How often the association is active, how long it stays active, and how often it lobbies without superstars and superstars lobby without it identify its organizing cost. A ceiling on its spending disciplines κ_A and ϕ_A. The difference in superstars' participation between quarters with and without an active association measures how the two sides respond to each other.

K.4 Bills

Bills without model agents. We measure the baseline passage probability P_ℓ^0 directly from bills on which no superstar and no industry association lobbied. We classify these bills with the procedure of Section 2.2: all that became law, every bill whose text passed inside another law, and a random sample of 4,000 others, weighted to the population. A bill counts

Table 16. Regulation moments: data and model

Moment	Data	s.e.	Model	Identifies
Compliance costs				
Compliance costs / output, band	0.0033 to 0.0096		0.0082	$\mathcal{C}$
Size elasticity of compliance costs, band	0.40 to 0.98		0.81	e_R
Superstars				
Lobbying in a quarter, leader	0.328	0.030	0.484	μ_R
Lobbying in a quarter, ranks 2–4	0.194	0.022	0.288	μ_R
Lobbying on log share of industry sales	0.149	0.013	0.163	σ_R
Still lobbying a year later, given the state	0.600	0.023	0.905	δ_R
Positions asking for higher costs	0.533	0.089	0.671	e_R
Log dollars on log sales, relative to the leader	0.545	0.115	1.268	θ, ϕ_R
Leader's dollars per lobbying quarter / value added, ceiling		2.0×10^{-4}	6.5×10^{-5}	κ_R
Associations				
Active in an industry-quarter	0.558	0.031	0.897	μ_A
Active next quarter, given the state	0.658	0.021	0.975	δ_A
Association lobbies, no superstar	0.244	0.024	0.175	μ_A, σ_A
Superstar lobbies, no association	0.134	0.018	0.083	μ_A, σ_A
Superstars lobbying: association active − inactive	0.042	0.027	−0.004	κ_A
Positions asking for higher costs	0.343	0.029	0.273	e_R
Dollars / industry value added, ceiling		4.3×10^{-5}	6.8×10^{-5}	κ_A, ϕ_A

Notes: Superstars headquartered in the United States and associations in the census of industry associations, 2008–2022, on industry-quarters with at least six Compustat firms, weights equal across industries. Compliance band: Trebbi et al. (2026), $103 billion in 2014, over business value added and one minus the capital share; the lower bound multiplies it by 0.345, the share of their compliance cost serving our four instruments. Elasticity band: rule-level studies, including Dahl et al. (2018) and Ewens et al. (2024), and occupation-based measures from Trebbi et al. (2026). Positions: each position counts once, by the share of its asks that raise costs. Appendix L gives the definitions.

as passed if it became law or if its core regulatory provision is in a law of the same Congress. Table 17 shows the result. Without a model agent, 3.42 percent of raising bills pass (95 percent interval 2.21 to 4.95) and 7.67 percent of lowering bills (4.07 to 13.61). Lobbying by other clients changes these rates little. With a model agent alone the rates are 4.4 and 8.3 percent; with model agents and other clients together, 9.0 and 16.0 percent. Big bills attract every kind of lobbyist, so these differences are not the effect of lobbying. They show why the model must fit the level of passage as well as its slope in lobbying.

Table 17. Passage of regulation bills by who lobbied

	Raising			Lowering		
Who lobbied	Rows	Bills	Passed	Rows	Bills	Passed
Nobody	64	441	2.7%	13	61	9.8%
Other clients only	149	507	4.0%	55	159	6.9%
Superstars or associations only	228	228	4.4%	84	84	8.3%
Superstars or associations, and other clients	1,341	1,341	9.0%	426	426	16.0%

Notes: Regulation bills, 2008–2022; rows read, bills after sample weights. Other clients: any client other than a superstar or an industry association that listed the bill before its outcome. Passed: law, or the core provision in a law of the same Congress.

Baseline. We estimate how bill features move the baseline on the 39,835 proposals without model agents, by a probit of passing on the sponsor's majority status, whether the sponsor chairs a referral committee, bipartisan sponsorship, original cosponsors, whether other clients lobbied, and Congress. Other clients raise the index by 0.076 (standard error 0.034). This gives each bill an index $\mathcal{I}_\ell$, and

$$P_\ell^0 \;=\; \Phi\Big(\sqrt{1+\sigma_Z^2}\,\big(\mathcal{I}_\ell + \mathcal{I}^0_{\text{type}}\big) + \sigma_Z\, Z_\ell\Big), \qquad Z_\ell \sim N(0,1), \tag{64}$$

where the shift $\mathcal{I}^0_{\text{type}}$ sets the average of P_ℓ^0 on the classified bills to the rates above for every value of σ_Z, the dispersion of the unobserved component Z_ℓ. Other clients are not agents in the model, so they enter only here, as a feature of the bill.

Simulation. We simulate every real proposal. Each industry that lobbied a proposal in the data is drawn from the stationary distribution, conditional on one of its firms being active, and forces add up across industries. A firm appears on the proposal if its dollars on it exceed a threshold, and it reveals its side with the probability observed in the data. Who lobbies each proposal is therefore simulated, in the same way as the passage of the proposal. On bills without model agents, $P_\ell = P_\ell^0$. At each parameter point, the repeal intensity is set by the ratio of compliance costs removed to costs added by the laws of the period, 0.050 (standard error 0.068), the listing threshold by the number of superstars and the presence of

associations on each industry's row of a proposal, and σ_Z by the fit to the regression below and to passage.

K.5 Untargeted moments

Table 18 compares moments that the estimation does not target.

Table 18. Untargeted moments

	Data	Model
Who lobbies on regulatory costs in an industry-quarter		
Both superstars and the association	0.314	0.722
Neither	0.308	0.020
Instruments a superstar lobbies on (share of quarters)		
None	0.661	0.466
Regulatory costs only	0.096	0.213
Tax only	0.040	0.114
Procurement only	0.038	0.062
All three	0.042	0.016
Industry structure and markups		
Industries with at most two superstars	0.111	0.071
Cost-weighted superstar markup	1.27	1.47

L Target moments

This appendix defines each targeted moment and its model counterpart. Table 14 reports the technology moments and Tables 15 and 16 the lobbying moments. All moments receive equal weights (Section 5.1).

L.1 Technology moments

Sample. The firm sample is the Compustat panel of firms headquartered in the United States, 2008–2022. An industry is a four-digit NAICS code, and an industry-year enters if it has at least six Compustat firms: 2,131 industry-years in 170 industries. The superstars of an industry-year are the firms among its four largest by sales with at least 10% of the leader's sales (7,752 firm-years of 1,082 firms); all other firms form the fringe. For moments that follow firms across years, a change in a firm's reported industry code is treated as a re-coding, not a move, when its segment data show the same business in both years. A firm-year is affected by a material acquisition if cash acquisitions or the sales of acquired firms exceed

10% of the firm's sales, or goodwill rises by more than 5% of sales. In the model, statistics of industries are computed over the stationary distribution of industry states, and statistics of firms over time from the equilibrium transition rates or a simulated panel. Table 19 lists the sources.

Table 19. Technology moments: sources and samples

Moments	Source	Years	Observations
Sales among superstars	Compustat	2008–2022	2,131 industry-years
Sales / cost of goods sold	Compustat	2008–2022	superstar firm-years
Payroll ratio	Economic Census	2017, 2022	132 industries
New superstars	Compustat	2008–2022	2,089 industry-years; 275 new superstars
CR4	Economic Census	2012, 2017, 2022	132 industries
Exit premium of loss-making firms	Compustat	2008–2022	61,266 fringe firm-years
Dispersion of fixed costs	Compustat	2008–2022	5,801 fringe firms
Firm exit rate	Business Dynamics Statistics	2008–2022	all four-digit industries
Business applications	Business Formation Statistics; Business Dynamics Statistics	2008–2021	all sectors
Growth	Fernald (2014), business output per hour	2008–2022	annual
Labor share	Karabarbounis and Neiman (2014); OECD sector accounts	2008–2022	annual
R&D intensity	National Science Foundation	2008–2022	annual

Sales among superstars. The share of industries with fewer than four superstars is the share of industry-years in which the fourth-largest firm sells less than 10% of the leader's sales. The leader's share is its sales over the superstars' total sales (median and 90th percentile). The ratio #2/#1 is measured in industry-years with at least two superstars (10th percentile, median, and shares below 0.3 and above 0.8), and #4/#1 in industry-years with four superstars (median). The model computes each statistic with the same conditioning over the stationary distribution.

Cost structure of the largest firms. The sales-to-cost ratio is the superstars' total sales over their total cost of goods sold, with each firm's dollars divided by the Compustat sales of its industry-year so that industries have equal size, as in the model; in the model, it is superstars' sales over their variable cost, which equals their cost-weighted markup divided by α. The payroll ratio is the payroll per dollar of receipts of the four largest firms of an industry over that of all its other firms in the Economic Census, averaged over 2017 and 2022 and across the 132 industries of our panel in the Census, weighted by industry-years; in the model, it is the superstars' labor cost per dollar of sales over the fringe's, which equals the inverse of the superstars' markup.

New superstars. A new superstar in year t is a superstar that was in the same industry in $t-1$ without being a superstar, has no material acquisition in t, and would remain a superstar if the firms that left the industry's four largest between $t-1$ and t were still present, with their sales grown at the industry's median rate. The entry rate is the number of new superstars per industry-year, overall and in the lowest and highest terciles of the previous year's #4/#1 (2,089 industry-years, 275 new superstars). A new superstar's landing is its real sales in t over the real entry bar of $t-1$, the larger of the fourth firm's sales and 10% of the leader's; its number of steps above the bar is the log landing divided by $\ln 1.27$, rounded, with landings below the bar set to zero. The step of 1.27 is the median sales ratio of the third to the fourth superstar in industry-years with four superstars. We target the shares landing zero, one, and two steps above the bar and the mean and standard deviation of log sales relative to the leader in t, excluding the two new superstars that become leaders. The model computes the same statistics exactly over one year of the continuous-time process, with sales grown by one year of aggregate growth so that landings are in real terms.

Gap dynamics. For each pair of superstars in year t, the sales ratio is the smaller firm's sales over the larger firm's. Pairs are tied (ratio at least 0.9), close (0.7 to 0.9), separated (0.4 to 0.7), or far (below 0.4). A pair moves *apart* if its ratio is below its value in t in both $t+1$ and $t+2$, and *closer* if it is above in both years. The sample keeps pairs whose firms remain superstars of the industry in $t+1$ and $t+2$ without material acquisitions: 3,825 of the 9,521 pairs with base years 2008–2020. The model computes these probabilities exactly from the transition rates, for every pair of superstars in every state, weighted by the stationary distribution and conditional on both firms remaining superstars.

Concentration. CR4 is the receipts share of the four largest firms in the Economic Census, averaged over 2012, 2017, and 2022; we take its median across the 132 industries of our panel present in all three Censuses, weighted by industry-years. In the model, it is the superstars' share of industry sales, averaged over the stationary distribution.

Fringe firms. The loss share is the share of corporate returns without net income among corporations with assets of \$1 to \$500 million, by minor industry of the IRS Corporation Source Book, 2008–2013, each industry-year weighted equally; in the model, it is the share of producing fringe firms whose operating profit net of compliance, breakthrough spending, and fixed cost is negative. The exit premium is the coefficient from a regression of an indicator that a fringe firm's last Compustat year is t and it was deleted for bankruptcy, liquidation, or other reasons (deletion codes 2, 3, and 10) on an indicator of negative operating income before depreciation in t, with industry-year fixed effects and industry-years weighted equally; the model runs the same regression on a simulated panel in which a firm exits when it dies or stops producing. The dispersion of fixed costs is the standard deviation across 5,801

fringe firms of the firm average, over at least three years, of the residual from regressing log selling, general, and administrative expenses net of R&D on log sales with industry-year fixed effects (slope 0.667); in the model, it is the same statistic for the log of the fixed cost plus compliance of producing fringe firms, net of its industry-year mean. The exit rate is firm deaths over firms in the Business Dynamics Statistics, summed over industries and averaged over 2008–2022; in the model, it is the share of producing fringe firms that die or stop producing within a year. Applications per employer firm are high-propensity applications in the Business Formation Statistics over employer firms in the Business Dynamics Statistics, averaged over 2008–2021, and the share that become employers is business formations within eight quarters over high-propensity applications. In the model, applications are entries into the pool, so the first moment is the entry flow per producing fringe firm and the second the share of new firms that produce within two years before dying.

Aggregates. Growth is the mean annual log growth of business output per hour of Fernald (2014) over 2008–2022; in the model, it is the growth rate (42). The labor share is the corporate labor share of Karabarbounis and Neiman (2014), extended with the OECD accounts of nonfinancial and financial corporations, averaged over 2008–2022; in the model, it is the wage bill over output times one minus the capital share. R&D intensity is business-funded R&D over GDP from the National Science Foundation, averaged over 2008–2022; in the model, it is superstars' R&D plus fringe breakthrough spending, over output.

Untargeted moments. The superstar markup is the production-function markup computed as in De Loecker et al. (2020), weighted by cost of goods sold relative to industry-year sales; in the model, superstars' markups are weighted by their variable cost. The patent elasticity is the coefficient from a Poisson regression of a superstar's granted patents, by filing year, on the log of its R&D over industry-year sales, with firm and year fixed effects, on superstar firm-years with positive R&D, 2008–2020 (2,356 firm-years of 380 firms). Patents are matched to firms with the match of Dyèvre and Seager, extended to recent filing years with that of Kogan et al. (2017). In the model, innovation is $z = (\mathrm{R\&D}/\chi)^{1/\phi}$, so the elasticity is $1/\phi$.

L.2 Lobbying moments

Tables 15 and 16 give the lobbying moments and their samples. Compustat superstars are headquartered in the United States; regulation and association moments cover 2008–2022, and tax and procurement participation is measured at the last quarter of the fiscal year, fiscal years 2008–2022. Every lobbying moment weights industries equally.

M From bills to the contest

This appendix describes how the enactment of bills maps into the regulatory fixed cost of the contest (13), and how the bill moments are simulated.

Rules add up. The regulatory cost of an industry is the stock of rules in force. Bills that raise it arrive at rate $\Lambda^{\uparrow}$, and an enacted bill ℓ of this type adds R_ℓ^+ to the stock. Bills that lower it arrive at rate $\Lambda^{\downarrow}$, and an enacted bill ℓ of this type removes a share R_ℓ^- of the stock. Rules lapse at rate δ^{rule}. Bill ℓ passes with probability P_ℓ, given by (48), and with probability P_ℓ^0 without lobbying. In the steady state, $\Lambda^{\uparrow}\mathbb{E}^{\uparrow}[R^+P] = \big(\delta^{\text{rule}} + \Lambda^{\downarrow}\mathbb{E}^{\downarrow}[R^-P]\big)R$, where $\mathbb{E}^{\uparrow}$ and $\mathbb{E}^{\downarrow}$ average over bills that raise and lower the cost, so that

$$\frac{R}{R_0} \;=\; \frac{\mathbb{E}^{\uparrow}[R^+P]}{\mathbb{E}^{\uparrow}[R^+P^0]} \cdot \frac{1+\mathcal{E}}{1+\mathcal{E}\,\mathbb{E}^{\downarrow}[R^-P]/\mathbb{E}^{\downarrow}[R^-P^0]}, \qquad \mathcal{E} \;=\; \frac{\Lambda^{\downarrow}\,\mathbb{E}^{\downarrow}[R^-P^0]}{\delta^{\text{rule}}}. \tag{65}$$

Here $\mathcal{E}$ measures how fast bills that lower the cost erode the stock relative to lapsing, and we fit $\mathcal{E} = 0.035$. Without lobbying, $P = P^0$ and $R = R_0$. The result is exact when bills arrive independently of the rules in force and the industry's lobbying is constant while a rule lasts.

How a participant splits its influence. Participant i puts a share $H_{i\ell}$ of its lobbying dollars on bill ℓ, with mean one across bills. With a lobbying cost of convexity ϕ_R, its influence on bill ℓ grows as $H_{i\ell}^{\theta/\phi_R}$. To first order in influence, the value of influence on bill ℓ for R/R_0 is

$$\mathcal{V}_\ell^{\uparrow} \;=\; \frac{\Lambda\, R_\ell^+}{\Lambda^{\uparrow}\,\mathbb{E}^{\uparrow}[R^+P^0]} \times \begin{cases} 1-P_\ell^0 & \text{push} \\ P_\ell^0 & \text{block} \end{cases}, \qquad \mathcal{V}_\ell^{\downarrow} \;=\; \frac{\mathcal{E}}{1+\mathcal{E}}\,\frac{\Lambda\, R_\ell^-}{\Lambda^{\downarrow}\,\mathbb{E}^{\downarrow}[R^-P^0]} \times \begin{cases} 1-P_\ell^0 & \text{push} \\ P_\ell^0 & \text{block} \end{cases},$$

where $\Lambda = \Lambda^{\uparrow} + \Lambda^{\downarrow}$, and $\mathcal{V}_\ell$ denotes $\mathcal{V}_\ell^{\uparrow}$ or $\mathcal{V}_\ell^{\downarrow}$ by the type of the bill. Let $\beta_{i\ell}$ be the participant's effectiveness on the bill, one for pushing and β^{anti} for blocking. The optimal split is $H_{i\ell} \propto (\mathcal{V}_\ell\beta_{i\ell})^{1/(1-\theta/\phi_R)}$. We scale influence so that the participant's influence on bill ℓ is $B_{i\ell} = B_i\,\varsigma_{i\ell}$, with $\varsigma_{i\ell} = \beta_{i\ell}H_{i\ell}^{\theta/\phi_R}/\mathbb{E}[\mathcal{V}\,\beta\,H^{\theta/\phi_R}]$ the share in equation (48), $B_i = \beta^{\text{pro}}b_{Ri}^{\theta}$ for a participant that lobbies for a higher cost, and $B_i = \beta^{\text{anti}}b_{Ri}^{\theta}$ for one that lobbies against. The association splits its influence in the same way, with its own convexity ϕ_A. Then $\mathbb{E}[\mathcal{V}_\ell B_{i\ell}] = B_i$, and to first order the bills give $R/R_0 - 1 = \beta^{\text{pro}}B^{\text{pro}} - \beta^{\text{anti}}B^{\text{anti}}$, which is the first-order expansion of the contest (13).

Pushing and blocking. A participant with influence B alone on a bill that raises regulatory costs changes its passage probability by $B(1-P^0)/(1+B)$ by pushing and by $-P^0\beta^{\text{anti}}B/(1+\beta^{\text{anti}}B)$ by blocking. Since P^0 is small, blocking matters only if β^{anti} is large. To first order, lobbying superstars raise passage on average only if $\beta^{\text{anti}} < \mathcal{S}^{\text{pro}}(1-P^0)/((1-$

$\mathcal{S}^{\text{pro}})P^0)$, where $\mathcal{S}^{\text{pro}}$ is the share of lobbying superstars that lobby for a higher cost. The association's dues determine its blocking influence rather than β^{anti}, because a larger β^{anti} with less intensity gives the same dues and the same blocking. Information on β^{anti} comes from association opposition on bills that raise regulatory costs and from bills that lower them, on which the roles flip.

N Welfare decomposition

Exact decomposition. Superstar i of an industry produces $y_i = q_i l_i^\alpha$, and the fringe of m^{a} identical firms, efficient internally, produces as one producer $Y_f = (m^{\text{a}})^{1-\alpha} q_f L_f^\alpha$. Industry output is $y_j = (\sum_i y_i^{(\eta-1)/\eta} + Y_f^{(\eta-1)/\eta})^{\eta/(\eta-1)}$, and $\ln Y_0 = \sum_S \mu(S) \ln y(S)$. Since every producer is of degree α in labor, $\ln Y_0 = \alpha \ln L + \ln \mathcal{A}^* + \ln E$ exactly, with

$$\ln \mathcal{A}^* \;=\; \sum_S \mu(S)\, \frac{1}{\upsilon} \ln \Big[\sum_{i=1}^{N(S)} \mathcal{A}_i(S)^\upsilon + \mathcal{A}_f(S)^\upsilon \Big], \qquad \upsilon = \frac{\eta - 1}{\eta - \alpha(\eta - 1)},$$

the productivity of a planner who allocates the same labor equally across industries (all have the same expenditure share and the same α) and within each industry in proportion to $\mathcal{A}_i^\upsilon$ and $\mathcal{A}_f^\upsilon$, given the industry states and the operating fringe. The term $\ln E = \ln Y_0 - \alpha \ln L - \ln \mathcal{A}^* \leq 0$ splits into a within-industry and an across-industry part. Welfare (46) is $W = \ln C_0/\rho + g/\rho^2 - D/\rho$ with $\ln C_0 = \ln Y_0 + \ln(C_0/Y_0)$, so that

$$100 \ln(1 + \text{CEV}) \;=\; 100 \Big[\alpha\, \Delta \ln L + \Delta \ln \mathcal{A}^* + \Delta \ln E + \Delta \ln(C_0/Y_0) + \frac{\Delta g}{\rho} - \Delta D \Big],$$

where Δ is the economy without lobbying minus the lobbying economy. The change in $\ln \mathcal{A}^*$ is split into the change within industry states, at the stationary distribution of the economy with lobbying, and the change due to the shift in the industry states, which itself splits into the superstars' configuration and the fringe's technology. The change in C_0/Y_0 is split by item of the resource constraint (44), and the change in growth by source (42).

Other preferences. For balanced-growth preferences with flow utility $[C\, e^{-D}]^{1-1/\epsilon_C}/(1 - 1/\epsilon_C)$, where ϵ_C is the elasticity of intertemporal substitution, our preferences are the case $\epsilon_C = 1$. The growth term is then $\frac{1}{1-1/\epsilon_C} \ln \frac{r-g}{r^{\text{ref}} - g^{\text{ref}}}$, where $r - g = \rho - (1 - 1/\epsilon_C)\, g$ is the gap between the interest rate and the growth rate. To first order, the growth term is $(g^{\text{ref}} - g)/(r^{\text{ref}} - g^{\text{ref}})$, the change in growth capitalized at the interest rate net of growth. Thus the curvature and the discount rate matter only through $r - g$, which equals ρ with log utility. To first order, the hours term is $[(1 - \tau_w)\, wL/C]\, (\ln L - \ln L^{\text{ref}})$, the change in

hours valued at the after-tax wage (equation (17)).

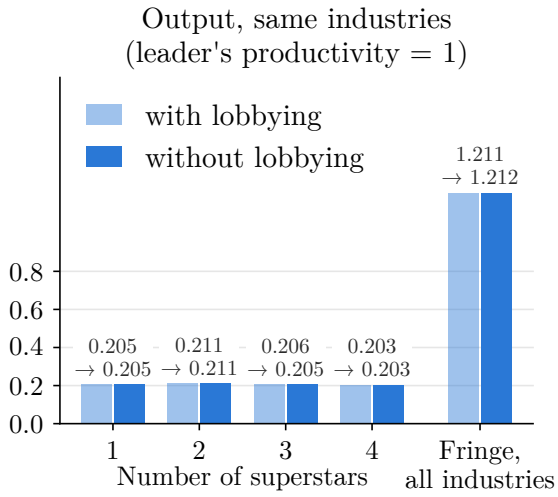


Figure 18. Output of superstars and the fringe, same industry states

Notes: Output of all superstars in an industry, by number of superstars, and fringe output in all industries, in units of the leader's quality: log average across industries, from the economy with lobbying to the economy without lobbying in the same industry states. At the estimate.